THE INCOMPLETENESS OF OBSERVATION

Finite-Observer Representations and Quantum Reconstruction

Author: Alex Maybaum
Date: April 2026
Status: DRAFT PRE-PRINT
Classification: Theoretical Physics / Foundations

ABSTRACT

An observer embedded in a finite deterministic system cannot access the complete state. This paper establishes an exact finite-horizon equivalence between three descriptions of what such an observer can record: finite stochastic laws (S), visible marginals of finite reversible deterministic systems (D), and finite-dimensional unitary systems read out by projective Born-rule measurement in one fixed basis at every step, from an initial state diagonal in that basis (Q_{fb}). Thus

$$S \iff D \iff Q_{\mathrm{fb}}$$

at the finite observable-law level. The quantum representation is exact, not merely approximate; it is also universal, so representability alone does not distinguish the memory-bearing sector from an ordinary Markov process. The Born dictionary $p = |U|^2$ is part of the definition of Q_{fb}: the equivalence exhibits the Born form and does not select the quadratic exponent, which is named as a frontier item (§3.1). The discriminating content is history readback: (C4) holds exactly when the visible law has positive conditional memory $M_t = I(X_{<t}; X_{t+1} \mid X_t)$ at some accessible time, and then every faithful deterministic completion contains readout-relevant hidden predictive memory. On one fixed finite reversible representative the full rooted visible process is indivisible somewhere in its recurrence cycle whenever some rooted map is not a permutation, since finite recurrence returns the visible map to the identity and a non-permutation stochastic matrix has no stochastic inverse. With a full-support visible prior this indivisibility is equivalent to a memory obstruction, $M_k > 0$ at some order k below the recurrence time, and that order may lie as late as one step before the recurrence time. Observational access is a separate requirement: (C4) asks for the obstruction at an accessible order, and recurrence does not guarantee one, since every non-deterministic visible step gives the indivisibility, including in laws that are Markov on every accessible window. Conditions (C1) and (C3) are then necessary diagnostics of the hidden mediation and its capacity; (C2-structural), the persistence of the stored record until readback, holds by reversibility in every bijective realization and constrains coarse-grained descriptions of the hidden sector, with the slow-bath inequality one sufficient physical mechanism.

For spatial OI systems, the coupling graph defined by one-step dynamical de-

pendence supplies exact causal cones: influence propagates at most one graph edge per update. Bell accounting is correspondingly sharp: a deterministic completion whose setting interventions, applied at one step, each read and write only their own wing, whose readouts lie outside the other wing's cone and record a ± 1 outcome in every run, and whose pre-setting ensemble is setting-independent is a local hidden-variable model, so $|S_{\mathrm{CHSH}}| \leq 2$ and quantum correlations are unreachable. Reproducing them requires ontic parameter dependence — targeted in the specific SM/GR branch through preparation-indexed state-dependent adjacency or an equivalent Bell-nonlocal response structure, with no operational signal — or else setting-dependent ontic ensembles, for which $S_{\mathrm{CHSH}} \leq 2 + 4\sqrt{2 \ln 2\, I_{\mathrm{ont}}}$ gives the model-independent necessary lower bound $I_{\mathrm{ont}} \geq 0.0309$ bits for the quantum value. Conversely, every finite stochastic law, and therefore every finite quantum record, admits a finite reversible realization; for fixed finite dimension, instrument menu, horizon, and accuracy ε, one finite OI construction reproduces the whole corresponding quantum experimental envelope to that accuracy. In this precise sense the stochastic description of embedded observation and its quantum representation are mathematically equivalent. Two logically separate completion questions remain beyond that finite-law result: whether the complete family of coherent interventions and composites is represented simultaneously by the standard quantum instrument theory (§3.4), and whether preparation-indexed Bell adjacency in the state-dependent cubic SM/GR graph supplies the required ontic parameter dependence with operational no-signaling (part of H-Bell, [Substratum §3.3]). Accordingly, the exact equivalence claim is at the finite observable-law level; universal Hilbert-space representability is not promoted into a uniqueness theorem for operational quantum mechanics entire. In particular one visible family can admit distinct coherent unitary lifts whose relating evolutions differ, so representability together with ordinary coherence does not by itself select the unitary evolution relating two times (§3.4 Remark). The specific nearest-neighbor cubic representative used for the Standard-Model sector is not by itself a Bell-violating completion under measurement independence; whether preparation-indexed Bell adjacency supplies the required ontic parameter dependence with operational no-signaling is part of the separate open hypothesis H-Bell of [Substratum §3.3].

1. INTRODUCTION

1.1 The Problem of Embedded Observation

An observer embedded in a deterministic universe cannot access the complete state: degrees of freedom beyond causal reach are permanently hidden, and the observer's description is obtained by marginalizing over the hidden sector. Whether a description obtained this way can be considered complete is the question Einstein, Podolsky, and Rosen posed of quantum mechanics [46]; it returns here with the opposite valence — incompleteness is derived as a structural property of embedded observation, and the finite observable laws an embedded observer records are identified exactly with fixed-basis unitary Born representa-

tions (§3.4); whether the full operational theory of quantum mechanics follows is the separate completion question stated there. This incompleteness is inaccessibility of the hidden sector, a statement distinct from the uncertainty relations of quantum theory (§3.4). The question is what this reduction imposes on the form of the observer's physical laws.

Prior work (QBism, relational QM, 't Hooft's cellular automaton [1]) takes observer-dependence as an interpretive starting point or derives it from specific microphysical models. This framework instead proves the finite observable-law correspondence directly: every finite stochastic law has both a finite reversible realization and an exact fixed-basis unitary/Born representation, and every finite reversible visible law lies in the same class (§3.4). History readback then identifies the nontrivial sector by forcing hidden predictive memory in every faithful completion. The remaining distinction between this exact stochastic–quantum equivalence and *full operational quantum mechanics* is confined to the simultaneous coherent-instrument and composite structure stated explicitly in §3.4.

1.2 The Starting Point

The framework begins from a starting point usually stated in two words: *observation occurs*. An observer records distinguishable outcomes of interactions with a system not wholly under the observer's control. This is the cogito of Descartes made precise — not as philosophy, but as a mathematical constraint. As §1.2 makes explicit, that starting point is not a single indubitable fact but resolves into two axioms, one indubitable and one a substantive posit; the two-word phrase is the informal name for the pair.

The axiom has structural content beyond contingent occurrence. Substrata that do not admit embedded observers exist as mathematical objects with dynamics; they do not produce physics in the present framework's sense. Applied to our universe, the starting point thus carries a further commitment: that our universe lies in the observer-admitting subset of substrata — substrata whose bijection structure satisfies the conditions C1–C4 below for some partition. This is the C1–C4 selection condition, not a consequence of the first axiom: a momentary registering with no persistent record satisfies the first axiom and fails C2. This is a structural commitment about the framework's domain, not a contingent fact about whether someone happens to be looking.

The phrase "observation occurs" can be stated with more precision than a single undifferentiated axiom. Distilled, it is *two axioms*. **Axiom 1** is the evidential floor: tokened differentiation occurs — concrete differentiated content is *registered*, a this-not-that tokened from a locus against a remainder; equivalently, *embedded representation occurs*. It is indubitable, since to doubt it is itself to register a distinction from a locus — the cogito made precise, which is irreducibly perspectival: a registering from a locus, not a free-floating fact, and not a thinking subject — the primitive asserts no I (companion methodology paper,

§2.3). Read this way, the primitive already contains three things that therefore need no separate derivation: a differentiation (the this-not-that), a locus that registers it (the observer of the Definition below), and the remainder it is registered against. The indubitability is the warrant available to whatever locus raises the doubt; the Definition covers every registering locus, and for a detector or a horizon that a registering occurs is a physical fact, not an indubitable one. The proper-part decomposition $V \subsetneq S$ — the observer embedded in a whole it does not exhaust — is accordingly constitutive of the primitive rather than a downstream posit: a registering that left no remainder would be the whole representing itself, not an observation. From Axiom 1, by analysis of the words "occurs" and "registered," the relational structure, the existence of a dynamics, determinism and per-moment finiteness of the visible configuration space (Lemma 1) follow as lemmas; the bijective (reversible) representative is a reconstruction choice motivated by observed near-unitarity, and total finiteness of that representative is a posit (status Remark, §1.2; ledger §4.5) — the package is not a set of independent assumptions, and it is not all theorems. One thing does not follow: that the differentiation *recurs* — that there is more than one moment for the dynamics to act across. This is **Axiom 2**: differentiation recurs, the temporal domain being the minimal such recurrence. Axiom 2 presupposes Axiom 1 (there is no recurrence of differentiation without differentiation) but is not derivable from it: a structure containing a single tokened differentiation and nothing further satisfies Axiom 1 while containing no recurrence, so no analysis of Axiom 1 alone can yield Axiom 2 — non-derivability by countermodel. The framework's foundation is therefore described not as a single axiom but as a *two-axiom observation base*: one indubitable claim, and one substantive posit that the first does not yield. The two invoke a single principle — tokened differentiation — and are thematically one; they are numerically two. The Definition and Lemmas below unpack this structure.

The perspectival reading of the primitive is a commitment, and it is worth stating what the alternative would cost. One could instead read the evidential floor *thinly* — as bare differentiated content with no locus occupying a side — in which case the embeddedness $V \subsetneq S$ does not follow and would have to be added as a third, independent axiom. The framework adopts the perspectival reading: the floor the cogito secures is already a registering, not a free-standing fact, so embeddedness belongs to the primitive rather than being posited beside it. This choice has a payoff beyond economy of axioms. On the perspectival reading the foundation's incompleteness is an *internal* consequence of the primitive, not an external import: an embedded representation cannot close over the whole that contains it, since a complete self-inclusive representation would have to represent its own representing without remainder, collapsing the locus into the whole. This is the same form of diagonal that limits formal systems (Gödel [49]) and embedded inference devices (Wolpert [50]); those results corroborate the internal statement rather than supply it. The choice between the perspectival primitive and the thin-plus-third-axiom formulation is presentational — it bears only on how the foundation's commitments are counted and changes no result

downstream of the structure (S, φ, V) — but the perspectival statement is the more faithful to the indubitable floor the cogito secures, which is why it is adopted here.

We ask: what is the minimal mathematical structure consistent with this fact?

Definition. An *observation* is a triple (S, φ, V): a total system S, a dynamics $\varphi : S \to S$, and an observer $V \subsetneq S$ — a proper subsystem with finitely many distinguishable internal states, coupled to the complement $H = S \setminus V$ through φ.

This definition formalizes three features implicit in the concept of observation: there is a whole (S), the observer is not the whole ($V \subsetneq S$), and the observer registers changes (coupling through φ). From this definition, three structural properties follow. The dynamics φ itself is nothing exotic, and its form need not be posited: it follows from a single innocuous premise, that evolution *composes* — advancing the state by t and then by s equals advancing it by $t + s$. This composition (semigroup) law is Cauchy's functional equation, and it forces the evolution to be a one-parameter family generated by a fixed rule [45]; in the continuous case this is the *phase flow* foundational to classical mechanics [44], and in the discrete case relevant here it collapses to iteration of a single transition map, $x_t = \varphi^t(x_0)$. Thus the state-transition structure is not an assumption particular to this framework but a consequence of consistency of time evolution — the same structure standard physics rests on. What distinguishes the present setting is not the existence of such a map but finiteness: the observer distinguishes finitely many visible configurations (Lemma 1), and the map acts on a finite S under the total-finiteness posit (§1.2 Remark, §4.5 ledger); finiteness, not the dynamics, is the load-bearing commitment.

Lemma 1 (Finiteness). *The observer has finitely many distinguishable internal states, so the visible configuration space $\mathcal{C}_V$ is finite, with a discreteness scale ϵ providing a finite minimal cell volume.* Any observer bounded by a finite-area surface can couple to only finitely many modes across that boundary; independent support comes from holographic entropy bounds [2].

Lemma 2 (Causal partition). *The observer is a proper subsystem: $V \subsetneq S$. The complement $H = S \setminus V$ is the hidden sector.* This partition is not a modeling choice but the definition of embedded observation: an observer that could access all of S would have nothing external to observe. The product decomposition

$$\Gamma = \Gamma_V \times \Gamma_H, \qquad H_{\text{tot}} = H_V + H_H + H_{\text{int}}$$

follows, with cross-partition correlations entering only through H_{int}. The product decomposition is idealized; §4.5 addresses approximation quality.

Lemma 3 (Determinism, reversibility, and the selected measure). *φ is a function with a unique successor for every state (determinism) and, as the reversible representative of §1.2, a bijection: distinct states have distinct successors and*

predecessors (injectivity — reversibility). On each finite cycle of the reversible representative, the invariant probability measure is unique and uniform; globally, the counting measure is selected among invariant measures as the maximal-entropy one — a selection principle, not uniqueness from invariance alone.

Remark (the continuum analogue). The same distinction appears in the continuum: Liouville measure is invariant but is not, in general, the unique absolutely continuous invariant measure.

Remark (the measure as a realization datum). Lemma 3 fixes the canonical measure — invariant, maximal-entropy — and the framework's baseline transition matrices (§2.3) and the physical realization use it. It does not preclude structured preparations: a realization in the sense of §3.4 carries its own fixed hidden prior μ_H as part of the realization datum, and the emergent law is a function of $(\varphi, \text{partition}, \mu_H)$ — the same bijection and partition under different priors generally yield different visible laws (trivially: the two-state swap under uniform versus point-mass hidden priors). Partition-relativity statements are therefore relative to the declared measure; the counting measure is the default, not a constraint on representations.

Remark (status of the early commitments). Of the structure above, only finite visible resolution is a theorem (Lemma 1). The Definition and the two axioms do not fix the cardinality of S. The bijective substratum is a representation choice — the minimal recurrent bijective representative of the hidden dynamics, defined here for a finite system only: for a map f on a finite set X it is the restriction of f to its eventual image $E = \bigcap_{n\geq 0} f^n(X)$, the set of states lying on cycles of f, on which f is a bijection of finite order, $f^L = \mathrm{id}$ on E for some $L \geq 1$; no construction is given for an infinite S. Total finiteness of that representative is a premise of the results that use it, carried as item (ii) of the posit ledger (§4.5), the total-finiteness posit; observed near-unitarity is the empirical input motivating the choice. The exact return $\varphi^L = \mathrm{id}$ is a property of the finite representative; Axiom 2 asserts that differentiation recurs, not that every microstate returns. The physical reading of the characterization theorem — the identification of nature with a finite reversible substratum — is accordingly conditional on that representation choice, an empirically motivated reconstruction rather than a pure derivation; the law-level equivalence of §3.4 itself holds for every finite-horizon law. A finite factor adjoined to the hidden sector, prepared independently in a normalized state and evolved by its own permutation, with product dynamics, leaves every visible transition probability unchanged, because the visible coordinate of the product map does not depend on the added coordinate and the added prior has total mass one; the cardinality of such a deep sector is unobservable at the fixed visible/hidden interface. A deep sector that the dynamics couples to the remaining degrees of freedom, and an infinite hidden sector, are outside that statement, and this Remark makes no claim about the observability of their cardinality. Of the two finiteness statements, the following results use only that of Lemma 1: the stochastic-inverse lemma and the readback lemma (accessible conditional memory) of §2.3, and

the equivalence of §3.4 with its process and intervention dilations, in which a finite hidden space is constructed, not assumed. The finite representative is used by the recurrence step of §2.3 and the results resting on it (recurrence-scale P-indivisibility, the CP-indivisibility theorem of §3.2, the all-time periodicity of a fixed realization in §3.4), by the eventual-image and surjectivity steps of the determinism argument below, by the counting measure of Lemma 3 and the uniform hidden prior that defines the baseline transition matrices, by the finite hidden space of the §3.2 setup, by the corner-occupancy claim of §3.1, by the coupling-graph causal cone of §3.3, and by §4.6; the obstructions stated in §3.4 for a fixed finite realization bear on nature through it. Determinism (injectivity) holds by a two-pronged argument. For states differing in registered content, injectivity is the requirement that the dynamics preserve the observer's records — merging them would erase a distinction that has occurred. For states differing only in the hidden sector, a merge is either statistically observable or removable: a merge whose successor configurations differ visibly breaks the double stochasticity of the marginalized process at some step order (the marginal of a bijection under the uniform prior is doubly stochastic, since each set $\{x_j\}\times\mathcal{C}_H$ has exactly $|\mathcal{C}_H|$ preimages, §3.2, and unitary emergent statistics are unistochastic), so it is excluded wherever the observed statistics are of that doubly stochastic class under the uniform prior; a merge that leaves no statistical trace is removable for registered histories whose states lie in the *eventual image* E: any map on a finite set restricts to a bijection on E, at most one of two merged states lies in E, and on registered histories in E the non-injective description coincides with its restriction to E. That representative is a restriction, not a factor, and the measure Lemma 3 assigns it is the counting measure on E, which conditioned on the visible value is uniform on the fibre of E rather than on $\mathcal{C}_H$ and need not reproduce the uniform-prior statistics of the original description: on $\{0,1\}\times\{a,b,c\}$ with $f(0,a)=f(0,c)=(0,a)$, $f(0,b)=(1,a)$, $f(1,a)=(0,b)$ and $(1,b)$, $(1,c)$ fixed, f has the visible process of the bijection that differs from it only in sending $(0,c)$ to itself, whose uniform-prior one-step matrix is $\left(\begin{smallmatrix}2/3 & 1/3\\ 1/3 & 2/3\end{smallmatrix}\right)$, while E omits $(0,c)$ and the counting measure on E gives $\left(\begin{smallmatrix}1/2 & 1/2\\ 1/3 & 2/3\end{smallmatrix}\right)$, which is not doubly stochastic. Either horn delivers a bijective substratum for the dynamics governing registered history in its scope, the second for registered states in E; the two prongs use different inputs — the first empirical (the observed statistics, in the uniform-prior doubly stochastic scope), the second structural (finiteness, for registered states in the eventual image). On a finite set, injectivity implies surjectivity, so φ is a bijection. As for the measure: on the orbit of the actual microstate, the uniform measure is the unique φ-invariant measure; the selection among orbits is fixed by maximal entropy compatible with the observer's records — and the atypical short orbits of the idealized linear dynamics (fixed points, low-period cycles) are precisely observer-lacking substrata — an orbit of period p makes at most p states dynamically accessible, so for small p no partition of it can satisfy the high-capacity condition C3 — admitting no C1–C4 partition, hence outside the scope of every theorem below. The invariant counting measure, in the continuum limit, becomes the Liouville measure on Γ:

$$\frac{\partial \rho}{\partial t} = \{H_{\text{tot}}, \rho\} \equiv \mathcal{L}\rho$$

Remark (reversibility as the conservative assumption). The bijectivity of φ — equivalently, that the substratum conserves information — is not an exotic commitment peculiar to this framework but the orthodox assumption of mainstream physics, imported here rather than invented. Unitary evolution in quantum mechanics *is* information conservation: a pure state evolves from a unique pure state, so the dynamics is reversible and the past is in principle recoverable from the present [42]. The classical counterpart is Liouville's theorem, the conservation of phase-space volume recovered above. That the physics community treats information conservation as effectively non-negotiable is shown by the black-hole information paradox, in which the *apparent* violation of unitarity by Hawking evaporation is regarded as a paradox to be resolved rather than a discovery to be accepted [43, 42]; the no-hiding theorem [42] makes the underlying conservation statement rigorous. The framework therefore assumes no more than standard physics already does. What is non-standard here is not the assumption of reversibility but the claim to *recover* it, below, from the two-pronged argument of Lemma 3; a reader unwilling to grant that recovery may simply take reversibility as the standard postulate it is elsewhere, with no cost to the emergence results.

These properties contain no quantum postulates. The framework proposes that quantum mechanics emerges for an embedded observer under the conditions below. What is established is the finite-horizon representation equivalence of §3.4, which holds for every finite stochastic law whether or not the conditions hold; the selection by (C4) of the laws with memory, every faithful deterministic completion of which carries hidden predictive memory; and, within complex matrix kinematics taken as given, the characterization of exact finite operational quantum mechanics by the five completion conditions of §3.4, none supplied by bare finite OI.

1.3 Conditions on the Hidden Sector

Companions and independence. The four conditions below are diagnostics of a realization rather than hypotheses of the characterization: the equivalence of §3.4 quantifies over none of them (Remark, *the conditions are diagnostics, not hypotheses*), (C1) follows from (C4) in any faithful realization, (C3) holds in every faithful realization by the capacity bound of §3.4, and (C2) is a physical-regime premise. They are stated and used because they are what one can check of a candidate partition — a cosmological horizon, a laboratory apparatus, an equilibrium fluctuation — where the bare existence claim cannot be checked directly. Apart from the inputs collected in the posit ledger (§4.5), every result uses only these, the two axioms, the hypotheses of the conditional results, stated where they enter, and the external results scoped in item (v) of the Remark *scope of the equivalence* (§3.4). Their cosmological realization — the horizon as the

hidden sector, with its capacity, its timescales, and its bidirectional read-write coupling — is developed in companion work and is nowhere consumed by a proof here; the realization clauses below are physical statements of that picture, not premises.

(C1) Non-zero coupling. $H_{\text{int}} \neq 0$: hidden degrees of freedom affect the visible evolution at some accessible step — the conditional law of some visible transition depends on the hidden state. A non-permutation one-step matrix T is the cheapest sufficient *witness* (§2.3), not an equivalent: influence can hide at step one and surface later (the delayed-revival family, realized by the §3.4 construction; `review4_probes.py`). (Status: doubly anchored. Necessity — observed non-Markovian dynamics force C1: with no hidden influence at any accessible step, every visible transition is a fixed function of the visible present and the process is Markov (§3.4); directly observed memory effects in open-system dynamics certify the antecedent. Bell violation supplies a separate spatial-composite anchor only on the adopted ontic-parameter-dependent branch of §3.3; it does **not** by itself imply temporal P-indivisibility. Realization — at the cosmological horizon the ADM boundary Hamiltonian and the constraint structure supply a *bidirectional* coupling stronger than $H_{\text{int}} \neq 0$. Bidirectionality is a property of the physical realization, not part of C1's defining content: a one-way influenced process can be P-indivisible — see the Remark at the non-permutation-witness lemma, §3.4. Dependency — C1 is *not independent of C4*: a readback gap requires hidden mediation, so any realization satisfying C4 satisfies C1, and zero-coupling realizations exhibit no gap at any order, certified by exhaustive enumeration at $(n_V, |\mathcal{C}_H|) = (2,2), (2,3), (3,2)$ in `primitive_probes.py`. C1 is retained as a separately checkable condition — the adopted Bell branch independently requires Bell-relevant cross-wing coupling, while C4 already forces hidden mediation — not as an independent postulate.)

(C2) Memory persistence. Stated in two named forms, because the corpus needs both and they are not equivalent. **(C2-structural):** a hidden record of visible history, once written, persists to the readback that uses it — the retention that clause (b) of the universal hidden-memory theorem shows every memory-bearing completion must contain (§3.4). In a bijective realization reversibility supplies this retention for every law, Markov laws included (§3.4, remark after that theorem), so as a condition (C2-structural) bears on a coarse-grained description of the hidden sector, whose effective channel between couplings can degrade the record before readback (fast-bath lemma, §2.3; physical-memory theorem, §3.4). **(C2-slow):** the timescale mechanism — the hidden sector's *coarse-grained mixing time* τ_B, the timescale on which the effective channel degrades a stored record of visible history, is long compared to the accessible window: $\tau_S \ll \tau_B$, the *inverse* of the Markovian regime. (C2-slow) implies (C2-structural) under the uniform-erosion estimate of §2.3 — a record eroding at rate $1/\tau_B$ survives windows $k\tau_S \ll \tau_B$ essentially intact — but not conversely: what is required is that records persist, not that the hidden sector evolve slowly. Slow evolution is one *sufficient mechanism* for persistence (a record that stays put survives), but fast autonomous hidden dynamics with conserved structure

serve equally — a shift-register bath rotating five cells per visible step produces a maximal order-2 readback gap, because its circulation forgets nothing and routes the record back within the window, so that (C4) holds while (C2-slow) fails (`fastbath_probes.py`). Unqualified "(C2)" in this paper means the persistence requirement, with the form named wherever the distinction bites. The same probe certifies the complementary boundary: persistence alone does not supply (C4) — with the return time pushed outside the accessible window the identical non-forgetting register yields an exactly i.i.d. visible process. (C4)'s operative demand on the hidden sector is thus persistence *together with in-window routing*; (C2) names the persistence half. (Status: **a physical-regime premise, not a structural condition** — and the distinction is load-bearing. C1, C3 and C4 are the structural conditions: the characterization's equivalence runs over those three (§3.4), and the existential realization satisfies C2 trivially with a static bath. What C2 supplies is physical: it is the mechanism by which a stored record survives to be read in realistic models, so that the readback gap C4 asserts survives rather than being suppressed by erosion (§2.3); and in the other direction, in the coupling-step model of the physical-memory theorem (§3.4) — next visible state determined by the current visible and hidden states, any hidden map between couplings — observed conditional memory $M_k = I(X_{<k}; X_{k+1} \mid X_k) > 0$ at an accessible order k forces some realized hidden conditional at that coupling time to lie at total-variation distance at least $\varepsilon_{\min}(M_k, n_V)$ (§3.4) from the model's reference prior π_H — the contrapositive of the physical-memory theorem of §3.4, in which ε is the realized distance and no mixing hypothesis enters — while the timescale form, a lower bound on bath persistence, follows from it under the hidden-sector mixing hypothesis. Note also that C2 does not imply C4: a slow bath can store without reading, certified by a C1–C3 realization with identically zero conditional memory on arbitrarily long horizons (`review3_probes.py`), which is why C4 is a separate condition and not a consequence. ETH, C2's physical motivation — a well-supported conjecture rather than a theorem — is discussed at the remark in §3.4. Realization — $\tau_S/\tau_B \sim 10^{-32}$ at the horizon.)

(C3) Sufficient memory capacity. The hidden sector has enough distinguishable states to encode the observed conditional past–future information: $\log_2 |\mathcal{C}_H| \geq I^*$, the per-process bound of §3.4. (Status: necessity — the data-processing bound, $m \geq 2^{I^*}$, for any realization. Realization — at the horizon the margin is astronomical: $A/\epsilon^2 \sim 10^{122}$ hidden modes at one nat per boundary mode against $\sim 10^{80}$ baryons, a hierarchy $N_H \gg N_V$ with the bath imperturbable on laboratory timescales; the hierarchy and the imperturbability belong to this realization clause, not to the condition. Dependency — C3 does not depend on C4: it holds in every faithful realization, Markov ones included, because the whole visible future is a function of the present visible and hidden states, so data processing gives $\log_2 |\mathcal{C}_H| \geq I^*$ (Theorem (C3 necessity), §3.4). Exhaustive enumeration over the realizations with a readback gap at $(n_V, |\mathcal{C}_H|) = (2,2), (2,3), (2,4), (3,2)$ checks the per-order form $\max_k M_k \leq \log_2 |\mathcal{C}_H|$ and finds it attained at $(2,2)$ and $(3,2)$ (`primitive_probes.py`, uni-

form prior, orders 1 and 2, floating point). Attainment is exact: at $(2, 2)$ the swap $\varphi(v, h) = (h, v)$ under the uniform prior has $X_{k+1} = X_{k-1}$, so $M_k = 1$ bit $= \log_2 |\mathcal{C}_H|$, and since $M_k \leq I^* \leq \log_2 |\mathcal{C}_H|$ it attains $I^* = \log_2 |\mathcal{C}_H|$, so C3 is tight rather than slack. It is retained as a separately checkable condition.)

(C4) History readback. History-sensitive hidden mediation: on accessible windows, hidden degrees of freedom carry information about the visible past into future visible conditionals — at some order, two visible histories with the same current state induce different next-step laws, mediated through the hidden state. The condition is operational; it does not by itself assert a causal write-then-read cycle — a pre-sampled hidden variable revealed by the history satisfies it (the response-table construction, §3.4). (Status: necessity — immediate given observed non-Markovianity, since any realization of a non-Markov process carries differing past-conditionals. Realization — in the cosmological realization the mediation *is* a causal read-write cycle: the interaction reads the same boundary degrees it writes, a property to be independently demonstrated there; this is what C1's redefinition relocated here. Separation — (C1)–(C3) alone do not imply (C4): a tape-and-ledger realization of the fair coin satisfies all three with identically zero conditional memory on arbitrarily long accessible horizons (`review3_probes.py`; §2.3).)

Theorem statement. The claims divide into three exact layers and one explicitly conditional extension. **(1) Representation equivalence:** on every finite accessible horizon, the stochastic laws S, their finite reversible deterministic realizations D, and fixed-basis unitary/Born representations Q_{fb} are the same class, $S \iff D \iff Q_{\text{fb}}$ (§3.4). **(2) Readback content:** C4 is not needed for the existence of a quantum representation; it is what makes the OI sector nontrivial. C4 forces conditional memory in the visible law and readout-relevant hidden predictive memory in every faithful completion (§3.4); indivisibility somewhere in the recurrence cycle of a fixed finite reversible representative follows from any non-deterministic visible step, including in laws that are Markov on every accessible window, and does not require C4; with a full-support visible prior it is equivalent to memory at some order below the recurrence time, which an observer reaches only when that order is accessible, as C4 asserts (§2.3). C1 follows as a necessary diagnostic of the mediation, and C3 holds in every faithful realization by the capacity bound; C2-structural, the persistence of the record to readback, holds by reversibility in every bijective realization and constrains coarse-grained descriptions of the hidden sector, with C2-slow one physical mechanism. **(3) Reverse finite realization:** every finite stochastic law, including every finite quantum record, has a finite reversible OI realization; bounded quantum instrument families are uniformly realizable to arbitrary finite accuracy (§3.4). **(4) Full operational extension:** one common *standard quantum instrument/composite theory* for all coherent interventions is obtained, on a finite carrier with complex matrix kinematics taken as given, exactly under the five completion conditions of §3.4. Under the substratum's stated interventions (its finite bijective read-write dynamics and the phase intervention), (iv), full reversible control, is not supplied (kernel: `substratumGen_not_control`

in `OIBridge/StructuralClosure.lean`, on the two-state carrier); whether an extended OI mechanism supplies it, and which of the other conditions the substratum supplies, are open. Bell is a separate constraint rather than one of those conditions: for a response-complete deterministic completion that records a ± 1 outcome in every run, measurement independence plus ontic parameter independence would force $|S_{\mathrm{CHSH}}| \leq 2$, so the Bell-violating branch adopted in §3.3 retains measurement independence and relinquishes ontic parameter independence with operational no-signaling required.

1.4 Partition-Relativity

Lemma (partition-relativity under the canonical invariant measure). *Under the canonical measure of Lemma 3, the emergent description is uniquely determined by the partition; any parameters of the emergent theory depend only on the geometric and thermodynamic properties of the partition boundary. For the abstract representation theorems the prior is a third datum: canonically $T = T(\varphi, \mathcal{P}, \mu_{inv})$, in general $T = T(\varphi, \mathcal{P}, \mu_H)$ (§1.2, Remark on the measure as a realization datum).*

Proof. By Lemma 3, the Hamiltonian flow ϕ_t is unique. By Lemma 2, the partition is fixed. By Lemma 3, the observer uses the Liouville measure μ — the maximal-entropy invariant measure on the full phase space. (An embedded observer with incomplete knowledge of the total energy averages over energies with a smooth prior, recovering Liouville on a phase-space band — the continuum form of Lemma 3's maximal-entropy selection; singular invariant measures concentrated on atypical short orbits correspond to observer-lacking substrata and are outside scope by the same lemma.) Letting π_V denote projection onto the visible sector, the marginalized transition probabilities

$$T_{ij}(t_2, t_1) = \int_{\Gamma_H} \delta_{x_j}[\pi_V(\phi_{t_2 - t_1}(x_i, h))]\, d\mu(h)$$

are therefore uniquely determined by three inputs: dynamics (ϕ_t), partition (Γ_V, Γ_H), and measure (μ). Since ϕ_t and μ are fixed by the definition and lemmas, the stochastic process — and hence any emergent quantum description (§3.1) — depends only on the partition. □

1.5 The four layers

The results organize into four layers, and every claim should be read at its layer. **Layer 1 — reversible dilation** (§2.2, §3.4): exact mathematics, no quantum claim —

$$\{\text{finite-horizon visible laws}\}$$
$$= \{\text{visible marginals of finite reversible hidden-state systems}\},$$

that is $S \iff D$, Markov laws included, so the existence of a realization does not distinguish non-Markovian laws (kernel: `S_imp_D` for $\subseteq$; `D_imp_Qfb`

with `Qfb_imp_S` for $\supseteq$; assembled in `finite_horizon_equivalence`). Non-Markovianity is instead recorded in every realization (§3.4, unavoidable hidden predictive memory). Here a faithful realization is a system as in class (D): a bijection φ of $\mathcal{C}_V \times \mathcal{C}_H$ with an initial law on $\mathcal{C}_V \times \mathcal{C}_H$, which may correlate the hidden and visible coordinates, under which the visible trajectory $(x_0, \dots, x_K)$ has the given law. In every faithful realization the next-step law after a positive-probability history p ending at x is the pushforward $(f_x)_{\#}\mu_p$ of its hidden posterior μ_p through $f_x(h) = \pi_V(\varphi(x, h))$, and next-step laws a total-variation distance δ apart force posteriors at least δ apart (kernel: `pushforward_identity`, `distinguishability_floor`, for a joint law of history, present and hidden state; reading a realization's time-t law as that joint law is a step not formalized). Hence a finite-horizon law has a memory witness at t — two positive-probability histories ending at the same x_t with different next-step laws, equivalently $M_t := I(X_{<t}; X_{t+1} \mid X_t) > 0$ by the standard vanishing criterion for conditional mutual information, which is not formalized — exactly when every faithful realization carries readout-relevant hidden memory at t, a pair of such posteriors with different pushforwards: the forward direction is the pushforward identity applied in each realization, the converse the same identity applied in one realization, which exists (kernel: `S_imp_D`). In a bijective realization run from time 0 the visible past is a function of (X_t, H_t), so $I(X_{<t}; H_t \mid X_t) = H(X_{<t} \mid X_t)$ for every law, Markov laws included; the readout-relevant part of that memory is what distinguishes non-Markovian laws, and it is the visible witness carried through the pushforward identity. Over deterministic completions without the bijectivity requirement, the least hidden predictive information $I(X_{<t}; H_t \mid X_t)$ is $I(X_{<t}; X_{>t} \mid X_t)$, attained by the canonical predictive quotient, a staged non-injective predictor (§3.4, variational remark); over bijective realizations it is $H(X_{<t} \mid X_t)$, as above. The equivalence $S \iff D$ and the universal hidden-memory statement are the paper's core. **Layer 2 — memory and capacity** (§2.3): (C4) forces positive conditional memory; the (C3) information bound; the physical-memory theorem ((C2): persistence form without a mixing hypothesis, slow-bath form conditional on it). **Layer 3 — quantum representation** (§3.1, §3.4): the imported correspondence (Q1) and the in-house uncompressed form (Q2), scopes separated. **Layer 4 — physical and operational realization** (§3.2, §4, companions): interventions (classical comb closed, §3.4; quantum instruments characterized by five completion conditions within complex matrix kinematics, §3.4, their physical sourcing open), coherent preparation, Bell and free-choice structure, H-spin' and H-χ', the cosmological realization — where the named hypotheses live. A Layer-1 result does not by itself establish any Layer-4 claim.

2. EMERGENT STOCHASTICITY AND P-INDIVISIBILITY

2.1 Emergent Stochasticity

The total system evolves deterministically, but the visible sector alone is stochastic. The observer knows x but not h; different hidden states h_k compatible with

the same x send it to different futures x'_k. Transition probabilities are obtained by marginalizing over the Liouville measure (§1.4). This makes the framework a hidden variable theory; consistency with Bell's theorem is addressed in §3.3.

2.2 The Slow-Bath Regime

The Markovian limit requires $\tau_B \ll \tau_S$ [3]. Condition (C2-slow) inverts this: $\tau_S \ll \tau_B$. A slow bath must be distinguished from a static field. (C1) states that the hidden state affects visible transitions; in the slow-bath mechanism the coupling also acts in the other direction, continuously: each visible-sector transition imprints on the hidden sector. Because the hidden sector is slow (not static), imprints persist without thermal overwriting. On subsequent transitions, coupling reads back stored correlations, producing history-dependent transition probabilities — the non-Markovian regime [3].

2.3 P-Indivisibility

By Lemma 1 the visible sector is finite; under the total-finiteness posit (§1.2 Remark, §4.5 ledger) the hidden sector is also finite — configuration spaces $\mathcal{C}_V$ and $\mathcal{C}_H$ (the discrete counterparts of Γ_V and Γ_H). On these finite sets:

Theorem. *Let $\mathcal{C}_V$ and $\mathcal{C}_H$ be finite sets with $|\mathcal{C}_V| \geq 2$, and let $\varphi : \mathcal{C}_V \times \mathcal{C}_H \to \mathcal{C}_V \times \mathcal{C}_H$ be a bijection. Define:*

$$T_{ij} = \frac{|\{h \in \mathcal{C}_H : \pi_V(\varphi(x_i, h)) = x_j\}|}{|\mathcal{C}_H|}$$

and the k-step matrix $T^{(k)}_{ij}$ by applying φ^k. If T is not a permutation matrix, then the process is P-indivisible.

Proof. Uses total variation distance $d(p,q) = \frac{1}{2}\sum_k |p_k - q_k|$. P-divisibility $\implies$ $d(p(t), q(t))$ non-increasing for all initial p, q [4, 5] (a stochastic divisor is a total-variation contraction); the proof below uses only this direction, in contrapositive form. The converse is *false*: monotonicity constrains the divisor only on the image directions $(\delta_i - \delta_j)T(1,0)$, not on the extreme points of the zero-sum ball, and explicit stochastic pairs exist whose unique divisor $T(1,0)^{-1}T(2,0)$ carries a negative entry — no stochastic divisor, hence P-indivisible — while d is non-increasing for every p, q; one such pair is strictly contracting at $n = 2$ and one lies in the doubly stochastic class at $n = 3$ (`translation_probes.py`). Monotonicity is therefore a necessary, not sufficient, condition for P-divisibility, and is not used as a criterion anywhere below.

Step 1 (Recurrence). φ bijective on a finite set $\Rightarrow \exists N : \varphi^N = \mathrm{id}$. Thus $T^{(N)} = I$ and $d(\delta_i T^{(N)}, \delta_j T^{(N)}) = 1$ for $i \neq j$.

Step 2 (Strict contraction). T not a permutation $\Rightarrow \exists i, j, l$: $T_{il} > 0$ and $T_{jl} > 0$. Then:

$$d(\delta_i T, \delta_j T) = \frac{1}{2} \sum_k |T_{ik} - T_{jk}| < 1$$

Step 3. $d(\delta_i T^{(1)}, \delta_j T^{(1)}) < 1 = d(\delta_i T^{(N)}, \delta_j T^{(N)})$: non-monotonic, hence P-indivisible. □

The theorem requires only bijective dynamics (Lemma 3) and a non-permutation one-step matrix T, under the uniform hidden prior that defines T; (C1) acting at the first step supplies one, and (C1) acting only at later steps does not (§1.3). Lemma 1 gives the finite visible space; the hidden space is finite under the total-finiteness posit (§1.2, §4.5).

Lemma (stochastic inverse). *If a finite square stochastic matrix has a stochastic left or right inverse, then it is a permutation matrix.*

Proof. Suppose $AB = I$ with A and B stochastic. For $i \neq j$, $0 = (AB)_{ij} = \sum_k A_{ik} B_{kj}$ is a sum of nonnegative terms. Hence each k with $A_{ik} > 0$ has the kth row of B supported only on column i; stochastic normalization makes that row the basis vector e_i. Two positive entries in one row of A would therefore make two rows of B identical, contradicting invertibility. Thus every row of A has one unit entry, and invertibility makes the occupied columns distinct. The right-inverse case is symmetric. □

Theorem (history readback + finite recurrence implies indivisibility). *Let φ be a bijection of $\mathcal{C}_V \times \mathcal{C}_H$ of order L and μ_H a hidden prior shared by every root, and define the rooted visible maps $\Gamma_t(a, j) = \mu_H\{h : \pi_V \varphi^t(a, h) = j\}$ for every $a \in \mathcal{C}_V$, roots of visible probability zero included. The family is P-divisible on $\{0, \dots, n\}$ if $\Gamma_t = \Gamma_s \Lambda_{s,t}$ with $\Lambda_{s,t}$ stochastic for all $s < t \leq n$. Let the root be drawn from a visible prior and the hidden state from μ_H, independently. If two positive-probability visible trajectories $(x_0, \dots, x_k) \neq (x'_0, \dots, x'_k)$ have $x_k = x'_k$ — in particular if a (C4) gap (§1.3) holds at order k — then some Γ_s with $1 \leq s \leq \min(k, L-1)$ is not a permutation, and for every $n > s$ with $\Gamma_n = I$, $n = L$ among them, no stochastic Λ satisfies $\Gamma_n = \Gamma_s \Lambda$: the rooted family is not P-divisible on $\{0, \dots, n\}$.*

Proof. Because φ is a permutation of a finite set, it has finite order L, so $\Gamma_L = I$ and $\Gamma_{t+L} = \Gamma_t$. If $\Gamma_1, \dots, \Gamma_{\min(k, L-1)}$ were all permutations, then by periodicity every Γ_t with $t \leq k$ would be one, $\Gamma_t(a, \cdot) = \delta_{\sigma_t(a)}$ with σ_t a bijection, and the visible configuration at each time $t \leq k$ would be σ_t of the root, μ_H-almost surely. Two positive-probability histories with the same endpoint x_k would then share the root $\sigma_k^{-1}(x_k)$ and coincide. Hence some Γ_s, $1 \leq s \leq \min(k, L-1)$, is not a permutation, possibly only through a row of a root of visible probability zero; the lemma applies to the square matrix, which is why every row is kept. A stochastic Λ with $\Gamma_n = \Gamma_s \Lambda = I$ would be a stochastic right inverse of Γ_s, which the lemma forbids. □ (kernel: the lemma is `isPermMatrix_of_stochastic_inverse` in `OIBridge/EquivalenceChain.lean`; the final clause, that the rooted

family is not P-divisible on $\{0, \dots, n\}$, follows from two distinct rows of Γ_s sharing positive mass and $\Gamma_n = I$ by the overlap-and-return theorem below, `pIndivisible_of_rooted_identity_return_of_overlap`; the step from two positive-probability histories with a common endpoint to a non-permutation rooted map, the step from a non-permutation square stochastic matrix to two distinct rows sharing positive mass (a square stochastic matrix whose rows have pairwise disjoint supports is a permutation), and $\Gamma_L = I$ are proved on paper)

Scope. This is a global recurrence-cycle result. It does **not** say that C4 forces P-indivisibility on every accessible short-time window; the XOR control remains a counterexample to that stronger statement. With a full-support visible prior the hypothesis and the conclusion coincide on the representative: a readback gap at some order below L — two positive-probability histories with a common endpoint and different next-step laws, equivalently $M_k > 0$ for some $k < L$ — a non-permutation Γ_s for some $0 < s < L$, and failure of P-divisibility on $\{0, \dots, L\}$ are equivalent. Without a gap below L the law on $\{0, \dots, L\}$ is Markov with kernels Q_k, $\Gamma_t = Q_0 \cdots Q_{t-1}$, and $I = \Gamma_L = \Gamma_s(Q_s \cdots Q_{L-1})$ makes each Γ_s a permutation by the lemma; if every Γ_s is a permutation, $\Gamma_s^{-1}\Gamma_t$ is a stochastic divisor. Full support enters only in the first of these steps: otherwise a non-permutation row may belong to a root of visible probability zero while the realized law has no gap. For every visible prior, a non-deterministic visible step — $P(X_{t+1} \mid X_t = x)$ not a point mass for some positive-probability x — makes some Γ_s a non-permutation, so the recurrence-cycle indivisibility holds for every representative with such a step, laws that are Markov on every accessible window included. Under a full-support prior, the gap it is equivalent to may lie at any order below the recurrence time; (C4) and condition (ii) of §3.4 require an accessible order, $k\tau_S \ll t_R$. The equivalence is a property of the representative's rooted law over its whole recurrence cycle; an observer's statistics are its restriction to accessible windows, and on them the memory is witnessed only by a gap at an accessible order. What (C4) adds within accessible windows is the positive conditional memory of the readback lemma below.

Theorem (overlap and return). *Let $\mathcal{C}_V$ and $\mathcal{C}_H$ be finite sets, φ a bijection of $\mathcal{C}_V \times \mathcal{C}_H$, and μ_H a probability distribution on $\mathcal{C}_H$, not required to have full support, from which the hidden state is drawn whatever the root, with rooted visible maps $\Gamma_t(a, j) = \mu_H\{h : \pi_V\varphi^t(a, h) = j\}$ for every $a \in \mathcal{C}_V$. If $\Gamma_n = I$ for some n, and for some $s < n$ two distinct roots $a \neq b$ and a visible value x have $\Gamma_s(a, x) > 0$ and $\Gamma_s(b, x) > 0$, then the rooted family is not P-divisible on $\{0, \dots, n\}$.*

Proof. Distinct rows of $\Gamma_n = I$ are at total variation distance 1. Two rows sharing positive mass at x are at distance at most $1 - \min(\Gamma_s(a, x), \Gamma_s(b, x)) < 1$. Right multiplication by a stochastic matrix does not increase the total variation distance between two rows, so no stochastic Λ satisfies $\Gamma_n = \Gamma_s\Lambda$, and the pair $s < n$ violates P-divisibility. □ (kernel: `pIndivisible_of_rooted_identity_return_of_overlap`, through

`c4r_of_rooted_identity_return_of_overlap`, in `OIBridge/RecurrenceHorizon.lean`; the matrix form `pIndivisible_of_identity_return_of_overlap` assumes only that Γ_s is row-stochastic)

Routed readback. The realization clause of (C4) (§1.3), a visible-history record written into the hidden state and routed back into later visible conditionals, has two formalizations on the rooted maps above, which differ in the hidden variable that carries the record at the storage time. Both fix roots $a \neq b$, times $0 < w \leq s < t \leq K$ and a visible value x, and require four clauses: a write, some seed h with $\mu_H(h) > 0$ for which $\varphi^w(a,h)$ and $\varphi^w(b,h)$ have different hidden components; a store, $\Gamma_s(a,x) > 0$, $\Gamma_s(b,x) > 0$ and different root-conditioned hidden laws at (s,x) for the two roots; a read, those two hidden laws placed beside x and evolved for $t-s$ steps giving different visible laws at t; and a reappearance, $\Gamma_t(a,\cdot) \neq \Gamma_t(b,\cdot)$. In the storage-time reading the hidden law is that of the hidden state at the storage time, $\mathrm{Law}(H_s \mid X_s = x, X_0 = a)$; in the seed-conditioned reading it is that of the initial hidden state, $\mathrm{Law}(H_0 \mid X_s = x, X_0 = a)$. Under either reading, the rooted family is not P-divisible on $\{0,\dots,n\}$ for every $n > s$ with $\Gamma_n = I$, and so for some n, because (Γ_t) is periodic and $\Gamma_0 = I$ (kernel: `routedAtStorage_forces_return_indivisibility`, `routedAtStorage_forces_indivisible_somewhere` in `OIBridge/PhysicalC4StorageReadback.lean` for the storage-time reading; `routed_forces_return_indivisibility`, `routed_forces_indivisible_somewhere` in `OIBridge/PhysicalC4Discharge.lean` for the seed-conditioned reading; the period and the root identity are `rootedMap_periodic` and `rootedMap_zero` in `OIBridge/RootedClassification.lean`). Both proofs use only the two positivity conditions of the store clause, so both conclusions are instances of the overlap-and-return theorem, and the horizon reached is a return of the rooted map, not an accessible window. Neither reading implies the other: each holds at $K = 2$ on an exhibited finite reversible realization on which the other fails at $K = 2$ (kernel: `storageReadback_not_implies_routedReadback`, `routedReadback_not_implies_storageReadback`). Whether the clause holds at a physical cut, in either reading, is not established (§1.3).

Genericity. The theorem's precondition — that T is not a permutation matrix — is a sufficient first-step *witness* of (C1) — neither necessary nor equivalent to the global condition (§1.3; the delayed-revival family hides its influence at step one, `review4_probes.py`) — and is generic among bijections on $\mathcal{C}_V \times \mathcal{C}_H$ in a quantifiable sense. A bijection φ on $\mathcal{C}_V \times \mathcal{C}_H$ produces T equal to a permutation matrix if and only if φ has the form $\varphi(x_i,h) = (\varphi_V(x_i), \sigma_{x_i}(h))$ — the V-dynamics is independent of h. Counting these: $|\mathcal{C}_V|!$ choices for φ_V times $(|\mathcal{C}_H|!)^{|\mathcal{C}_V|}$ choices for the family $\{\sigma_{x_i}\}$, giving a total of $|\mathcal{C}_V|!\,(|\mathcal{C}_H|!)^{|\mathcal{C}_V|}$ bijections with T a permutation, out of $(|\mathcal{C}_V|\,|\mathcal{C}_H|)!$ total. The fraction failing the precondition is therefore

$$P_{\mathrm{perm}}(n_V, n_H) = \frac{n_V!\,(n_H!)^{n_V}}{(n_V n_H)!}, \quad n_V := |\mathcal{C}_V|,\ n_H := |\mathcal{C}_H|.$$

Stirling's approximation gives the asymptotic decay $\log P_{\mathrm{perm}} \sim -n_V n_H \log n_V$ as $n_H \to \infty$ with n_V fixed — equivalently, $P_{\mathrm{perm}} \sim n_V^{-n_V n_H}$. For any $n_V \geq 2$, the fraction of bijections failing C1 at the T-level decays exponentially in the hidden-sector capacity n_H. At modest physical scales — say $n_V = 10$, $n_H = 100$ — the fraction is $\sim 10^{-981}$; at the scales C3 invokes, it is astronomically smaller. The theorem is therefore non-vacuous on two independent grounds: the precondition excludes a structurally distinguished class of decoupled bijections (those whose V-dynamics have no h-dependence), and that class is a vanishing fraction of the configuration space as C3 is engaged. C3 (high-capacity hidden sector) and C1 (non-trivial coupling) are independent conditions in statement, but C3 is the *engine* of C1's genericity: enlarging n_H shrinks the measure of C1-violating bijections super-exponentially fast.

Relation to prior work. The P-indivisibility phenomenon is well-documented in the open-systems literature: reduced dynamics need not preserve positivity [6]; information backflow witnesses divisibility failure [4, 5], and divisibility can fail without it (§2.3, §3.2); process-tensor Markovianity provides an alternative characterization [7]; non-Markovian reduced dynamics arise in the slow-bath regime [8]. The Bylicka-Johansson-Acín equivalence between CP-divisibility and distinguishability monotonicity [9] narrows the gap between divisibility and non-Markovianity, but only for *invertible* reduced dynamics; the marginal $T(t)$ here is generally non-invertible, so the load-bearing property for the general argument remains P-indivisibility of the marginal process. Milz et al. [10] confirm CP-divisibility does not imply Markovianity in the process-tensor sense.

Continuous-time extension. The Hamiltonian flow on finite-dimensional phase space preserves Liouville measure on compact energy surfaces. $T_{ij}(t)$ is continuous with $T(0) = I$. With the full-support Liouville measure as hidden prior, (C1) makes $T(t)$ depart from the permutation class at some $t_1 > 0$. If the flow then returns near-totally at a common time — some $t_R > t_1$ at which, for every root x_i, the set of hidden states with $\pi_V(\varphi_{t_R}(x_i, h)) = x_i$ has measure $> 1-\delta$ — then for small δ the trace distance between rooted laws is non-monotonic on $[0, t_R]$ and the continuous-time family is P-indivisible. This common-time return is the continuous-time counterpart of $\Gamma_L = I$, which holds exactly for a bijection of finite order (theorem above). Poincaré recurrence does not supply it for a general Hamiltonian flow: it returns almost every state to its cell at a state-dependent time, and for a mixing flow $T_{ii}(t)$ tends to the normalized Liouville measure of the cell of x_i, which is less than 1 whenever another cell has positive measure, so near-total return fails at every sufficiently late time. Continuous-time P-indivisibility for general Hamiltonian flows is open.

The recurrence argument establishes P-indivisibility across the recurrence cycle. The readback lemma below shows that a readback gap at an accessible order

produces conditional memory on observable timescales; that is a property of the multi-time law and does not by itself make the rooted family P-indivisible on those windows (Scope, above). A sufficient further condition is a revival of distinguishability within the window: two roots whose rooted laws coincide, or are closer, at an earlier time and differ more at a later one.

Physical motivation (storage under C2). The coupling H_{int} writes visible-sector information into the hidden sector at rate $\sim 1/\tau_S$; between interactions the hidden sector's effective relaxation at rate $\sim 1/\tau_B$ (the coarse-grained channel's mixing; hypothesis (EM), §4.6) erodes it as $e^{-\tau_S/\tau_B}$ per step, so on windows $k\tau_S \ll \tau_B$ the stored history survives essentially intact. Storage, however, is not readback: what is stored must also be *read*. The lemma below makes the reading quantitative, and the separation remark shows why it is a genuinely additional requirement.

Run backwards, the same estimate bounds what the failure of (C2) costs. Since the equivalence of §3.4 does not quantify over (C2), its failure carries no consequence for the characterization except through that bound.

Lemma (fast bath erases readback). *Suppose the coarse-grained channel mixes fast relative to the visible step, $\tau_B \ll \tau_S$, and erodes the stored record uniformly, by $e^{-\tau_S/\tau_B}$ per step, across the history pairs at issue. Then for any $k \geq 2$ the order-k readback gap obeys*

$$\delta_k \;\leq\; \delta_0\, e^{-(k-1)\tau_S/\tau_B}.$$

A fast, uniformly erasing bath therefore suppresses the readback gap exponentially in τ_S/τ_B at every order $k \geq 2$.

Remark (what this lemma does not give). The readback lemma below is a lower bound on $I(X_{<k}; X_{k+1} \mid X_k)$ given a gap; an upper bound on δ_k cannot be inserted into it to bound I from above. The corresponding upper bound on conditional mutual information is available, but by the continuity route — uniform total-variation closeness through the Fannes–Audenaert estimate — and it is proved where it is used, at the physical-memory theorem of §3.4. The statement above is confined to the gap. It does not make the gap vanish at any finite τ_S/τ_B, and it does not bound the gap at order $k = 1$.

Remark (why this matters for the exclusion arguments). Since (C2) sits outside the equivalence (§3.4), its failure excludes nothing by the characterization alone. It also matters which reading of (C2) is in play. Under the *evolution* reading — fast hidden dynamics — failure is compatible with (C4) holding at full strength: a shift-register bath rotating five cells per visible step supports a maximal order-2 gap (`fastbath_probes.py`), which is why (C2) is stated as persistence rather than slowness (§1.3). It is erosion of the stored record, under the uniform erosion hypothesis of the lemma above, that suppresses the readback gap (C4) requires. In the exclusion arguments below — the equilibrium phase, and the anti-Boltzmann-brain corollary of §4.6 — this lemma bounds what fast mixing leaves of the readback: fast mixing erodes the stored record

before it can be read, and the readback gap that (C4) asserts, which is what the characterization registers, is suppressed exponentially in τ_S/τ_B at every order $k \geq 2$. The exclusion itself rests on reading (C2), in the C1–C4 selection condition (§1.2), in its timescale form (C2-slow), which $\tau_B \ll \tau_S$ contradicts by definition when τ_B is the mixing time of the coarse-grained description (§4.6, Scope (ii)); for the persistence form (C2-structural) and for (C4), the fast-bath chain gives suppression, not exclusion.

Lemma (accessible conditional memory from readback). *Suppose the visible process has a gap at order k in the accessible window, as (C4) asserts: two visible histories $p \neq p'$ of length k, values of $X_{<k} = (X_0, \ldots, X_{k-1})$ followed by a common present state $X_k = c$, each with $\Pr[X_{<k} = p,\, X_k = c] \geq p_0$, induce next-step laws at total-variation distance at least δ. Then*

$$I(X_{<k};\, X_{k+1} \mid X_k) \;\geq\; \frac{p_0\, \delta^2}{\ln 2} \;>\; 0.$$

Proof. Condition on the common present state c and let $w_a = \Pr[X_{<k} = a \mid X_k = c]$ be the conditional weights of the length-k histories, P_a their next-step laws, and $\bar{P} = \sum_a w_a P_a$. The conditional mutual information given $X_k = c$ is the mixture divergence $\sum_a w_a\, D(P_a \| \bar{P})$, and Pinsker's inequality gives $D(P_a \| \bar{P}) \geq \frac{2}{\ln 2}\, \mathrm{TV}(P_a, \bar{P})^2$ in bits. By the triangle inequality $\mathrm{TV}(P, \bar{P}) + \mathrm{TV}(P', \bar{P}) \geq \delta$, so $\mathrm{TV}(P, \bar{P})^2 + \mathrm{TV}(P', \bar{P})^2 \geq \delta^2/2$. Multiplying by $\Pr[X_k = c]$ converts conditional weights to path probabilities, each at least p_0; hence $I(X_{<k}; X_{k+1} \mid X_k) \geq \frac{2}{\ln 2} \cdot p_0 \cdot \frac{\delta^2}{2} = p_0 \delta^2 / \ln 2$. □ (Checked on a gap family, the revival family and a small-gap parametric family, with the coin silent, in `c4_backflow_probes.py`: the laws and gaps in exact rational arithmetic, the information quantities and the inequality in floating point.)

Remark (why (C4) is a separate condition). (C1)–(C3) admit realizations with zero conditional memory throughout arbitrarily long accessible horizons: the fair coin realized by the §3.4 dilation construction — the tape influences, the ledger records, and the two never meet — has $T^{(1)} = \frac{1}{2}J$ (so (C1) holds), a static bath ($\tau_B = \infty$), vast capacity, and $I(X_{<t}; X_{>t} \mid X_t) = 0$ exactly at every horizon time (`review3_probes.py`). Influence plus storage plus capacity, without readback, is noise.

Role of (C2)–(C4). The finite-recurrence argument and the readback lemma are independent: the former establishes P-indivisibility from a non-permutation one-step witness and finiteness alone, at the recurrence time; the latter locates observable conditional memory inside accessible windows, which does not by itself make the rooted family P-indivisible there. (C2) keeps stored history alive between couplings; (C3) provides the room to store it; (C4) is the reading of it — and only the reading produces memory an observer can measure.

Weak-coupling and fast-bath regimes. Known P-divisible systems with non-zero coupling are not counterexamples to the recurrence theorem, whose

hypotheses they do not meet. The Jaynes-Cummings model with Lorentzian spectral density is P-divisible when dissipation dominates coupling, and the Davies-Merkli Markovian convergence theorem applies in the Born-Markov limit; both couple to a continuum reservoir, so the finite recurrence the theorem uses is absent. Both sit in the fast-bath regime ($\tau_B \ll \tau_S$), where (C2) fails in its timescale form. The recurrence theorem uses no form of (C2), deriving recurrence-cycle P-indivisibility from a non-permutation rooted map and finite order alone; in particular the timescale form of (C2) is not necessary for it in a finite bijective realization, where (C2-structural) holds by reversibility (§1.3). What the §3.4 physical-memory theorem bounds is conditional memory, not P-divisibility: under the mixing hypothesis, for $\varepsilon \leq 1-1/n_V$, a fast bath leaves at most $2[\varepsilon \log(n_V-1)+h(\varepsilon)]$ of it, with ε the residue of relaxation at the coupling time.

2.4 Explicit Construction: A Six-Hidden-State Model

Visible sector: $x \in \{0,1\}$. Hidden sector: $h \in \{1,\ldots,6\}$. Dynamics: the permutation

$$\sigma = (0{,}1 \leftrightarrow 1{,}1)\ (0{,}2 \leftrightarrow 1{,}2)\ (0{,}3 \leftrightarrow 0{,}4)\ (0{,}5 \leftrightarrow 0{,}6)\ (1{,}3 \leftrightarrow 1{,}4)\ (1{,}5 \leftrightarrow 1{,}6)$$

satisfying (C1)–(C4) — the flip depends on hidden components that the dynamics itself conditions on the past, so readback is present. Since $\sigma^2 = \mathrm{id}$: at $t = 1$, two of six hidden states flip x, giving

$$T(1,0) = \begin{pmatrix} 2/3 & 1/3 \\ 1/3 & 2/3 \end{pmatrix}$$

At $t = 2$ ($\sigma^2 = \mathrm{id}$): $T(2,0) = I$. A Markov chain would predict continued mixing: $T(1,0)^2 = \left(\begin{smallmatrix} 5/9 & 4/9 \\ 4/9 & 5/9 \end{smallmatrix}\right)$. The actual dynamics show complete *un-mixing*. Computing the intermediate propagator:

$$\Lambda(2,1) = T(2,0) \cdot [T(1,0)]^{-1} = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$$

Negative entries — no valid stochastic matrix exists. **P-indivisible.** □

Independent check (process-tensor non-Markovianity). The same model also fails the Pollock et al. [7] multi-time Markov condition without invoking $T(t)$-invertibility: $P(x_2 = 0 \mid x_1 = 1, x_0 = 0) = 1$ versus $P(x_2 = 0 \mid x_1 = 1) = 1/3$. Conditioning on x_0 changes the one-step prediction from $1/3$ to 1 — a memory effect computed directly from multi-time conditional probabilities, valid regardless of $\det T(t)$. The non-Markovian character is therefore confirmed under both the divisibility-based criterion (Rivas et al. [4,

5]; Bylicka et al. [9] in the invertible case) and the process-tensor criterion (Pollock et al. [7] in the general case).

The mechanism. The hidden sector acts as a memory register. The transpositions $(0,h) \leftrightarrow (1,h)$ for $h \in \{1,2\}$ flip x while preserving h, so a state at $x=1$ with $h \in \{1,2\}$ after step 1 carries the record "I was at $x=0$." Step 2 reads this record and flips x back — a return of the earlier visible state, here a revival of distinguishability, which no memoryless process produces.

---

3. THE EMERGENCE OF QUANTUM MECHANICS

3.1 The Stochastic-Quantum Correspondence

By Barandes' correspondence [11, 12], any finite-configuration member of the source's indivisible-stochastic-process class — its term of art for the tuple class, which includes Markov chains — embeds into a unitarily evolving quantum system on $\mathcal{H}$ of dimension $\leq n^3$; (T) is what admits the framework's processes to that class. The transition matrices themselves remain stochastic; what develops negative entries at fine time resolution is the candidate intermediate divisor Λ, which is what P-indivisibility asserts cannot be made stochastic; the Stinespring dilation theorem guarantees a dilated Hilbert space and unitary $\tilde{U}(t)$ whose unistochastic matrix $|\tilde{U}(t)|^2$ marginalizes over the ancilla index to $T_{ij}(t)$ — transition probabilities are exactly Born-rule probabilities. *The ancilla is not optional for this framework's own processes.* Marginalizing a bijection under the uniform hidden prior yields a doubly stochastic one-step matrix (§3.2), and for $n \geq 3$ doubly stochastic does not imply unistochastic: enumerating the construction's own realizations at $n_V = 3$, $|\mathcal{C}_H| = 2$ gives twenty-one distinct one-step matrices, all doubly stochastic, of which six — including $\begin{pmatrix} 1/2 & 1/2 & 0 \\ 1/2 & 0 & 1/2 \\ 0 & 1/2 & 1/2 \end{pmatrix}$ — fail the row-orthogonality condition necessary for unistochasticity (`translation_probes.py`). An undilated $T = |U|^2$ representation is therefore unavailable on the framework's class, and the translation hypothesis (T) must be read against the ancilla-dilated statement — not against the bare one. What that costs is exactly one ancilla factor.

Lemma (one-step ancilla dilation). *For any stochastic T on n states, the map $W|i\rangle = \sum_j \sqrt{T_{ij}}\,|j\rangle \otimes |i\rangle$ is an isometry $\mathbb{C}^n \to \mathbb{C}^n \otimes \mathbb{C}^n$ whose ancilla-marginal is T: $\sum_\alpha |\langle j,\alpha|W|i\rangle|^2 = T_{ij}$. It extends to a unitary on $\mathbb{C}^n \otimes \mathbb{C}^n$.*

Proof. $\langle Wi|Wi'\rangle = \delta_{ii'} \sum_j T_{ij} = \delta_{ii'}$, so W is an isometry, and the marginal is $\sum_\alpha |\sqrt{T_{ij}}\,\delta_{\alpha i}|^2 = T_{ij}$; an isometry onto an n-dimensional subspace of an n^2-dimensional space extends to a unitary. □ (Certified exactly on all six non-unistochastic realizations, with the completion carried out explicitly for the displayed witness as a 9×9 unitary over the algebraic numbers, in `tdilate_probes.py`.)

So the obstruction above is priced rather than fatal: total dimension n^2 suffices at a single step, and at every time separately: extending $|i\rangle \otimes |a_0\rangle \mapsto W|i\rangle$, for one fixed ancilla configuration a_0 and W built from $T(t)$, to a unitary $U(t)$ gives $\mathrm{Tr}[P_j U(t)(|i\rangle\langle i| \otimes |a_0\rangle\langle a_0|)U(t)^\dagger] = T_{ij}(t)$, the form of the §3.4 definition on dimension $n^2 \leq n^3$. Representability is not what (T) is about. **What (T) asserts is tuple instantiation**, and the source's membership conditions are the tuple's own — a finite configuration space, transition maps normalized and trivializing at the conditioning time, a prior, and the observable algebra — not a divisibility property. Divisibility failure is generic in the class rather than a condition of entry, so P-indivisibility is not what admits the framework's processes to it. What remains genuinely open is downstream: whether the ancilla-dilated representation can be made compatible with the passive $\hat{H}$ across horizons (§3.4). The proof requires a finite-dimensional configuration space (Lemma 1); Calvo [13] extends the correspondence to infinite dimensions, but this is not required here. Pimenta [14] distinguishes P-divisible and P-indivisible stochastic-quantum dynamics: the correspondence is non-trivial only in the P-indivisible regime, which is precisely the regime produced by §2.3. An independent derivation using only Stinespring's dilation theorem [15] and standard partial-trace properties [16] is given in §3.2; either route alone suffices for the representation of the transition data; the operational completion — physical preparation of coherent states, noncommuting interventions, the full instrument algebra — is a separate bridge: on a finite carrier, with complex matrix kinematics taken as given, it holds exactly under five completion conditions, none supplied by bare finite OI (§3.4); the one-step coherence theorem of §3.2 establishes non-entanglement-breaking one-step coherence for the linear lattice realization under the $w < \kappa$ criterion; under the substratum's stated interventions, (iv), full reversible control, is not supplied (§3.4), and whether an extended OI mechanism supplies it, and which of the other four conditions the substratum supplies, are open.

Scope of the correspondence (what it does and does not deliver). Because the compressed-representation clause rests on this external result — the equivalence (ii) $\iff$ (iii) is proved in-house (§3.4), and the fixed-basis representation's existence is internal (the equivalence theorem), neither resting on it —, we state its scope precisely rather than treat it as a black-box equivalence. The correspondence is a published theorem (Barandes, *Philosophy of Physics* 3(1):8, 2025 [12]; proof in [11]) and is genuinely load-bearing here, but four scope conditions are carried throughout the framework and should be read as standing qualifications on every downstream statement that leans on the compressed representation. (i) *Directionality and non-uniqueness.* The map is many-to-one in both directions; the complex matrix Θ with $T_{ij} = |\Theta_{ij}|^2$ is guaranteed to exist but is not unique (an entrywise phase freedom analogous to a gauge potential), so the correspondence does not single out a unique quantum system from a stochastic process. (ii) *Scaffolding versus full QM.* The stochastic data directly supply only configuration-diagonal random variables; non-commuting observables enter as emergent patterns ("emergeables"), and spin enters by dilating the Hilbert space and imposing a rotation representation rather than being read

off the stochastic side. Composite systems are on a different footing. For a spatial OI constitution the coupling graph is defined by one-step dependence, so its iterates give exact graph causal cones: before two cones meet, a remote intervention cannot dynamically alter the local substratum state. That kinematic locality does **not**, by itself, prove local tomography or that all visible composites and coherent interventions belong to one common tensor-product quantum instrument category. The operational-completion characterization of §3.4 takes the complex matrix kinematics as given, the tensor-product composite and with it local tomography included, and deriving that kinematics is open; within it, whether the available interventions on a finite carrier and its composites are exactly the quantum instruments is characterized by the five completion conditions of §3.4. Bell setting-independence is logically separate: the framework retains the measurement-independent branch and, for quantum Bell violation, gives up ontic parameter independence as stated in §3.3. The bare correspondence therefore reconstructs the fixed-basis/Born representation and quantum kinematic scaffolding, while the spatial architecture independently supplies causal propagation. Promotion to the full standard operational quantum theory is the coherent-instrument/composite bridge; compatibility of the specific local lattice realization with Bell-required ontic parameter dependence is the independent H-Bell bridge. Its unification with the staggered-to-Dirac spin structure of the lattice substratum — whose spinor decomposition is keyed to the lattice's bipartite partition — is the open step. One premise of that step, the survival of the bipartite grading under the trace-out at the linear level, is derived in [SM §4.8.1]. (iii) *Restricted process class.* The representation $T_{ij} = |\Theta_{ij}|^2$ imposes substantive structure on the admissible processes; in particular Doukas (2026) shows the differentiable form forces the first-order rate matrix to vanish, excluding ordinary continuous-time Markov chains from the *differentiable* form — a restriction on the continuous-time reading, not on the discrete-time tuple class, whose membership conditions (above) include Markov chains — so "any stochastic process" is narrower than it sounds for that reading, and within the differentiable reading the qualifying regime is precisely the indivisible one §2.3 produces, not the generic one. The restriction also runs the other way. Doukas's accompanying bound — that marginalization over a *finite* environment produces no first-order dissipative term, secular leakage requiring an infinite reservoir — is, in continuous-time form, the same structural fact as the second-order estimate of §3.2 for the discrete substratum: a finite deterministic substratum observed by marginalization cannot generate first-order leakage, and its emergent dynamics is thereby confined to the near-unitary corner of process space, displaced from exact unitarity only by the residual dissipation $\mathcal{D}$. What appears as a restriction on the correspondence's process class is, under the present architecture, the mechanism's signature: the framework occupies the corner Doukas identifies not by selection but by structural necessity, conditional on the total-finiteness posit (§1.2 Remark, §4.5 ledger). This is explanatory unification rather than corroboration — near-unitarity is an empirical input elsewhere in the framework — but it converts scope condition (iii) from a qualification into a prediction: fundamental dynamics of continuous-time Markov type would require an actually

infinite hidden sector, and the finite-ϵ displacement $\mathcal{D}$, whose bath-motion term is $\mathcal{O}((\tau_S/\tau_B)^2)$ (§3.2; $\sim 10^{-64}$ for the horizon ratio of (C2)'s realization clause), is the corner-occupancy claim's falsifiable residue. (iv) *Reception.* The result is recent and located at the physics/philosophy boundary. Its mathematical core has since been independently reworked and generalized: the lift structure and its divisibility mechanics by Doukas [22], the countable-dimension extension by Calvo [13], the divisible/indivisible classification by Pimenta [14], and the Tsirelson-bound result in a mainstream physics venue [24, forthcoming in Phys. Rev. A] (a result for the causal-local indivisible construction of [24], not for generic P-indivisibility). The strongest interpretive readings (deflation of the wave function, of entanglement) remain contested philosophical overlays on the mathematical core, which is all the framework requires. None of (i)–(iv) undercuts the framework's use of the correspondence to obtain Born-rule statistics and unitary evolution from the §2.3 process; they bound what may be claimed to follow *from* it.

Independent corroboration of the trace-out mechanism. The technical move from a deterministic substrate to non-Markovian visible dynamics via the trace-out of inaccessible degrees of freedom — central to both routes — is now established at theorem level in the open systems literature. Brandner [17, 18] proves that for autonomous linear evolution equations, integrating out inaccessible degrees of freedom (whether external environments or internal subsystems) yields well-defined non-Markovian visible dynamics in a controlled weak-memory regime, with explicit error bounds and a convergent perturbation scheme. This is the mathematical apparatus the present construction relies upon, derived independently for general open systems. Direct experimental demonstrations of the mechanism include Mehl et al. [19], who showed that hidden slow degrees of freedom in a colloidal system produce non-Markovian visible dynamics violating naive Markovian fluctuation theorems, and Gröblacher et al. [20], who observed non-Markovian Brownian motion in a macroscopic micromechanical oscillator. On the quantum side, Kim [21] shows that monitored quantum systems are formally quantum hidden Markov models, with a rigorous correspondence to classical hidden Markov models for setups involving Haar-random unitaries and measurements — the same formal structure as the stochastic-quantum bridge. The framework's central technical move is therefore standard rather than speculative.

From phase space to configuration space. The transition probabilities T_{ij} are projections of the full phase-space flow onto the configuration sector of Γ_V, with momenta and hidden-sector degrees of freedom absorbed into the Liouville marginalization. The resulting process lives on the discrete configuration space of Lemma 1.

Continuous spectra and the finite-substratum commitment. Lemma 1 fixes the visible configuration space as finite, and the Barandes correspondence as applied here produces an emergent Hilbert space of finite dimension. Standard quantum mechanics makes heavy use of continuous-spectrum observ-

ables — position on $\mathbb{R}^3$, momentum, continuous-spectrum scattering — which strictly require infinite-dimensional Hilbert spaces and unbounded self-adjoint operators. The framework's position on this is that continuous spectra are the infrared limit of the discrete spectrum produced by Lemma 1, not a deeper reality that the framework fails to capture. The quantitative separation is extreme: for a visible sector bounded by the cosmological horizon, of radius L_H, and an assumed discreteness scale ϵ of the order of the Planck length l_p, the number of distinguishable positions is $|\mathcal{C}_V| \sim (L_H/\epsilon)^3 \sim 10^{183}$, and no independent continuous-spectrum prediction is needed or claimed. Calvo [13] extends the Barandes correspondence to countable-infinite dimensions for completeness, but the finite-dim correspondence is the load-bearing one for the framework — every result in §2–§3 other than the infinite-region completion (§3.2, §3.4), which is assembled from finite-region stages, runs on finite configuration spaces: Lemma 1 for the visible sector, with the total-finiteness posit (§1.2 Remark, §4.5) where a result also needs the hidden sector finite.

Three features are fixed at the representation level. When a continuous interpolation $U(t) = e^{-i\hat{H}t}$ is chosen (§3.2), it is generated by a Hermitian $\hat{H}$. The Born form $p = |U|^2$ is the dictionary of Q_{fb} (§3.4); its quadratic exponent is a frontier item. The action scale $\hbar$ enters when defining $\hat{H}(t) \equiv i\hbar\,\partial_t U \cdot U^\dagger$, converting dimensionless rates to energy units; its value requires additional physical input from the partition.

Phase uniqueness from continuous-time data. The relation $T_{ij}(t) = |U_{ij}(t)|^2$ discards phase information at any single time. Doukas [22] identifies this gap for discrete-time data. The resolution: continuous-time data $\{T_{ij}(t)\}_{t\in\mathbb{R}}$ provides strictly more information.

Lemma (phase-locking). *Let H be Hermitian on $\mathbb{C}^n$ with eigenbasis $\{|k\rangle\}$, eigenvalues $\{E_k\}$, and $V_{ik} = \langle i|k\rangle$. Assume $n \geq 3$, for both Hamiltonians being compared (G1) non-degenerate spectrum, (G2) non-degenerate energy gaps and (G3) $V_{ik} \neq 0$ for all i, k, and for at least one of them (G4): every relation $\omega_1+\omega_2 = \omega_3$ among the frequencies $\omega_{kl} = E_k - E_l$ of three different pairs of levels is a triangle identity $\omega_{kl} + \omega_{lm} = \omega_{km}$. Then $T_{ij}(t) = |\langle i|e^{-iHt}|j\rangle|^2$ for all i, j, t uniquely determines H up to an overall energy shift, basis phase conventions, and the antiunitary conjugation $H \to -H^*$ — the complex-conjugate representation, which satisfies $|\langle i|e^{-i(-H^*)t}|j\rangle|^2 = |\langle i|e^{-iHt}|j\rangle|^2$ identically and, for generic H, is not shift- or rephasing-related (its spectrum is negated).*

Proof. Write:

$$T_{ij}(t) = \sum_{k,l} V_{ik}\, V_{jk}^*\, V_{jl}\, V_{il}^*\, e^{-i(E_k - E_l)t}$$

Fourier extraction yields each frequency with its coefficient, not the ordered pair of levels that carries it, and (G2) alone does not identify the pairs: the spectra $\{0, 1, 4, 10, 12, 17\}$ and $\{0, 1, 8, 11, 13, 17\}$ have the same fifteen distinct

gaps without being translates or reflections of one another. (G4) identifies them. Let H', with levels E'_k and frequencies ω'_{kl}, reproduce the data. By (G3) every frequency of either Hamiltonian carries a non-zero coefficient ($a^{kl}_{ii} = |V_{ik}|^2|V_{il}|^2$), and distinct frequencies are linearly independent functions of t, so the two sets of frequencies coincide; by (G2) each ω_{kl} equals $\omega'_{k'l'}$ for exactly one ordered pair (k', l'), the image of (k, l). Take (G4) for H'; for H, exchange the two Hamiltonians, which preserves the form of the conclusion. Fix levels $o \neq a$ and let (p, q) be the image of (a, o). For a third level y, the images of (y, a), (a, o) and (y, o) are three different pairs of levels by (G2), and their frequencies satisfy $\omega_{ya} + \omega_{ao} = \omega_{yo}$; by (G4) they form a triangle, so the image of (y, o) has first index p or second index q. If the image (u, q) of some (y, o) had $u \neq p$ and the image (p, v) of some (y', o) had $v \neq q$, then $y \neq y'$, and the same step with y, y' in place of a, y would give (p, v) first index u or second index q, both excluded. Hence either every image of (y, o) has second index q, and E' contains the translate $E + E'_q - E_o$, or every one has first index p, and E' contains the reflection $-E + E'_p + E_o$. Both spectra have n levels, so $E'_{\tau(k)} = E_k + E_0$ for all k or $E'_{\tau(k)} = -E_k + E_0$ for all k, with τ a relabelling: under (G2) and (G4) the unlabelled frequency set fixes the levels up to translation and reflection. In the first case the image of (k, l) is $(\tau(k), \tau(l))$ and E_0 is the energy shift of the statement. In the second it is $(\tau(l), \tau(k))$, so each coefficient of H matches the complex conjugate of the corresponding coefficient of H': the simultaneous conjugation below, which is the $H \to -H^*$ alternative and satisfies (G1)–(G4) whenever H does. By (G2), the frequencies $\omega_{kl} = E_k - E_l$ are distinct for distinct pairs with $k \neq l$; since $T_{ij}(t)$ is real, the components at $\pm\omega_{kl}$ are conjugate-paired, and the simultaneous conjugation $a \to a^*$, $\omega \to -\omega$ (i.e. $V \to V^*$, $E \to -E$: the $H \to -H^*$ ambiguity of the statement) is invisible to the data. Fourier-transforming yields, up to that pairing:

$$a^{kl}_{ij} = V_{ik}\, V^*_{jk}\, V_{jl}\, V^*_{il}$$

The moduli $|V_{ik}|$ follow from $a^{kl}_{ii} = |V_{ik}|^2|V_{il}|^2$ (non-zero by (G3)): the rank-1 outer-product structure determines $|V_{ik}|^2$ up to a global scale, and unitarity $\sum_k |V_{ik}|^2 = 1$ fixes the scale. The phases carry a double-difference structure:

$$\arg(a^{kl}_{ij}) = (\varphi_{ik} - \varphi_{il}) - (\varphi_{jk} - \varphi_{jl})$$

The gauge freedom preserving all double differences is $\varphi_{ik} \to \varphi_{ik} + \alpha_i + \beta_k$ (physically irrelevant basis rephasing). Fixing $\varphi_{1k} = 0$, $\varphi_{i1} = 0$, each remaining phase is $\varphi_{ik} = \arg(a^{k1}_{i1})$, directly extracted from Fourier data. $H = V\,\mathrm{diag}(E_k)\,V^\dagger$ is unique up to energy shift, phase conventions and the antiunitary alternative (kernel: `twoBranch_of_triangleGeneric` and `twoBranch_of_triangleGeneric_source` in `OIBridge/PairIdentification.lean`, with (G4) for H' and for H respectively). □

The ancilla-marginal form of the reconstruction — where the visible transition data constrains a Hamiltonian on $\mathcal{H}_V \otimes \mathcal{H}_{\text{anc}}$ through the block projectors P_i — is governed by the linearised idempotency system, and admits a reformulation that makes its solution space transparent.

Lemma (ancilla-marginal reformulation). *Let $M_i = V^\dagger P_i V$ be the block projectors in the eigenbasis, θ real antisymmetric, and set $K_i := i\,(\theta \circ M_i)$ with $\circ$ the entrywise product. Then each K_i is Hermitian, and the linearised idempotency constraint $\sum_b (\theta_{ab} + \theta_{bc}) M_i[a,b] M_i[b,c] = \theta_{ac} M_i[a,c]$ holds for all a, c if and only if*

$$M_i K_i M_i = 0 \qquad \text{and} \qquad (I - M_i) K_i (I - M_i) = 0,$$

that is, if and only if K_i is purely off-diagonal in the splitting induced by M_i. In particular the coboundaries $\theta_{ab} = \alpha_a - \alpha_b$ satisfy it identically: $\theta \circ M = [\Delta, M]$ with $\Delta = \operatorname{diag}(\alpha)$, so $K = i[\Delta, M]$ and $MKM = i(M\Delta M - M\Delta M) = 0$ for every projector M and every V.

Theorem (ancilla-marginal phase-locking; certified block structures). *Fix (n, m_a) with $D = n\,m_a$. The kernel of the linearised system contains the $(D-1)$-dimensional coboundary space for every V, and for $(n, m_a) \in \{(2,2), (3,2), (2,3), (4,2), (2,4), (3,3), (3,4)\}$ it equals that space for all V outside a proper closed algebraic subset of $\mathrm{U}(D)$ — hence for Haar-almost-every $\hat{H}$. The reconstruction is therefore locally identifiable up to eigenbasis rephasing at those structures.*

Proof. Containment is the Lemma. For equality: $\dim\ker$ is upper semi-continuous in V and $\mathrm{U}(D)$ is irreducible, so it suffices to exhibit a single V at which the kernel has dimension exactly $D - 1$; the maximal-rank locus is then Zariski-open and dense. Such a V is given explicitly over the Gaussian rationals — a product of Pythagorean Givens rotations across all coordinate planes times a diagonal of unit-modulus Gaussian rationals — and the kernel dimension is computed there in exact arithmetic and equals $D - 1$ at each of the seven block structures (`phaselock_probes.py`). □

Remark (genericity is load-bearing). The hypothesis cannot be dropped: at $V = I$ the constraint is vacuous and the kernel is the whole parameter space (six of six at $D = 4$, fifteen of fifteen at $D = 6$), and a real Hadamard V degenerates likewise. *Remark (scope).* The semi-continuity argument runs per block structure, and no uniform-in-(n, m_a) statement is claimed.

Scope of the lemma under the ancilla-marginal form, and its extension. The lemma as stated assumes the strict form $T_{ij}(t) = |\langle i|e^{-iHt}|j\rangle|^2$ on an n-dimensional space — the trivial-ancilla case of the definition in §3.4. The emergent object is the ancilla-marginal (§3.1), and the lemma extends to it in a stronger form. Writing $U(t) = e^{-i\hat{H}t}$ on the dilated space of dimension $D = n\,m_a$ with eigenbasis $\{|a\rangle\}$ and configuration projectors P_i, the marginal

takes the exact form

$$T_{ij}(t) = \frac{1}{m_a} \sum_{a,b} e^{-i(E_a - E_b)t} \, \langle a|P_i|b\rangle \, \langle b|P_j|a\rangle,$$

so under non-degenerate gaps and (G4) for the *dilated* spectrum, Fourier analysis of the continuous-time visible data recovers, for each eigenvector pair (a, b), identified as in the lemma up to relabelling and the reflection that gives the antiunitary conjugate, the rank-one coefficient matrix $A_{ij}^{ab} = \frac{1}{m_a}\langle a|P_i|b\rangle\langle b|P_j|a\rangle$, hence the coherence vector $\langle a|P_i|b\rangle$ up to one phase per pair. The residual gauge is larger than the diagonal rephasing of the original lemma — one phase per pair, $D(D-1)/2$ parameters, under which $T(t)$ is exactly invariant — but the requirement that the reconstructed P_i be orthogonal projectors summing to the identity is a nonlinear constraint tying the pairs together, and it eliminates precisely the excess. The reduction can be shown directly. Writing a per-pair rephasing as the linearized perturbation $M_i[a, b] \mapsto M_i[a, b](1 + i\theta_{ab})$ with θ antisymmetric (where $M_i[a, b] = \langle a|P_i|b\rangle$), the linearized idempotency constraint $\delta(M_i^2) = \delta(M_i)$ reads $\sum_b (\theta_{ab} + \theta_{bc})\, M_i[a, b]\, M_i[b, c] = \theta_{ac}\, M_i[a, c]$ for all i, a, c. The diagonal coboundary $\theta_{ab} = \alpha_a - \alpha_b$ satisfies this identically: its left side telescopes to $(\alpha_a - \alpha_c) \sum_b M_i[a, b]M_i[b, c] = (\alpha_a - \alpha_c)\, M_i^2[a, c] = (\alpha_a - \alpha_c)\, M_i[a, c]$ using $P_i^2 = P_i$, which is the right side. This exhibits the $D - 1$ diagonal-gauge directions (the D shifts α_a modulo a global constant that acts trivially) as an analytic null space of the constraint, for every projector family. That these are the *only* null directions — that the constraint admits no pair-phase deformation beyond the coboundary — holds for generic $\widehat{H}$ at each block structure, by the genericity argument set out below; the constructive Fourier extraction has been demonstrated end-to-end. The marginal data therefore determine — generically, and at least locally — the *dilated* Hamiltonian and the configuration projectors up to an energy shift, diagonal phase conventions, and the antiunitary conjugation (statement above); the reconstruction reaches the dilation itself, not only an n-dimensional effective generator. The status of each component: the Fourier structure, its rank-one property, and the coboundary null directions are exact algebra; and the coboundary *exhausts* the null space for generic projector families, $\ker L_{\{P_i\}} = \{\theta_{ab} = \alpha_a - \alpha_b\}$, at each fixed (n, m_a). The families of n rank-m_a projectors summing to the identity are a single $U(D)$ orbit, hence an irreducible variety; $\dim\ker L$ is upper semi-continuous on it, its entries being polynomial in the family; and the coboundary lies in the kernel identically, so $\dim\ker L \geq D-1$ throughout. A single family with $\dim\ker L = D-1$ therefore confines the excess locus to a proper closed algebraic subset, of empty interior and Haar measure zero. Witnesses over $\mathbb{Q}(i)$ are certified in exact arithmetic at $(n, m_a) = (2, 2), (3, 2), (2, 3), (4, 2), (2, 4), (3, 3), (3, 4)$ (`phaselock_probes.py`); the argument runs shape by shape and supplies no statement uniform in D. The genericity conditions (G1)–(G4) transfer to the dilated spectrum, gaps, and coherence vectors. One caveat applies in sharpened form: the idealized permutation dynamics has dilated eigenphases at roots of unity — equally spaced, hence maximally gap-degenerate — so the lemma's genericity holds for the phys-

ical continuous-time dynamics away from that idealized limit, not at it; this is the same idealization boundary carried by the finite-ϵ displacement $\mathcal{D}$.

Conditions (G1)–(G3) fail only on a measure-zero set of Hamiltonians. A stronger argument applies whenever the partition itself breaks symmetries of H_{tot}. The genericity conditions concern the *effective* visible-sector Hamiltonian $\hat{H}_{\text{eff}}$ — the operator emerging after the trace-out — not the total Hamiltonian H_{tot}. The relevant distinction is whether a symmetry of H_{tot} *preserves the visible/hidden factorization* or not. A symmetry that fails to preserve the factorization — which includes the spatial symmetries an observer-centered partition breaks, since a partition centered on a specific observer maps boundary visible sites to hidden sites under translation, rotation about distant points, and, when the coarse-graining map determines a simultaneity foliation (the hypothesis under which the §3.5 lemma excludes boosts), boost — has no well-defined action on the visible factor alone, and the degeneracies it protects in H_{tot} are lifted in $\hat{H}_{\text{eff}}$. (By contrast, a symmetry that *does* factorize as $S_V \otimes S_H$ descends to the visible symmetry S_V of $\hat{H}_{\text{eff}}$, because the uniform-bath partial trace is invariant under any unitary on the hidden factor; such factorization-preserving symmetries are not lifted, and the visible-internal ones among them — those acting entirely within the visible sector and commuting with the trace-out — are precisely the gauge symmetries of the emergent description. Gauge degeneracies correspond to physically equivalent states and do not affect the phase-locking argument, which operates on gauge-inequivalent transition data.) Because the spatial symmetries of an observer-centered partition are factorization-breaking, $\hat{H}_{\text{eff}}$ satisfies G1 (non-degenerate spectrum) not by typicality in an abstract ensemble but because the partition acts as a symmetry-breaking mechanism. Condition G2 (non-degenerate energy gaps) is not symmetry-protected — gap coincidences are a codimension-one condition rather than a group-theoretic one — so it is not delivered by the partition mechanism and rests on genericity directly. G2 holds for generic $\hat{H}_{\text{eff}}$; it can fail only on a measure-zero set, and we are not aware of a structural feature of the substratum that forces such failure systematically. (In particular, single-cycle hidden dynamics does not: although the unitary cycle operator has equally-spaced phases on the unit circle, the Hermitian effective generator $\hat{H}_{\text{eff}}$ has real eigenvalues that are not equally spaced, and retains non-degenerate gaps after the trace-out.) Condition (G4) has the standing of G2. The triangle identities are the only relations $\omega_1 + \omega_2 = \omega_3$ among frequencies of three different pairs of levels that hold for every spectrum, so (G4) excludes only relations that are not forced, each a non-trivial linear equation on the levels; relations with four or more terms, including forced ones such as $\omega_{kl} + \omega_{mn} = \omega_{kn} - \omega_{lm}$, are not restricted. (G4) therefore fails only on finitely many hyperplanes of the spectrum, is not delivered by the partition mechanism, and is not derived from the substratum: it is a posited genericity hypothesis, and the lemma is conditional on it. It enters only the identification of level pairs.

3.2 Independent Derivation via Stinespring Dilation

An independent derivation of the emergent quantum description uses only Stinespring's dilation theorem [15] and standard properties of the partial trace [16], without invoking the stochastic-quantum correspondence of §3.1. The two routes are independent in their inputs and in the results they invoke (the correspondence's own proof also proceeds via Stinespring-type dilation, so the shared bedrock is Stinespring's 1955 theorem); either alone suffices.

Setup. The finite configuration spaces $\mathcal{C}_V = \{x_1, \ldots, x_n\}$ and $\mathcal{C}_H = \{h_1, \ldots, h_m\}$ (Lemma 1 for $\mathcal{C}_V$; $\mathcal{C}_H$ under the total-finiteness posit, §1.2, §4.5) embed into Hilbert spaces $\mathcal{H}_V = \mathbb{C}^n$ and $\mathcal{H}_H = \mathbb{C}^m$ via $|i\rangle \leftrightarrow x_i$ and $|k\rangle \leftrightarrow h_k$. This introduces no quantum postulates: it is the canonical identification of probability distributions on a finite set with diagonal density matrices.

Lemma (permutation unitarity). *Any bijection $\varphi : \mathcal{C}_V \times \mathcal{C}_H \to \mathcal{C}_V \times \mathcal{C}_H$ defines a unitary U_φ on $\mathcal{H} = \mathcal{H}_V \otimes \mathcal{H}_H$.*

Proof. Define $U_\varphi |i, k\rangle = |\varphi(x_i, h_k)\rangle$. Since φ is a bijection, U_φ permutes the orthonormal basis, hence is unitary. □

A finite permutation does not by itself generate a continuous flow: on a finite discrete configuration space the dynamics is the discrete group $\{U_\varphi^k\}$, and any continuous one-parameter interpolation is *additional structure*, chosen rather than derived. The depth-two structure of the substratum update supplies something weaker and derived: an exact strongly continuous path of $*$-automorphisms of the quasilocal algebra from the identity to the update's Heisenberg action, for which no one-parameter-group law is established and no generator is exhibited (kernel: `driveQ_isContinuousPath`, `driveQ_smul`, `driveQ_one_eq_heisQ` in `OIBridge/SecondOrderDrive.lean`). When such an interpolation is introduced — a strongly continuous one-parameter unitary group U_t with $U_1 = U_\varphi$, which always exists but is not unique (any branch of $\log U_\varphi$, differing by 2π shifts of eigenphases, defines one) — the exponential $U_t = e^{-i\hat{H}t}$ then supplies the continuous unitary interpolation for Hermitian $\hat{H}$; the framework's continuous-time statements are conditional on that choice, with the branch ambiguity part of the phase-retrieval gauge catalogued below.

Lemma (Reverse direction, scoped). *Any single one-step doubly stochastic matrix on a finite configuration space $\mathcal{C}_V$ can be realised as the one-step marginal of a deterministic bijection on $\mathcal{C}_V \times \mathcal{C}_H$ with uniform prior on $\mathcal{C}_H$, for some finite $\mathcal{C}_H$ — exactly when the transition probabilities are rational, and to arbitrary precision otherwise.* The restriction to doubly stochastic matrices is forced, not elective: a uniform-prior bijection marginal preserves the uniform distribution and is therefore doubly stochastic, so a merely row-stochastic matrix — one sending both states of a two-state space to the first, say — is not realisable in this form. The emergent statistics this converse serves lie inside the realisable class: ancilla-uniform marginals of unistochastic matrices are doubly stochastic. The scope is the matrix, not the process: realising prescribed *multi-time* statistics

— a family of joint distributions over trajectories — with a single bijection and a single uniform prior is a stronger demand and is not claimed by this lemma; where multi-time structure is used below, it enters through the dilation actually constructed, not through this converse.

Proof. Any marginal of a bijection with uniform prior is a doubly stochastic matrix, and by Birkhoff–von Neumann is a convex combination of permutation matrices. A bijection on $\mathcal{C}_V \times \mathcal{C}_H$ realising the required mixture is obtained by letting $\mathcal{C}_H$ enumerate the permutations; for multi-step processes, the construction extends by history dilation — made precise, with a structured prior, by the process-dilation theorem of §3.4. □

Remark. This is the correspondence for a single step with a uniform hidden prior: every such marginal of a bijection is doubly stochastic, and a doubly stochastic matrix is such a marginal exactly when its entries are rational, and is approximated by such marginals to arbitrary precision otherwise. The process-level correspondence, with a structured prior, is the finite-horizon process-dilation theorem of §3.4. It is *not* a full CPTP-channel correspondence — channels that create coherences from diagonal inputs have no permutation-unitary realisation with uniform ancilla. For the characterization theorem (§3.4), which is stated in terms of transition probabilities, this is the appropriate scope.

The quantum channel. The observer's ignorance of the hidden sector (Lemma 3) corresponds to $\rho_H = I_m/m$. The visible-sector quantum channel is

$$\Phi(\rho_V) = \mathrm{Tr}_H\left[U_\varphi\,(\rho_V \otimes \rho_H)\,U_\varphi^\dagger\right]$$

This is CPTP by a standard result [16, Theorem 8.1], with Kraus representation $\Phi(\rho_V) = \sum_{k,l} K_{kl}\,\rho_V\,K_{kl}^\dagger$ where $K_{kl} = m^{-1/2}\langle l|U_\varphi|k\rangle_H$. The triple $(\mathcal{H}_H, U_\varphi, \rho_H)$ is a *unitary realization* of Φ ([Structure] Definition 9.2). The reference state here is the uniform mixed state, so the triple is a unitary realization rather than a Stinespring dilation: with a pure reference the Stinespring dilation is the associated isometry $\psi \mapsto U_\varphi(\psi \otimes |0\rangle)$, not the unitary, and obtaining one from a mixed reference requires purifying ρ_H. Nothing below depends on the distinction; it is kept because the uniqueness statements of [Structure] Propositions 9.7a–9.7b do.

Theorem (Born-form representation of the transition probabilities). *The classical transition probabilities T_{ij} (§1.4) equal the Born-rule probabilities of Φ — a representation identity: it exhibits the Born form and does not select the quadratic exponent (§3.4).*

Proof. $P(j|i) = \langle j|\Phi(|i\rangle\langle i|)|j\rangle = m^{-1}\sum_{k,l}|\langle j,l|U_\varphi|i,k\rangle|^2$. Since U_φ is a permutation, $\langle j,l|U_\varphi|i,k\rangle = \delta_{(j,l),\varphi(i,k)}$. Thus $P(j|i) = m^{-1}\sum_k \delta_{j,\pi_V(\varphi(x_i,h_k))} = T_{ij}$. □

Remark (failure of (C1)). Condition (C1) does not suffice. For $|V| = |H| = 2$ and $\varphi(x,h) = (h,x)$ one has $T_{ij} = 1/2$ for all i,j, yet $\Phi(\rho) = I/2$

identically: a constant channel, hence measure-and-prepare. Exhaustive enumeration gives the same verdict for every (C1)-satisfying bijection at $|V| = |H| = 2$, and for 612 of 648 at $|V| = 2$, $|H| = 3$ (verification code: `papers/oi_lattice_code/coherence/`). The obstruction is structural: (C1) constrains the diagonal of Φ, whereas entanglement-breaking is a property of $\mathrm{id} \otimes \Phi$ and is not fixed by the action on single-system inputs.

What replaces (C1) is not a further condition on T but a condition on the partition, and it is established for a specific realization rather than for arbitrary bijections. Take the substratum of §1.2 on a lattice Λ with $q = 2$, state $(u, v) \in \mathbb{F}_2^\Lambda \times \mathbb{F}_2^\Lambda$ and the reversible nearest-neighbour update

$$u'_x = \sum_{z \sim x} u_z + v_x, \qquad v'_x = u_x,$$

the visible sector being the sites of a region $R \subseteq \Lambda$. Write M_{HV} and M_{VH} for the blocks of this update carrying visible input to hidden output and hidden input to visible output, $\kappa = \dim \ker M_{HV}$, $w = \operatorname{rank} M_{VH}$, and $s = \dim V = 2|R|$. The inner and outer boundaries of R are $\partial^- R = \{x \in R : \exists\, z \notin R,\ z \sim x\}$ and $\partial^+ R = \{y \notin R : \exists\, z \in R,\ z \sim y\}$.

Lemma 1 (boundary bound). $\operatorname{rank} M_{HV} \le |\partial^+ R|$ *and* $\operatorname{rank} M_{VH} \le |\partial^- R|$.

Proof. For $y \notin R$ the component $v'_y = u_y$ is a hidden input, and $u'_y = \sum_{z \sim y} u_z + v_y$ involves a visible input only if some $z \sim y$ lies in R, that is only if $y \in \partial^+ R$; so M_{HV} has non-zero rows only among $\{u'_y : y \in \partial^+ R\}$. Dually, for $x \in R$ the component $v'_x = u_x$ is a visible input, and u'_x involves a hidden input only if $x \in \partial^- R$. □

One update step transports information one lattice site, so the sectors resolve one another only across the cut. What the argument establishes is a factorization: each cross-block has non-zero rows only at the boundary sites named, so it factors through the corresponding boundary coordinates. That statement uses no property of the coefficient ring and holds verbatim over $\mathbb{Z}_q$; over a field — in particular over the $\mathbb{F}_2$ used here — it gives the displayed rank bounds, ordinary vector-space rank being the relevant notion there. For a general nearest-neighbour bijection the same support argument yields the weaker conclusion that the hidden output depends on the visible input only through $\partial^- R$. Since adjacency is symmetric, the same reading applies to the columns: M_{HV} reads visible input only through $\{u_x : x \in \partial^- R\}$ and M_{VH} only through $\{u_y : y \in \partial^+ R\}$, so each rank is in fact bounded by $\min(|\partial^- R|, |\partial^+ R|)$ — a sharpening that uses the symmetry where the row statement does not, and that Corollary 2 consumes.

Lemma 2 (channel normal form). *Write $A = M_{VV}$, $B = M_{VH}$, $C = M_{HV}$ for the blocks of the update, and L_{RR} for the adjacency of Λ restricted to R. Then: (i) A is invertible, with $A^{-1}(a_u, a_v) = (a_v,\ a_u - L_{RR}\, a_v)$, so the visible block alone induces the permutation (Clifford) unitary $V_A|i\rangle = |Ai\rangle$; (ii) $\Phi(|i\rangle\langle i'|)$*

vanishes unless $C(i-i')=0$; (iii) on the Weyl operators the channel acts, with no phase, as

$$\Phi(X^aZ^b) = [\,Ca=0\,]\,[\,B^TA^{-T}b=0\,]\;X^{Aa}\,Z^{A^{-T}b},$$

so that $\Phi = \mathrm{Ad}_{V_A}\circ\Phi_G$, where $\Phi_G = |G|^{-1}\sum_{g\in G} g\cdot g^\dagger$ is the uniform mixture over the group G of Weyl operators $X^\alpha Z^\beta$ with $\alpha\in\mathrm{im}(A^{-1}B)$ (translations) and $\beta\in\mathrm{im}\,C^T=(\ker M_{HV})^\perp$ (phases); (iv) G is abelian, of order $2^{\,s-\kappa+w}$.

Proof. (i) is the displayed inverse, read off from $v'_x=u_x$ and $u'_x=(L_{RR}u)_x+v_x$ on R; invertibility of A is what makes $i\mapsto Ai$ a relabelling. (ii) The hidden outputs of i and i' agree for some hidden input exactly when $C(i-i')=0$, and then for every hidden input. (iii) With the hidden input uniform, $\Phi(|i\rangle\langle i'|) = [\,C(i-i')=0\,]\;\mathbb{E}_{t\in\mathrm{im}\,B}\,|Ai+t\rangle\langle Ai'+t|$. Summing this against $X^aZ^b=\sum_i(-1)^{b\cdot i}\,|i+a\rangle\langle i|$ and substituting $i\mapsto i-A^{-1}t$ gives the displayed rule: the character sum over $\mathrm{im}\,B$ produces the second indicator and cancels every phase. The right-hand side is Ad_{V_A} applied to the Hilbert–Schmidt-orthogonal projection onto $\mathrm{span}\{X^aZ^b : Ca=0,\ B^TA^{-T}b=0\}$, and the uniform mixture over the symplectic complement of that index set — which is exactly G — is that projection. (iv) An element $X^\alpha Z^\beta$ with $\alpha\in\mathrm{im}(A^{-1}B)$, $\beta\in\mathrm{im}\,C^T$ commutes with another iff the cross-pairings $\alpha\cdot\beta'$ vanish; over all of G this is the single condition $CA^{-1}B=0$, which holds identically for this update: B feeds hidden u into the visible u'-components, A^{-1} carries the u'-subspace onto the v-subspace, and C reads only the visible u-components. The translation and phase families intersect only in the identity, so $|G|=2^{\mathrm{rank}\,B}\cdot 2^{\mathrm{rank}\,C}=2^{\,w+(s-\kappa)}$. □

Theorem (separability threshold). *Let G be an abelian subgroup of the Weyl group on s qubits, of order 2^t modulo phases. Then $\Phi_G=|G|^{-1}\sum_{g\in G} g\cdot g^\dagger$ is entanglement-breaking if and only if $t=s$.*

Proof. Modulo phases the Weyl operators form the symplectic space $\mathbb{F}_2^{2s}$, on which abelian subgroups are exactly the isotropic subspaces; isotropy bounds $t\le s$. Any isotropic subspace of dimension t is carried to $\langle Z_1,\dots,Z_t\rangle$ by a symplectic transformation (Witt's extension theorem for the symplectic space $\mathbb{F}_2^{2s}$), and every symplectic transformation is implemented by a Clifford unitary U. Hence $U\Phi_GU^\dagger$ is complete dephasing on t qubits tensored with the identity on the remaining $s-t$. Complete dephasing is measure-and-prepare and therefore entanglement-breaking, and a tensor product of channels is entanglement-breaking exactly when every factor is — separability of the joint Choi state survives the partial trace onto any single factor pair, so one entangled factor Choi rules out separability of the whole; the identity on a non-trivial factor is not entanglement-breaking. Since conjugation by a unitary preserves the property, Φ_G is entanglement-breaking precisely when $s-t=0$. □
(kernel: `entanglementBreaking_twirl`, `not_entanglementBreaking_twirl`, `exists_lagrangian_tuple` in `OIBridge/WeylLift.lean`; `isotropic_iff_commute` in `OIBridge/WeylTwirl.lean`; `separable_imp_ppt` in `OIBridge/Separability.lean`)

Corollary 1. *For one step of the displayed update on* $\mathbb{F}_2$ *—* $u'_x = \sum_{z\sim x} u_z + v_x$, $v'_x = u_x$, *with the hidden sector initially uncorrelated and maximally mixed — the visible channel* Φ *admits the Clifford-conjugated abelian Weyl-mixture normal form of Lemma 2 and is not entanglement-breaking if and only if* $w < \kappa$. Immediate from Lemma 2 and the Theorem with $t = s - \kappa + w$: entanglement breaking is invariant under composition with the unitary Ad_{V_A}, so Φ is entanglement-breaking exactly when Φ_G is, and $t = s$ reads $w = \kappa$ (Lemma 2 gives $w \le \kappa$, matching $t \le s$).

Corollary 2. *If* $\min(|\partial^- R|, |\partial^+ R|) < |R|$ *then* Φ *is not entanglement-breaking.*

Proof. $s = 2|R|$. Lemma 1's column sharpening bounds each of $\operatorname{rank} M_{HV} = s - \kappa$ and $\operatorname{rank} M_{VH} = w$ by $\min(|\partial^- R|, |\partial^+ R|)$, so $w + (s - \kappa) \le 2\min(|\partial^- R|, |\partial^+ R|) < 2|R| = s$, that is $w < \kappa$. □

Remark. For an exact cubic block of side $\ell \ge 2$ in $d = 3$: $|\partial^- R| = \ell^3 - (\ell - 2)^3$ and $|\partial^+ R| = 6\ell^2$, the inner boundary being the smaller of the two, so the hypothesis $\min(|\partial^- R|, |\partial^+ R|) < |R|$ reads $\ell^3 - (\ell-2)^3 < \ell^3$, that is $(\ell-2)^3 > 0$: the criterion holds exactly when the block has a non-empty bulk, first at $\ell = 3$ (at $\ell = 2$ every site lies on the cut and $8 = 8$); the threshold is structural, and a big-O statement alone could not name it. Within this single-step result the criterion carries no dependence on $|H|$, since the coupling reads only the boundary; the counterexample above is the degenerate case, every visible site lying on the cut so that the region has no bulk. Whether the same holds over many steps is not settled here: repeated interaction can propagate boundary information into the hidden bulk, and a multi-time statement would require a process-tensor argument rather than a single-channel one. The suggestion that coherence is a bulk quantity and decoherence an area quantity is at this stage an interpretation of the one-step result, not a theorem about the asymptotic dynamics.

Remark (scope). Lemma 1's factorization holds over any $\mathbb{Z}_q$, the displayed rank bounds being its reading over a field, and in weakened form the support argument covers arbitrary nearest-neighbour bijections. Lemma 2, the Theorem and both Corollaries are stated for $q = 2$; the symplectic and Clifford arguments carry over to prime q with $2^\bullet$ replaced by $q^\bullet$, while composite q requires module-theoretic treatment of rank and kernel and is not covered. All of this concerns the realization displayed above and is not a property of arbitrary finite bijections. For a general linear bijection the abelian property of Lemma 2(iv) can fail ($CA^{-1}B \neq 0$); the mixture is then over a non-abelian Weyl family, and entanglement breaking is governed by the isotropy of the surviving index set rather than by its order alone — not treated here. Two boundary markers on interpretation. Coherence preservation is a property of this realization, not an additional structural condition: (C1)–(C4) constrain coupling, timescales, and capacity, and neither contain nor imply a coherence requirement — the swap example satisfies (C1) while its channel is entanglement-breaking, and which realizations preserve one-step coherence is the geometric question answered above for the displayed update alone. And non-entanglement-

breaking is a statement about a single channel, not about quantum mechanics: it supplies neither the observable algebra, nor preparations and interventions, nor tensor composition, nor multi-time statistics — the operational reconstruction claims rest on the dilation and correspondence arguments, not on Corollary 1 — and the entanglement-breaking status of one step does not determine whether the visible multi-time process is P-divisible: the hidden sector can carry memory even when a step's channel is entanglement-breaking, as a two-state system with a two-state hidden register already shows. The enumeration and the boundary bounds are reproduced by the deterministic tests in `papers/oi_lattice_code/coherence/` (`coherence_enumeration.py`); the transfer rule and normal form of Lemma 2, the threshold Theorem exhaustively at $s = 2, 3$, and both directions of Corollary 1 — a negative partial transpose wherever $w < \kappa$, an explicit measure-and-prepare form wherever $w = \kappa$ — are verified for $q = 2$ and $q = 3$ by `coherence_theorem_tests.py` in the same directory; the strictness of the dephasing/entanglement-breaking containment and the compatibility of an entanglement-breaking step with a P-indivisible visible process are certified by `eb_pindivisibility_tests.py` there as well.

Lemma (diagonal preservation). *The channels* $\Phi_t(\rho) = \mathrm{Tr}_H[U^t(\rho \otimes \mu_H)\,U^{-t}]$ *with* U *a permutation unitary map computational-diagonal states to computational-diagonal states.*

Proof. A permutation unitary conjugates a diagonal state to a diagonal state; the partial trace of a diagonal state is diagonal. □ (Certified exactly for $t = 1, \ldots, 6$ in `c1_cp_scope_probes.py`.)

Theorem (CP-indivisibility, framework scope). *For the permutation-dilation family* $\{\Phi_t\}$ *above, the P-indivisibility of §2.3 implies CP-indivisibility: there exist* $t_2 > t_1 > 0$ *with no CPTP map* Λ *satisfying* $\Phi_{t_2} = \Lambda \circ \Phi_{t_1}$.

Proof. Suppose such a Λ exists. $M_{kj} := \langle j|\Lambda(|k\rangle\langle k|)|j\rangle$ is a stochastic matrix (positivity and trace preservation of Λ). By the Lemma, $\Phi_{t_1}(|i\rangle\langle i|) = \sum_k T_{ik}(t_1)\,|k\rangle\langle k|$, so $T_{ij}(t_2) = \langle j|\Lambda(\Phi_{t_1}(|i\rangle\langle i|))|j\rangle = \sum_k T_{ik}(t_1)\,M_{kj}$ — P-divisibility of the population process. The contrapositive gives the result. □

Remark (scope: P-indivisibility is not quantum non-Markovianity). For a general CPTP family the reduction fails: $\Phi_{t_1} = \mathrm{Ad}_H$ (Hadamard), $\Phi_{t_2} = \mathrm{id}$ is CP-divisible with a unitary intermediate map, while the computational-basis populations $\frac{1}{2}J \to I$ are P-indivisible — the intermediate map consumes coherences invisible to $T(t_1)$. Stochastic P-indivisibility must therefore not be equated with open-system non-Markovianity outside the diagonal-preserving class; the framework's channels are in that class, which is what licenses the theorem. (Certified exactly over $\mathbb{Q}(\sqrt{2})$ in `c1_cp_scope_probes.py`.)

By the Breuer-Laine-Piilo criterion [63], CP-indivisibility is witnessed by non-monotonic distinguishability, but the correct witness is *ancilla-assisted*: the entanglement of the system with a shielded, maximally entangled ancilla decreases monotonically under CP-divisible evolution, so its increase witnesses

CP-indivisibility [4], and for bijective (invertible) dynamical maps on $\mathbb{C}^d$, CP-divisibility is equivalent to monotonic non-increase of the trace distance between every pair of equiprobable states of the system and a $(d+1)$-dimensional ancilla [9]; ordinary system trace distance is a strictly weaker witness and can stay monotone on a CP-indivisible family. The classical analogue is exact and is certified here: a family can be P-indivisible while total-variation distance is non-increasing for every pair of visible distributions (`translation_probes.py`, §2.3). Backflow in the system alone is therefore sufficient evidence of indivisibility, never necessary.

Remark (three notions, kept apart). Three distinct things travel under nearby names, and this paper uses them in fixed senses. (1) **Accessible finite-horizon memory**, equivalently conditional past–future dependence in the visible law: this is the framework's own notion, what the characterization theorem is about, what $M_t > 0$ measures, and what (C1), (C3) and (C4) diagnose on a finite horizon. Where this paper writes "accessible non-Markovianity" it means exactly this and says nothing about a quantum channel. (2) **Stochastic P-indivisibility**: failure of stochastic divisibility of the visible population law, which on one fixed finite reversible representative is what the recurrence-scale theorem delivers — resting on recurrence together with a non-permutation rooted map, not on (C1), (C3) and (C4) alone. (3) **Generic quantum non-Markovianity**: an open-system notion whose content depends on which quantum divisibility or backflow criterion is adopted. No unconditional implication among the three is asserted here. The two preceding remarks give the exact conditions under which (2) constrains a quantum intermediate map, and the Hadamard family shows that outside the diagonal-preserving class (2) and (3) come apart.

Approximate unitarity. On observable timescales $t \ll \tau_B$, the hidden-sector state is approximately frozen (the slow-bath regime, (C2-slow); at the horizon the bath is also imperturbable on laboratory timescales, with the $N_H \gg N_V$ hierarchy of (C3)'s realization clause). At $t = 0$ the channel generator is exactly Hamiltonian, $d\Phi_t/dt|_{t=0}(\rho_V) = -i[\hat{H}_{\text{eff}}, \rho_V]$ with $\hat{H}_{\text{eff}} = H_V + \mathrm{Tr}_H[V_{\text{int}}(\mathbb{1}\otimes\rho_H)]$ the bath-averaged coupling; the dissipative part of the generator vanishes at $t = 0$. The bath average moves on the scale τ_B: with $\rho_H(t) = \rho_H + t\,\dot{\rho}_H + \mathcal{O}((t/\tau_B)^2)$ and $\dot{\rho}_H$ of order $1/\tau_B$, the bath-averaged coupling $\mathrm{Tr}_H[V_{\text{int}}(\mathbb{1}\otimes\rho_H(t))]$ changes at first order by $t\,\mathrm{Tr}_H[V_{\text{int}}(\mathbb{1}\otimes\dot{\rho}_H)]$, Hermitian because V_{int} and $\dot{\rho}_H$ are, so over $t \sim \tau_S$ the first-order effect of the bath's motion is a Hermitian drift of $\hat{H}_{\text{eff}}$ of order $(\tau_S/\tau_B)\,\|V_{\text{int}}\|$, not a dissipative term. At finite time the visible channel acquires two distinct dissipative contributions, which it is important to separate. (i) Even in the strictly frozen-bath limit ($\tau_B \to \infty$) the channel is not unitary: it is a mixture over the bath ensemble of bath-conditioned visible unitaries $e^{-i(H_V + \langle h|V_{\text{int}}|h\rangle)t}$, which dephases by the *spread* of those conditioned Hamiltonians — a contribution of order $\|V_{\text{int}}\|^2$ (the coupling strength), independent of τ_B. (ii) The dissipative correction from the bath's actual motion away from the frozen limit enters at order $(\tau_S/\tau_B)^2$. Consequently the dynamics is approxi-

mately the Schrödinger equation generated by $\hat{H}_{\text{eff}}$ to the extent that *both* the slow-bath condition $\tau_S \ll \tau_B$ *and* a weak-coupling condition (small conditioned-Hamiltonian spread, $\|V_{\text{int}}\|\,\tau_S \ll 1$) hold; of the two contributions, the bath-motion term is fixed by the timescale hierarchy at $\mathcal{O}((\tau_S/\tau_B)^2)$ ($\sim 10^{-64}$ for the horizon ratio of (C2)'s realization clause), while the conditioned-Hamiltonian spread is not — its magnitude for the cosmological partition awaits the substratum interaction structure. Where these hold, the phase-locking lemma (§3.1), under its genericity hypotheses (G1)–(G4), determines $\hat{H}_{\text{eff}}$ from continuous-time transition data up to the lemma's full gauge — an overall energy shift, the residual diagonal-unitary rephasing freedom that is physically trivial (basis convention), and the antiunitary conjugation $\hat{H} \to -\hat{H}^*$, which is not shift- or rephasing-related and is a genuine twofold ambiguity, not a convention; the residual dissipation $\mathcal{D}$, if measurable at sufficient precision, is a prediction of the framework rather than an approximation artifact.

Comparison of routes.

	Barandes route (§3.1)	Stinespring route (§3.2)
Input	Finite-configuration member of the source's class, under (T)	Bijection on $\mathcal{C}_V \times \mathcal{C}_H$ + marginalization
Bridge	Stochastic-quantum correspondence [11, 12]	Stinespring dilation [15] + partial trace [16]
Output	Dilated unitary whose ancilla-marginal Born probabilities reproduce T_{ij}	CPTP channel with $T_{ij} = \langle j\|\Phi(\|i\rangle\langle i\|)\|j\rangle$
Born rule	Representation dictionary of Q_{fb}, not selected (§3.4)	Partial trace structure (above)
Scope	The source's class, Markov special cases included	Processes from marginalized bijections

The Barandes route is more general (any finite-configuration member of the source's class); the Stinespring route requires only textbook results [15, 16] and delivers additional structure at three separated levels. *Level one (generic, proved):* the tensor product $\mathcal{H}_V \otimes \mathcal{H}_H$, a unitary representation of the enlarged basis dynamics, and a CPTP reduced channel — the representation of the basis statistics. *Level two (specific, proved):* non-entanglement-breaking one-step coherence for the linear lattice realization under the $w < \kappa$ criterion (§3.2) — not generic: the swap-class dilations are entanglement-breaking. *Level three (characterized conditionally):* coherent preparation, noncommuting interventions, and the instrument algebra — the operational bridge, which on a finite carrier, with complex matrix kinematics taken as given, holds exactly under the

five completion conditions of §3.4, none supplied by bare finite OI; under the substratum's stated interventions, (iv), full reversible control, is not supplied (§3.4), and whether an extended OI mechanism supplies it, and which of the others the substratum supplies, are open (§3.2 checklist). Within level one, the frozen-limit error analysis stands: the leading departure from the frozen-limit description enters at $\mathcal{O}(\tau_S/\tau_B)$ as an absorbable Hermitian drift of $\hat{H}_{\text{eff}}$, with genuine non-unitarity $\mathcal{D}$ entering through the conditioned-Hamiltonian spread and the second-order bath-motion term $\mathcal{O}((\tau_S/\tau_B)^2)$ (§3.2); for the horizon ratio of (C2)'s realization clause the two orders are $\sim 10^{-32}$ and $\sim 10^{-64}$. The two routes agree on the configuration-transition probabilities $T_{ij}(t)$; equivalence of the complete operational observable structure is part of level three. The internal permutation-unitary route carries (Q2) for realized processes (the fixed-$\hat{H}$ lemma, §3.4), with the Barandes route supplying the compressed form, (Q1), under (T); the Stinespring route supplies the level-one representation of (iii) independently.

Dependency scope of the correspondence. The equivalence theorem of §3.4 fixes the framework's exposure to the correspondence precisely, and it is narrower than earlier versions of this paper claimed. *In-house:* the finite-horizon equivalence $S \iff D \iff Q_{\text{fb}}$ — existence of a fixed-basis unitary Born representation for every finite-horizon law, the framework's own processes included, by the permutation unitary of any realization — together with the fixed-$\hat{H}$ lemma, the phase-locking gauge, and the universal hidden-memory theorem. *Imported, under (T):* the source's $\leq n^3$ dilation bound for its finite-configuration class, which the per-time form on dimension n^2 (§3.1) does not need, and the interpretive apparatus that travels with it (basis independence, the general-observable reading). *Additional, with neither route supplying it:* the operational instrument algebra and coherent preparation — supplied exactly by the five operational completion conditions of the characterization of §3.4, none of which bare finite OI provides — and representation consistency across growing horizons (the obstruction theorem of §3.4). A failure of the correspondence would therefore cost the framework the interpretive apparatus, and nothing else: existence of the representation, its per-time form on dimension n^2 (§3.1), the equivalence, and the necessity results are internal. P-indivisibility appears nowhere in this division of labor — it gates neither membership in the source's class nor the representation's existence (§3.4, Remark) — and its role is confined to nontriviality at the recurrence scale and to the Bell-class constructions of §3.3.

Operational-reconstruction route: target conditions and the outstanding bridge. Both routes above are constructive: they exhibit quantum mechanics as a description of the marginalized process. The reconstruction theorems [Hardy, arXiv:quant-ph/0101012 (2001); Masanes and Müller, New J. Phys. **13**, 063001 (2011); Chiribella, D'Ariano and Perinotti, Phys. Rev. A **84**, 012311 (2011)] derive quantum theory from operational axiom sets. The finite physical layer supplies two useful ingredients: every fixed finite representative has finite predictive dimension, and the substratum dynamics admits

reversible dilations. Neither statement by itself discharges the corresponding hypothesis for an **exact operational completion**. The classical-dimension obstruction below forces any literal continuum-QM endpoint beyond one fixed finite carrier into a compatible closure/limit, and stagewise finiteness does not prove that this completion retains finite predictive rank. The purification, continuous-transitivity, and tomographic-locality axioms of the cited reconstructions are hypotheses of their continuum endpoint, which the classical-dimension obstruction places beyond any fixed finite carrier; they are not what the finite characterization uses. On a fixed finite carrier and its composites the intervention side is settled as a characterization rather than a derivation (§3.4; kernel: `exactAll_iff_physical_general`, with necessity `physical_of_exactAll` and sufficiency `exactAll_of_physical_general`; `main_result`; no premise beyond its stated setting): for every nonempty finite visible system, with complex matrix kinematics taken as given — operations act on the complex matrices indexed by the system's configurations and, at each ancilla level, by configuration–ancilla pairs, and a derivation of that kinematics from operational premises is open — an operational theory's available outcome families on the system and on every positive ancilla level are exactly the normalized finite Kraus instruments — the full instrument algebra, complete positivity of every branch, noncommuting composition, and consistency across composites — if and only if five conditions hold: valid probabilities, trivial-ancilla consistency, inert spectators, full reversible control, and iterated composition. Each condition is independent of the other four and of the observation process: for each there is a qubit theory realizing the same finite C1–C4 process with its visible readout that satisfies the other four and fails it. Bare finite OI therefore does not select quantum mechanics; the route classifies the OI-compatible completions by those conditions. The intervention dilation's classical rung — permutation instruments reproducing the comb statistics $P(X_{k+1} \mid X_{\leq k}, A_{\leq k})$ — is constructed (Theorem, §3.4) and reaches the intervention statistics of any finite instrument set (Leggett-Garg violations included, §3.4); the general form of (T) is instantiation-checked for representative finite processes, not established for the framework's realizations at arbitrary horizon. (T) alone licenses the imported stochastic–quantum representation [11, 12] — P-indivisibility gates neither membership nor representation (§3.4, Remark) — and P-indivisibility does not select the nonclassical branch of these reconstructions either: a classical stochastic system can be P-indivisible. The route is therefore a classification conditional on the five completion conditions, not an independent derivation.

3.3 Bell Inequality Violations

The framework is a deterministic hidden-state theory, so its Bell content must be stated at the correct level. Bare finite OI representability is too broad to select the quantum Bell set: if the two wings are allowed to share globally coupled setting-dependent hidden dynamics, the same reversible response-table construction that realizes arbitrary finite stochastic laws also realizes PR-box

behavior. That fact is not in tension with the exact unitary/Born representation of §3.4; a global stochastic law can be represented unitarily while failing the *local subsystem* conditions of standard quantum mechanics. The selector in a Bell experiment is therefore causal locality, not representability.

Coupling-graph causal cone. Write the spatial substratum as $S = \prod_{i \in V} S_i$, with V and each S_i finite, and define the coupling graph G_φ by one-step dependency: $i \sim j$ whenever the ith output component of φ depends on the jth input component for some state (symmetrized for the spatial graph). Then, by induction, $(\varphi^k(s))_i$ depends only on initial components in the graph ball $B_{G_\varphi}(i,k)$. Thus an intervention that reads and writes only region A cannot alter, k updates later, any component outside $B_{G_\varphi}(A,k)$; in particular it cannot alter a readout region R with $d(A,R) > k$. Finiteness of V is used in the first step, where insensitivity to each single component outside the dependency set makes the ith output a function of the components in it. This result uses the full spatial OI constitution — site factorization plus the graph read from the actual dynamics — but does not require the stronger simple-cubic nearest-neighbor restriction used elsewhere to obtain a particular dispersion relation. For a class of states $K \subseteq S$, write G_φ^K for the graph of the dependences that occur at states of K: $i \sim j$ whenever some $s \in K$ and some s' differing from s only in the jth component have $(\varphi(s))_i \neq (\varphi(s'))_i$ (symmetrized as above); then $G_\varphi = G_\varphi^S$, and G_φ^{ref} denotes this graph for the reference vacuum/uniform state class. For a linear update (ledger item (vii), §4.5) the difference $(\varphi(s'))_i - (\varphi(s))_i$ depends only on $s' - s$, so $G_\varphi^K = G_\varphi$ for every non-empty K; a graph that differs between state classes requires an update outside (vii).

The causal-cone statement is pointwise in the pre-setting microstate. It must be distinguished from **operational setting screening**. In a closed deterministic universe the setting generators may be statistically correlated with preparation-relevant hidden variables; conditioning on a setting can then reweight the hidden ensemble and create apparent remote statistical dependence even though no influence crossed the graph cone. **Theorem (Bell ceiling for a screened local completion).** *Let Ω be a pre-setting variable that fixes each wing's response to every setting. Suppose (i) the dynamics is deterministic; (ii) the setting operation is $u_a^A \times u_b^B \times \mathrm{id}$ on $S_A \times S_B \times S_{V \setminus (A \cup B)}$, on disjoint regions A and B at one step, each intervention reading and writing only its own region; (iii) wing A's outcome is a ± 1-valued function of the state on a readout region R_A k_A updates after the setting, with $d(B, R_A) > k_A$, and symmetrically $d(A, R_B) > k_B$; and (iv) the interventions act on one common pre-setting ensemble, so $p(\Omega \mid a, b) = p(\Omega)$ — the condition under which the completion is called screened. If the protocol records a ± 1 outcome in every run, there are deterministic local response functions $q_a(\Omega)$, $r_b(\Omega)$ with*

$$P(q, r \mid a, b) = \sum_{\Omega} p(\Omega)\, \mathbf{1}[q = q_a(\Omega)]\, \mathbf{1}[r = r_b(\Omega)],$$

so the model is Bell-local and $|S_{\mathrm{CHSH}}| \leq 2$.

Proof. By (ii) the post-setting state outside B does not depend on b, and by the cone lemma and (iii) the A-readout depends only on components within distance k_A of R_A, none of them in B; so the A-readout is independent of the B-intervention at fixed Ω, and conversely; determinism makes the conditionals point masses; the common ensemble supplies measurement independence. Pointwise $q_0(r_0 + r_1) + q_1(r_0 - r_1) \in \{-2, +2\}$; average. □ The pointwise bound over the sixteen deterministic response tables is checked exhaustively in `equivalence_recovery_probes.py` (check 11).

This is Bell's theorem applied to the framework's own hypotheses. **A deterministic completion whose wings satisfy (ii) and (iii) and whose response-relevant pre-setting variable satisfies (iv) cannot reproduce quantum Bell correlations at all.** The Barandes–Hasan–Kagan bound is not contradicted: their causal-independence condition constrains the stochastic law and is strictly weaker than factorization, which is what lets a causally local indivisible process occupy the quantum interval. A deterministic response-complete completion supplies more than they assume, and the surplus costs precisely the interval between 2 and $2\sqrt{2}$.

Determinism is constitutional here and the protocol records a ± 1 outcome in every run, so a completion reproducing quantum Bell correlations gives up at least one of two remaining hypotheses: (ii) and (iii), which with the cone lemma secure ontic parameter independence, or measurement independence, (iv). The framework gives up the first. Measurement independence is retained as a posit about the initial condition, not derived: when the setting generators are part of the system, the settings are functions of the full pre-setting microstate, and (iv) fails for that microstate whenever more than one setting pair occurs; Ω is therefore the response-relevant variable, and (iv) states that it is uncorrelated with the setting generators' outputs.

(a) Ontic parameter dependence — the option taken. At the microstate level, one wing's response depends on the remote setting. Operational no-signaling is an additional requirement on the averaged outcomes, not a consequence of that dependence. In the general finite response construction, the composite realizing the no-signaling PR law explicitly carries the joint history in each wing. For the specific SM/GR representative, whether the required response dependence can be realized with no-signaling is part of H-Bell. [SM §3.1] already uses a state-dependent coupling graph $G(x)$; the proposed realization is **preparation-indexed adjacency**, with a geometric bounded-range vacuum/uniform reference graph and bounded-degree long edges or an equivalent nonlocal response map supplied by a Bell preparation.

Corollary (Bell–lattice obstruction and state-dependent escape). *If the Bell-relevant degrees of freedom are governed entirely by the nearest-neighbor cubic wave-equation update used in [SM], the setting interventions are applied at one step to disjoint wing regions, each reading and writing only its own, and each wing's ± 1 outcome, recorded in every run, is read from a region whose graph distance from the other wing's setting region exceeds the number of updates*

from setting to readout, then (ii) and (iii) hold, the theorem above applies to every measurement-independent response-complete deterministic completion and $|S_{\mathrm{CHSH}}| \leq 2$. Hence the **reference nearest-neighbor graph alone** cannot supply option (a) for such wings. The graph cone advances one edge per update, a Euclidean reach of $1/\|\hat{n}\|_1$ lattice spacings per update in direction $\hat{n}$ (from 1 along the axes to $1/\sqrt{3}$ along the body diagonals), while the long-wavelength wave speed derived in [SM §4.1] for its isotropic lift is $1/\sqrt{3}$ lattice spacings per update in every direction. A clock and length calibration rescales both speeds by the same factor, so it does not make them equal off the body diagonals; which of the two, if either, is the physical c depends on which propagation sets the emergent light cone and is not derived here. Wings separated by less than the graph cone over the experiment's duration fall outside the Corollary, including any that are spacelike with respect to an emergent light cone lying strictly inside the graph cone in some direction. If c is the wave speed, these include wings that are spacelike with respect to c: a setting site and a remote readout site at Euclidean distance d lattice spacings along $\hat{n}$, Δt updates apart, with $\Delta t/\sqrt{3} < d \leq \Delta t/\|\hat{n}\|_1$; the Corollary covers wings outside the graph cone, which contains the wave cone in every direction. This is not a no-go for the state-dependent-graph framework: [SM §3.1] permits $G(x)$ to depend on the state, and its mod-q area-law proof is explicitly for uniform initial conditions. A measurement-independent Bell completion may therefore use preparation-indexed nonlocal edges while leaving the reference uniform/vacuum graph local; such edges are dependences present at some states and absent at others, so the update of this branch lies outside ledger item (vii) (§4.5). What remains to be proved for this branch is the part of H-Bell that this analysis bears on: that the prepared graph family supplies the required response dependence while preserving operational no-signaling.

(b) Ontic measurement dependence — recorded, not adopted. If instead the pre-setting ensemble is allowed to depend on the settings, then with $Z = (A, B)$ uniform and $I_{\mathrm{ont}} = I(\Omega; Z)$ in bits, data processing, Pinsker's inequality and Jensen's inequality give

$$\boxed{S_{\mathrm{CHSH}} \leq \min\left\{4,\ 2 + 4\sqrt{2 \ln 2\, I_{\mathrm{ont}}}\right\},}$$

so reaching the quantum value requires $I_{\mathrm{ont}} \geq (3 - 2\sqrt{2})/(8 \ln 2) = 0.0309409\ldots$ bits. This is a **model-independent necessary lower bound from the stated inequality, not an achievable cost and not claimed tight**. For comparison, Hall and Branciard [57] construct the optimized causal measurement-dependent simulation in their CHSH model class at approximately 0.080 bits for the maximal quantum violation; the approximately 0.046-bit optimum corresponds instead to their retrocausal causal structure. The framework's one-way constitution does not adopt that retrocausal branch. Measurement dependence is therefore retained as a quantified alternative but not adopted.

Response-completeness requirement for the information bound. The

quantitative measurement-dependence estimate must use a response-complete ontic variable and therefore has Bell-local reference baseline 2. If Λ is coarser than the response-relevant microstate, $I(\Lambda; Z)$ alone does not control CHSH: a constant coarse Λ has $I(\Lambda; Z) = 0$ while omitted response-relevant data remains setting-correlated and a local deterministic model reaches the algebraic maximum 4 (check 11). Thus the stated information bound is a bound in terms of the response-complete Ω, not an arbitrary preparation-level coarse graining.

Operational no-signaling requires that changing a local setting leave the remote wing's outcome distribution unchanged. It holds when the finite response construction realizes a target law with that property, such as the PR law; its realization by the proposed SM/GR prepared graphs is an open H-Bell obligation. Neither ontic parameter dependence nor an indivisible reduced law alone establishes it. Removing the screened local-intervention restrictions admits the unrestricted OI behavior class, including the PR negative control.

3.4 The Characterization Theorem: Necessity of the Conditions

The logical structure is easiest to state by keeping representation and selection separate. The finite-horizon theorem below proves internally that S, D, and Q_{fb} are the same observable-law class; the existence of a unitary/Born representation therefore requires neither C4 nor indivisibility. C4 supplies the nontrivial content: universal readout-relevant hidden predictive memory. The converse process-dilation theorem constructs a finite OI realization for every finite stochastic target. The remaining question is not whether a quantum representation exists, but whether an entire controlled family of preparations, coherent instruments, and composites is represented simultaneously by the standard local operational quantum theory.

Dilation existence. Any single one-step doubly stochastic matrix on a finite configuration space can be realized as marginalization of a deterministic process on a larger state space — furnishing Lemmas 1–3. The scope is the one scoped at the lemma above: the matrix, not prescribed multi-time trajectory statistics.

Lemma (non-permutation witness). *If the one-step matrix T is not a permutation, the marginalized process is P-indivisible — and in particular hidden-to-visible influence (C1) is present at the first step.*

Proof. The indivisibility is §2.3's recurrence theorem; and a non-permutation T means some visible transition has a nondegenerate conditional law, which requires hidden dependence. □

Remark (witness, not equivalence). The converse fails: the delayed-revival family $X_1 = X_0$, X_2 uniform and independent, $X_3 = X_0$ has $T^{(1)} = I$ — a permutation — while the family is P-indivisible ($T^{(3)} = T^{(2)}\Lambda$ is impossible: $\frac{1}{2}J\,\Lambda$ has identical rows for every stochastic Λ), and it is realized exactly by the §3.4 dilation construction (`review4_probes.py`). C1 is therefore formulated globally — influence at some accessible step (§1.3) — with the non-permutation

one-step matrix its cheapest witness. The witness is not an equivalence: the reverse direction would assume the semigroup property $\Lambda = T^{(k_2-k_1)}$, which hidden memory is precisely there to break.

Remark (influence, not bidirectionality). The equivalence is with hidden-to-visible influence, which is C1's defining content (§1.3); it does not force bidirectional coupling. $\varphi(v,h) = (v \oplus h,\ h)$ on two visible and two hidden states has $T^{(1)} = \frac{1}{2}J$ and $T^{(2)} = I$ — P-indivisible — while the hidden update $h' = h$ is independent of v: the coupling is strictly one-way. Bidirectionality is a property of the physical realization (the bidirectional horizon coupling of §1.3), recorded at C1's realization clause there. (Certified in `c1_cp_scope_probes.py`.)

Physical motivation (ETH dephasing). Assume the hidden-sector Hamiltonian H_H satisfies the eigenstate thermalization hypothesis (ETH) [17, 18]. Then in the fast-bath regime ($\tau_B \ll \tau_S$), the hidden sector dephases to its diagonal ensemble between coupling events, so the marginal dynamics on $\mathcal{C}_V$ approaches the homogeneous Markov benchmark T^k on accessible timescales, within the bound of the quantitative theorem below, $\|T^{(k)} - T^k\|_{\mathrm{TV}} \leq (k-1)\,\varepsilon$; contrapositively, a departure δ from that benchmark at step order $k \geq 2$ forces $\varepsilon \geq \delta/(k-1)$. This is the physical mechanism behind the mixing hypothesis; the load-bearing necessity statement is the quantitative theorem below, whose contrapositive makes the requirement operational. The dephasing picture imports from ETH a conditional-distribution equilibration that ETH's few-body form does not by itself deliver (Remark, *Status of ETH*), which is why it is carried as motivation rather than theorem.

The mechanism: between coupling events (separated by τ_S), the hidden sector evolves under its own Hamiltonian H_H. The substratum dynamics is reversible, so there is no relaxation in the strict sense — finite-dimensional Hamiltonian flow has purely imaginary spectrum and Poincaré recurrence at t_R exponentially large in the system size. What does occur, under the ETH assumption, is *dephasing*: starting from any distribution $\mu_H(\cdot|x_i)$ conditioned on the previous visible state, the off-diagonal matrix elements in the energy basis decay on the timescale $\tau_B \sim \hbar/\Delta E$, where ΔE is the hidden sector's typical energy spread. After dephasing, the diagonal ensemble agrees with the microcanonical distribution to corrections exponentially small in the hidden-sector size, so for any few-body visible-sector observable O_V and any time $\tau_S \gg \tau_B$:

$$\left|\langle O_V\rangle_{\mu_H(\tau_S|x_i)} - \langle O_V\rangle_{\mathrm{eq}}\right| \lesssim \tau_B/\tau_S$$

In the fast-bath regime ($\tau_B \ll \tau_S$) this is parametrically small: from the visible sector's perspective the hidden sector "looks" thermalized at every coupling event regardless of its post-coupling state, even though the underlying trajectory remains reversible and quasi-periodic with recurrence only at $t \sim t_R$. Each single-step transition matrix is therefore computed against nearly the same effective equilibrium distribution as the reference T. Under the quantitative theorem's hypotheses — the reference prior π_H defining T, and every realized

post-relaxation hidden conditional within ε of π_H in total variation — its bound $\|T^{(k)} - T^k\|_{\mathrm{TV}} \le (k-1)\,\varepsilon$ holds on accessible timescales $t \ll t_R$. Its specialisation $\varepsilon = 0$, an exact reset, gives $T^{(k)} = T^k$ for every k, so $T^{(t)} = T^{(s)}\,T^{t-s}$ with a stochastic factor and the family is P-divisible in the sense of §2.3; for $\varepsilon > 0$ the bound limits the departure from the benchmark but does not by itself determine P-divisibility.

Contrapositively, within the mixing class this analysis presupposes — hidden dynamics that relax toward the reference prior between couplings — a departure δ from the homogeneous Markov benchmark at step order $k \ge 2$ requires a residue $\varepsilon \ge \delta/(k-1)$ at some coupling time before step k: the visible sector must couple back into the hidden sector before dephasing has relaxed the record that far. (Fast *non-mixing* hidden dynamics evade the premise, not the theorem: conserved circulation retains records with no timescale separation, `fastbath_probes.py`, §1.3.) The continuous-time extension to general Hamiltonian flow replaces the collision-model picture with the full ETH/dephasing condition; the relevant machinery for chaotic open quantum systems in the weak-memory regime is established in [3, 8, 17, 18].

Theorem (C2 necessity, quantitative mixing form). *Model one coupling step as the uniform-prior bijection marginal of §3.2's construction, and between coupling events let the hidden sector evolve by any map on the hidden conditional. Suppose the mixing hypothesis holds: with π_H the* reference prior defining T^* — the uniform hidden prior of the model above — every realized post-relaxation hidden conditional, given the visible record to date, lies within ε of π_H in total variation. Then the k-step visible marginal satisfies*

$$\|T^{(k)} - T^k\|_{\mathrm{TV}} \;\le\; (k-1)\,\varepsilon$$

in the max-row total-variation norm, and the constant is optimal: no bound $C\varepsilon$ with $C < 1$ holds at $k = 2$. Contrapositively, departure of magnitude δ from the homogeneous Markov benchmark at step order k forces $\varepsilon \ge \delta/(k-1)$ — the benchmark form; the operative contrapositive in condition (ii)'s currency is the physical-memory theorem's $\varepsilon \ge \varepsilon_{\min}(M_k, n_V)$ below.

Proof. For $k = 2$: with $\mu'_{x_0x_1}$ the post-relaxation conditional given the visible record (x_0, x_1) and $\Delta = \mu'_{x_0x_1} - \pi_H$,

$$T^{(2)}_{x_0x_2} - (T^2)_{x_0x_2} = \sum_{x_1} T_{x_0x_1} \sum_h \Delta(h)\, P_\varphi(x_2 \mid x_1, h),$$

where $P_\varphi(x_2 \mid x_1, h) \in \{0, 1\}$ is the bijection's visible step. Taking the row's total variation and the triangle inequality, $\frac{1}{2}\sum_{x_2} |\cdot| \le \sum_{x_1} T_{x_0x_1} \cdot \frac{1}{2}\sum_h |\Delta(h)| \sum_{x_2} P_\varphi(x_2 \mid x_1, h) = \sum_{x_1} T_{x_0x_1}\,\mathrm{TV}(\mu'_{x_0x_1}, \pi_H) \le \varepsilon$ — pushing two hidden distributions through the same channel contracts total variation. For general k, telescope through the hybrids $A_j = T^{(j)}\,T^{k-j}$: $A_1 = T^k$,

$A_k = T^{(k)}$, and $\|A_{j+1} - A_j\| = \|(T^{(j+1)} - T^{(j)}T)\,T^{k-j-1}\| \le \|T^{(j+1)} - T^{(j)}T\| \le \varepsilon$, since right-multiplication by a stochastic matrix contracts the norm and the middle difference is the $k = 2$ argument applied at slot $j + 1$. Summing the $k - 1$ differences gives the bound. Optimality: the two-hidden-state revival bijection of §3.4's corollary, taken with zero relaxation, has $\varepsilon = \frac{1}{2}$ (its post-step conditionals are point masses, maximally far from uniform at $m = 2$) and $\|T^{(2)} - T^2\| = \frac{1}{2}$, saturating the bound. □

Remark (operational form of $\tau_S \ll \tau_B$). Define the *conditional mixing time* $\tau_B(\varepsilon_0)$ as the least inter-coupling relaxation time after which every realized hidden conditional lies within ε_0 of π_H in total variation. (The theorem's bound $\|T^{(k)} - T^k\|$ measures departure from the homogeneous Markov benchmark; it is not itself a measure of P-indivisibility — two-notions Remark, §3.4.)

Theorem (physical memory: C2, process form). *In the coupling-step model of the previous theorem — given the whole visible past and $H_k = h$, X_{k+1} has law $K(x_k, h; \cdot)$, with π_H the reference prior — and under the conditional-mixing hypothesis, that every realized hidden conditional lies within ε of π_H in total variation at the coupling time, every history-conditional next-step law satisfies $\|P(X_{k+1} \mid X_{\le k}) - \bar{K}_{x_k}\|_{\mathrm{TV}} \le \varepsilon$ with $\bar{K}_{x_k} := \sum_h \pi_H(h)\,K(x_k, h; \cdot)$; hence the departure $\|P(X_{k+1} \mid X_{\le k}) - P(X_{k+1} \mid X_k)\|_{\mathrm{TV}} \le 2\varepsilon$ uniformly in the history, and for $\varepsilon \le 1 - 1/n_V$,*

$$I(X_{<k}; X_{k+1} \mid X_k) \;\le\; 2\,[\varepsilon \log(n_V - 1) + h(\varepsilon)]\,,$$

h the binary entropy (for $n_V = 2$ the bound is $2h(\varepsilon)$). Contrapositive: observed conditional memory $M_k := I(X_{<k}; X_{k+1} \mid X_k) > 0$ at an accessible order k forces $\varepsilon \ge \varepsilon_{\min}(M_k, n_V) > 0$ — the hidden sector has not mixed by coupling time k. Here $\varepsilon_{\min}(M, n_V)$ is the least $\varepsilon \in [0, 1 - 1/n_V]$ at which the right-hand side of the display equals M; the right-hand side increases on that interval from 0 to $2 \log n_V$, and $M_k \le \log n_V$, so $\varepsilon_{\min}(M_k, n_V)$ exists and lies below $1 - 1/n_V$. This is (C2)'s necessity in the currency of condition (ii), in two forms. Taking ε as the realized maximum of $\mathrm{TV}(P(H_k = \cdot \mid x_{\le k}), \pi_H)$ over positive-probability histories $x_{\le k}$, the hypothesis holds by definition, so the contrapositive needs no mixing hypothesis: this is the persistence form (if that maximum exceeds $1 - 1/n_V$ the conclusion holds outright, since $\varepsilon_{\min} < 1 - 1/n_V$). The mixing hypothesis enters when ε is read as the residue of relaxation toward π_H over the inter-coupling time, which gives the timescale (slow-bath) form.

Proof. The hidden conditional given any visible past is within ε of π_H by hypothesis; pushing through the one-step visible kernel contracts total variation, giving the first display. $P(X_{k+1} \mid X_k)$ is a mixture of history-conditionals, hence also within ε of $\bar{K}_{x_k}$, giving the second. For the information bound, write $I = \mathbb{E}[H(P(\cdot \mid X_k)) - H(P(\cdot \mid X_{\le k}))]$ and apply Fannes–Audenaert continuity to each term against the common reference $\bar{K}_{x_k}$. □ (Certified against exact conditional mutual information on a parametric relaxation family, `c2_mixing_probes.py`.)

Remark. The matrix-power bound above measures departure from the homogeneous Markov benchmark and is blind to conditional memory — the XOR

family has $T^{(k)} = T^k$ exactly with one full bit of it (`review4_probes.py`). The process form is the statement that carries the necessity of (C2) for physical hidden sectors — without a mixing hypothesis in its persistence form, conditional on it in its timescale (slow-bath) form; (C2) is not part of the characterization's (iii).

Remark (coherent extension). The equivalence holds between the operational, dilated, and fixed-basis descriptions and is exact at that level; it does not by itself determine a coherent extension of the fixed-basis theory, and at least one conservative coherent extension preserves the fixed-basis theory exactly as a quotient while adding coherent structure not fixed by the equivalence. *Remark ((C2)'s role: two theorems).* The characterization's realization clause needs (C2) only in its limit form — the canonical dilation's bath is static, $\tau_B = \infty$ — so the structural equivalence is carried by (C1), (C3), (C4), with (C2) automatic in that limit. The physical content of (C2) is the memory theorem above: observed conditional memory $M_k > 0$ at an accessible order k keeps some realized hidden conditional at that coupling time at least $\varepsilon_{\min}(M_k, n_V)$ from π_H, and under the conditional-mixing hypothesis it lower-bounds the bath's mixing time. The structural equivalence and the physical-memory bound are distinct results with distinct hypotheses; (C1)–(C4) are not four necessities of equal standing — (C2)'s timescale necessity is the conditional bound above; its persistence form is the bound's realized-ε contrapositive.

Remark (passive scope). The theorem concerns the closed multi-time law $P(X_0, \ldots, X_K)$. Conditioning on interventions — $P(X_{k+1} \mid X_{\leq k}, A_{\leq k})$ — requires the instrument-augmented realization $(S, \varphi, \mu_H, \{\mathcal{I}_a\})$, which is not constructed here: the *intervention dilation* is a named target of the operational bridge (§3.2) — and its classical rung is constructed below.

Theorem (intervention dilation: the finite classical action-labelled comb). *Let a finite action alphabet $\mathcal{A}$ and a finite-horizon action-labeled family $\{P(x_{k+1} \mid x_{\leq k}, a_{\leq k})\}_{k<K}$ be given. There exist a finite set S, a bijection φ, a fixed prior μ_H, and bijections $\{\mathcal{I}_a\}_{a\in\mathcal{A}}$ on S such that, for every action sequence, alternating $\mathcal{I}_{a_k}$ then φ from the initial ensemble reproduces the target conditional law at every step — the classical process-tensor (comb) statistics — with one fixed realization serving all action sequences and cross-history consistency automatic.*

Proof (construction). Extend the response-table realization: contexts $c = (x_{\leq k}, a_{\leq k})$; hidden variable $u = (u_c)_c$ with independent components $u_c \sim P(\cdot \mid c)$ under the product prior μ_H; state $= (x, \text{history ledger}, \text{action ledger}, u, \text{clock})$. $\mathcal{I}_a$ writes a into the action ledger's next blank slot — injective on blank-bearing states, completed to a bijection on the finite remainder as in the base construction. φ reads the ledgers and u, outputs $x' = u_{c(\text{state})}$, appends x to the history ledger, and advances the clock — bijective by the base theorem's ledger argument. For any action sequence, the conditional law of X_{k+1} given $(x_{\leq k}, a_{\leq k})$ is exactly $P(\cdot \mid x_{\leq k}, a_{\leq k})$: precisely the coordinate u_c is read, and its marginal is the target. One table, one prior: consistency across histories and across action

sequences is automatic. □ (Certified exactly — all four action sequences, forty-eight conditionals in exact arithmetic, both instruments and φ verified bijections on $|S| = 124{,}416$ at $n_V = |\mathcal{A}| = K = 2$ — in `intervention_probes.py`.)

Remark (scope of the closure). The theorem closes the classical rung of the intervention bridge: classical action labels, permutation instruments, comb statistics of the visible configurations. Its reach is wider than it may appear. Because the instruments are *invasive* — they write to the action ledger — the construction is not bound by the non-invasiveness assumption behind temporal inequalities, and it reproduces quantum instrument statistics that violate them. A worked case is certified exactly (`lg_comb_probes.py`): sequential σ_z measurements on a qubit with $\pi/3$ rotations between them give $C_{12} = C_{23} = 1/2$, $C_{13} = -1/2$ and a Leggett-Garg value $K = 3/2 > 1$, and the realization — twelve contexts, one fixed prior, permutation instruments — reproduces every conditional and returns $K = 3/2$. The open remainder is therefore not the numbers.

Proposition (no uniform finite realization). A fixed finite S with a fixed prior and a fixed bijection supports only finitely many distinct action-labelled conditional families, whereas the instrument space of any non-trivial system is a continuum. Hence no fixed finite realization carries the whole instrument algebra: the theorem above is a statement per finite instrument set, and $|S|$ grows with the set chosen. □

Remark (what the remainder is). What stays open on the operational bridge is accordingly the instrument *algebra* rather than the statistics: multilinearity of the process tensor over the full instrument space, complete positivity of the induced slot maps, composition of noncommuting interventions, and the requirement that the intervened description use the same Hilbert space and the same $\hat{H}$ as the passive one (§3.2).

That last clause can be settled, and it is settled negatively at fixed dimension.

Proposition (no horizon-independent intervened representation). *Fix a representation dimension d, a Hamiltonian $\hat{H}$, and the slot structure. An instrument with n_{out} outcomes contributes $n_{out}d^4 - d^2$ real parameters per slot, so the intervened statistics reachable by CP instruments on that fixed representation are the image of a polynomial map from a space of dimension LINEAR in the horizon K. The action-labelled comb family with a given passive marginal has dimension growing like $(n_V|\mathcal{A}|)^K$. Hence for K beyond a small threshold the reachable set has measure zero in the family, and since the theorem above realizes the entire family, generic action-labelled combs are not hostable on any fixed passive representation.* □

The threshold is not asymptotic. At $n_V = d = |\mathcal{A}| = 2$ the counts are 70 against 168 at $K = 3$ and 310 against 224 at $K = 4$; at $|\mathcal{A}| = 3$, 244 against 252 then 1524 against 336; at $n_V = d = 3$, 438 against 1404 then 2868 against 1872. Keeping pace forces $d \sim (n_V|\mathcal{A}|)^{K/4}$ — concretely $d \geq 3$ at $K = 4$, 4 at $K = 6$, 8 at $K = 8$, 14 at $K = 10$ (`repconsistency_probes.py`).

Remark (the per-horizon qualifier is load-bearing). The characterization is stated per finite horizon (§3.4), and (T) gates the quantum representation per horizon likewise. The Proposition shows that this is not a technical convenience awaiting a better proof: a horizon-independent representation of the *intervened* process does not exist for generic combs, because the intervened statistics carry exponentially more information than any fixed representation can encode. The remaining question at Layer 4 is therefore not which fixed Hilbert space to choose — no such choice works — but how the representation must grow with the horizon, and whether that growth can be made compatible with the passive $\hat{H}$. Three successor problems replace it, and the framework must eventually choose among them: (a) identify the *quantum-compatible subclass* of action-labelled combs — those a fixed representation can host — and argue that physical interventions fall in it; (b) accept $d = d(K)$, a representation dimension growing with the horizon, and characterize the growth; or (c) allow $(\hat{H}, \mathcal{H})$ to depend on the intervention context, at the cost of the single-Hamiltonian reading of the passive dynamics. Each is a definite mathematical programme; none is supplied here. The quantitative-mixing theorem's contrapositive (§3.4) reads as a timescale statement in the homogeneous-Markov benchmark: memory of magnitude δ at step order k requires the coupling to return before the bath's conditional mixing to $\varepsilon < \delta/(k-1)$ completes — the benchmark form. For condition (ii)'s currency the operative statement is the physical-memory theorem's $\varepsilon \geq \varepsilon_{\min}(M_k, n_V)$ (§3.4); the benchmark inequality is not the general process-memory bound.

Remark (role of ETH, restated). The timescale (slow-bath) form of the necessity direction rests on the mixing hypothesis above — an operational, in-principle-measurable condition — not on ETH itself. ETH is the physical case for the hypothesis: dephasing to the diagonal ensemble is the mechanism by which a generic chaotic hidden sector drives ε small between coupling events (previous theorem). A hidden sector failing ETH but satisfying the mixing bound still delivers the necessity direction; a hidden sector satisfying ETH for few-body observables while retaining conditional-distribution memory (conserved quantities, slow modes) is exactly the case the quantitative form guards against. The bound, its optimality, and the fact that the revival counterexample violates the hypothesis maximally are certified deterministically by `papers/oi_lattice_code/foundations/c2_mixing_probes.py` *(preregistered ensemble: 21{,}000 exact rational evaluations, zero violations; all three controls green).*

Remark (Status of ETH). *The eigenstate thermalization hypothesis is a well-supported conjecture, not a theorem, for generic chaotic many-body Hamiltonians. It is proved for specific models (random matrix ensembles, certain integrable-chaotic transitions) and supported by extensive numerical evidence across many systems [17, 18]. For the OI framework's hidden sector — the cosmological-horizon complement, a generic chaotic many-body system — ETH is the standard assumption. The timescale (slow-bath) form of the C2 necessity direction is conditional on the quantitative mixing hypothesis (the theorem*

above); ETH is the physical case for that hypothesis rather than the load-bearing assumption, so relaxations or failures of ETH bear on how the hypothesis is argued for the framework's hidden sector, not on the theorem that uses it. The rounded bijection of [SM §4.2] behaves chaotically at the sizes measured, according to spectral-continuity, thermalization and cycle-consolidation diagnostics; whether the realized hidden sector satisfies the mixing hypothesis is open. The C1 and C3 necessity theorems do not depend on ETH.

Remark (Consequence for the characterization theorem). *Within the characterization below, (C2) is not part of (iii); the mixing hypothesis enters only the timescale (slow-bath) form of (C2)'s necessity for physical hidden sectors (the process-form theorem above; ETH its physical motivation). The equivalence's other content — the readback direction, the dilation existence, and the representation clause — does not depend on ETH.*

Theorem (C3 necessity). *Let $m = |\mathcal{C}_H|$. The non-Markovian mutual information satisfies:*

$$I(X_{<t}; X_{>t} \mid X_t) \leq \log_2 m$$

Proof. The total system is deterministic: $X_{>t}$ is a function of (X_t, H_t). Conditioning on X_t: $X_{<t} \to H_t \to X_{>t}$ is a Markov chain. By data processing:

$$I(X_{<t}; X_{>t} \mid X_t) \leq I(X_{<t}; H_t \mid X_t) \leq H(H_t \mid X_t) \leq \log_2 m \quad \square$$

Corollary (per-process capacity bound). Any deterministic-bijection realization of a process whose conditional past–future information reaches $I^* = \sup_t I(X_{<t}; X_{>t} \mid X_t)$ requires $m \geq 2^{I^*}$: the hidden sector must be at least as large as the conditional memory it carries. The bound is process-quantitative and does not reduce to the visible count — P-indivisibility alone does not force $m \geq n$: a two-hidden-state, three-visible-state bijection already exhibits full distinguishability revival ($m = 2 < n = 3$), so the necessity content of (C3) is the capacity to hold the *observed* conditional memory, not a bound set by n. Sustained conditional memory at rate $I_0 > 0$ per coupling event over K events requires $m \gtrsim 2^{KI_0}$ *under the additional hypothesis* that the per-event contributions accumulate without hidden-state reuse or compression; that independence hypothesis is not a consequence of P-indivisibility, and without it only the per-process bound above is forced. The framework's qualitative (C3) ("hidden sector large enough") is the conservative restatement of the per-process bound. $\square$

The revival counterexample and the doubly-stochastic obstruction of §3.2 are certified by the deterministic probes in `papers/oi_lattice_code/foundations/(c3_dilation_probes.py)`*; the process-dilation construction below, the one-way-coupling counterexample, and the diagonal-preservation lemma are certified in* `process_dilation_probes.py` *and* `c1_cp_scope_probes.py` *in*

the same directory. (kernel: `c3_necessity`, `card_hidden_ge_two_pow_Istar`, `c3_necessity_and_capacity` in `OIBridge/C3Necessity.lean`)

Theorem (finite-horizon process dilation). *Let $\mathcal{S}$ be a stochastic process on $\mathcal{C}_V$ ($n = |\mathcal{C}_V| \geq 2$) whose conditional kernels $P(x_{k+1} \mid x_0 \dots x_k)$ ($0 \leq k < K$) at positive-probability histories are rational, on a finite accessible horizon K. Then there exist a finite $\mathcal{C}_H$, a bijection φ on $\mathcal{C}_V \times \mathcal{C}_H$, and an initial hidden prior μ_H independent of the visible initial state, such that the visible marginal process of φ from the initial law $P_0 \otimes \mu_H$, P_0 the target's initial law, reproduces $\mathcal{S}$'s full multi-time joint law for all $k \leq K$. If the target process is non-Markovian, the construction necessarily contains hidden-to-visible influence at some accessible step — hence satisfies (C1); a non-permutation first-step matrix is a sufficient first-step witness only. The realization realizes (C2) in the static-bath form ($\tau_B = \infty$ between couplings, hence $\tau_S \ll \tau_B$), and satisfies (C3) with $|\mathcal{C}_H| \geq 2^{I^*}$. Rationality tracks the prior, not the mechanism: for a finite realization whose prior μ_H is rational and independent of the visible initial state, each $P(x_{\leq k} \mid x_0)$ with $P_0(x_0) > 0$ is a finite sum of μ_H-masses, so the conditional kernels at positive-probability histories are rational; conversely every law with rational conditional kernels is realized in this form. The rational class is that of the kernels; the visible initial law enters only through P_0 and may be irrational.*

Extension (arbitrary kernels — the response-table form). The rationality restriction is a property of the uniform-tape presentation, not of realizability. At every context $c = x_{\leq k}$ ($0 \leq k < K$) fix a kernel kern_c equal to $P(\cdot \mid x_{\leq k})$ where that history has positive probability and arbitrary elsewhere. Replace the tape by a response coordinate $u = (u_c)_c$ indexed by the horizon's finitely many history contexts, with product prior $\mu(u) = \prod_c \mathrm{kern}_c(u_c)$ and deterministic step $x_{k+1} := u_{c(x_{\leq k})}$; the ledger and clock complete the bijection exactly as in the proof below, with $|\mathcal{C}_H| = n^N(n{+}1)^K(K{+}1)$, $N = \sum_{k=1}^{K} n^k$ the number of contexts ($\prod_c |\mathrm{supp}\,\mathrm{kern}_c|\,(n{+}1)^K(K{+}1)$ if each u_c ranges over the support of kern_c). Conditioning on the visible past exposes u only at visited contexts: $\{X_{\leq k} = x_{\leq k}\} = \{X_0 = x_0\} \cap \bigcap_{j<k}\{u_{x_{\leq j}} = x_{j+1}\}$ is determined by X_0 and the coordinates at the contexts $x_{\leq 0}, \dots, x_{\leq k-1}$, each distinct from $x_{\leq k}$ because contexts of different lengths differ; the prior is a product and independent of X_0, so $u_{x_{\leq k}}$ is independent of that event and $P(X_{k+1} = y \mid x_{\leq k}) = \mathrm{kern}_{x_{\leq k}}(y)$. The chain rule then reproduces the full multi-time joint for arbitrary real kernels (a rational instance certified in `review4_probes.py`, P-G). The characterization below therefore quantifies over general finite-horizon laws.

Proof (constructive). Let D be a common denominator of the conditional kernels $P(x_{k+1} \mid x_0 \dots x_k)$ at positive-probability histories and of a rational kernel, for instance a point mass, fixed at every context of probability zero. Take $\mathcal{C}_H = (\mathcal{C}_V \cup \{\square\})^K \times \mathbb{Z}_D^K \times \mathbb{Z}_{K+1}$ — a history ledger, a randomness tape, and a clock — with μ_H uniform on $\{\text{ledger} = \square^K,\ \text{clock} = 0\} \times \mathbb{Z}_D^K$. On well-formed states with clock $c < K$ the step is $(x,\ (x_0 \dots x_{c-1}),\ \mathbf{s},\ c) \mapsto (Q_{c+1}(x_0 \dots x_{c-1}, x;\, s_{c+1}),\ (x_0 \dots x_{c-1}, x),\ \mathbf{s},\ c+1)$, where Q's tape fibers have

sizes $D \cdot P(\cdot \mid \text{past})$ (the conditional-quantile representation). The step is injective structurally — the ledger's last entry recovers the pre-image regardless of the kernels — and extends to a bijection of the finite product, since an injective map f from a subset A of a finite set X into X extends to a permutation of X ($|X \setminus A| = |X \setminus f(A)|$; `exists_perm_extending`, `OIBridge/Equivalence.lean`). Clock-$K$ and ill-formed states are assigned by this completion, and every prior-supported state at times $0, \ldots, K-1$ is well-formed with clock below K, so the K steps up to time K are all given by the step formula; the realized process after time K is unconstrained. Marginalizing over the uniform tape reproduces each conditional exactly, hence the full joint by the chain rule. When $\mathcal{S}$ is non-Markov on the horizon, the realization inherits (C4): distinct visible pasts with differing target kernels condition the ledger differently and are read back through Q exactly. (C1): if the target is non-Markovian, hidden-to-visible influence is present at some accessible step (the global argument, as in the characterization proof); a deterministic first kernel need not give a permutation marginal — many-to-one visible steps are realizable, the ledger retaining global reversibility. (C2): the ledger and tape have no autonomous dynamics between couplings — a static bath, the slow-bath limit. The map φ is fixed, with the target's time dependence carried by the hidden clock and the visible past recorded exactly by the ledger; φ depends on K and on the target. (C3): $\log_2 |\mathcal{C}_H| \geq I^*$, consistent with the per-process bound above. Certified exactly on P-indivisible targets — including the §3.2 revival family and a non-Markovian kernel set with $T^{(2)} \neq (T^{(1)})^2$ — by `process_dilation_probes.py` (full-joint equality in exact rational arithmetic; permutation control silent). □ (kernel: `S_imp_D` in `OIBridge/Equivalence.lean`, for the realization of the full multi-time law from an initial law on $\mathcal{C}_V \times \mathcal{C}_H$ that may correlate the hidden and visible coordinates; the independence of μ_H from the visible initial state rests on the construction above and the cited probes and is not formalized)

Remark (prior structure). The realization uses a structured prior — uniform on the initialized ledger/clock sector times the tape. The uniform-prior one-step lemma of §3.2, with its doubly stochastic scope, is the special case and is unchanged: double stochasticity is a property of uniform-prior marginals, not of deterministic realizability as such.

Definition (unitarily evolving QM). A stochastic process $\mathcal{S}$ on a finite configuration space $\mathcal{C}_V = \{x_1, \ldots, x_n\}$ is *mathematically equivalent to unitarily evolving quantum mechanics* if there exists a Hilbert space $\mathcal{H} = \mathcal{H}_V \otimes \mathcal{H}_{\text{anc}}$ of total dimension $\leq n^3$, a unitary $U(t)$ on $\mathcal{H}$ for each time t (a time-indexed family, with no composition law or fixed generator required), and orthogonal projectors $P_i = |i\rangle\langle i| \otimes \mathbb{1}_{\text{anc}}$ onto the configuration sectors, such that the transition probabilities are the ancilla-uniform marginal of the dilated Born-rule probabilities: $T_{ij}(t) = \mathrm{Tr}\big[P_j\, U(t)\, (|i\rangle\langle i| \otimes \rho_{\text{anc}})\, U(t)^\dagger\big]$ for a fixed ancilla state ρ_{anc} on $\mathcal{H}_{\text{anc}}$ (a diagonal ρ_{anc} suffices throughout; the uniform ancilla $\rho_{\text{anc}} = I/m_a$ recovers $\frac{1}{m_a} \sum_{k,l} |\langle j, l|\, U(t)\, |i, k\rangle|^2$ — a special case, and the one that forces doubly stochastic T; general priors, as the response-table construction of §3.4 requires,

need the general form).

Two objects. The definition covers two distinct targets, delivered by separate results. **(Q1) Barandes-representable:** the process corresponds to a unitarily evolving quantum system in the source theorem's own language [11, 12] — the $\leq n^3$ dilation bound is the source's result for the source's finite-configuration class, under its hypotheses. What connects the framework to the source is an *embedding*, and stating it as such is more accurate — and less demanding — than the identification of criteria it replaces. The source's object is a tuple: a configuration space, target and conditioning times, transition maps $\Gamma(t \leftarrow t_0)$ carrying first-order conditional probabilities, an initial distribution, and a random variable [11, 12]. Divisibility failure is *generic* within that class but is not what defines membership in it — a Markovian-homogeneous dynamical system is already a unistochastic process and sits inside the class — and the theorem is correspondingly broad: every finite-configuration process of that form can be regarded as a subsystem of a composite unistochastic process, on a Hilbert space of dimension $\tilde{N} \leq N^3$.

The framework's side supplies $\mathcal{C}_V$, the accessible times, the per-horizon transition family $T(t \leftarrow t_0)$, the visible prior, and the visible random variable. **Translation hypothesis (T) is the claim that this data instantiates the source's tuple, so that its theorem applies.** The instantiation is checked directly for representative processes: the tuple is constructed from marginalized bijections at $(n_V, |\mathcal{C}_H|) = (2,2), (2,3), (3,2)$ and the source's structural requirements — stochasticity of every $\Gamma(t \leftarrow t_0)$, trivialization at $t = t_0$, marginal consistency with the deterministic dynamics, and well-definedness of the random variable — hold in every case (`tuple_probes.py`). Two things follow. First, (T) is a structural check rather than an unproved equivalence between two notions of indivisibility, and it is correspondingly weaker than the corpus previously asked it to be. Second, the source theorem does not apply only to a "P-indivisible subclass": it applies to the class, with P-indivisibility governing whether the resulting representation is *nontrivially* non-Markovian rather than whether the representation exists. What (T) still does not license is transporting the multi-time content of (C4) into the source's framework, whose dynamical data is first-order by construction. **(Q2) autonomous OI representation:** a fixed Hermitian $\widehat{H}$ with the ancilla-marginal form above, exact at integer horizon times, delivered in-house by the fixed-$\widehat{H}$ lemma on an uncompressed space of dimension $n \cdot |\mathcal{C}_H|$. Condition (i) of the characterization means (Q1); the in-house chain proves (Q2); time-homogeneity and continuous-t interpolation belong to (Q2)'s autonomous form and are not imported claims (§3.1). (This is the form the correspondence delivers — the visible matrix is the ancilla-marginal of the dilated unistochastic matrix, per the scope statement in §3.1. The strict form $T_{ij}(t) = |U_{ij}(t)|^2$ on visible indices is the trivial-ancilla special case $m_a = 1$ and only that case: a doubly stochastic marginal need not be unistochastic at every t, let alone realized by a single one-parameter unitary group on $\mathcal{H}_V$.) This definition captures the Hilbert space, unitary dynamics, and the Born rule.

Additional quantum-mechanical structures — the tensor product decomposition for spatially separated visible-sector subsystems, state update, and the measurement formalism — are addressed in the remarks following the theorem.

Lemma (fixed-$\hat{H}$ form for realized processes). *Let a process arise as in (iii) below: the marginal of an iterated fixed bijection φ with a fixed hidden prior. Then the fixed-$\hat{H}$ ancilla-marginal form of (i) holds constructively at every horizon time: take U_φ the permutation unitary of φ on $\mathcal{C}_V \times \mathcal{C}_H$ and $\hat{H} := i \log U_\varphi$; for the uniform hidden prior, $T^{(k)}_{ij} = \frac{1}{|\mathcal{C}_H|} \sum_{h,l} |\langle j, l|U^k_\varphi|i, h\rangle|^2$ exactly, and for the §3.4 construction's structured prior the same identity holds with the ancilla prepared uniform on the initial register slice (final sum over the full space); and for a general fixed prior μ_H — the response-table construction's product prior included — with $\rho_{anc} = \sum_h \mu_H(h)\, |h\rangle\langle h|$, so that $T^{(k)}_{ij} = \mathrm{Tr}[P_j\, U^k_\varphi(|i\rangle\langle i| \otimes \rho_{anc}) U^{k\dagger}_\varphi]$ exactly; $U(t) = e^{-i\hat{H}t}$ matches U^k_φ at integer times.*

Proof. $|\langle j, l|U^k_\varphi|i, h\rangle|^2 = \mathbb{1}[\varphi^k(i, h) = (j, l)]$; summing l over the final space and h uniformly over the prepared set counts exactly the defining fraction of $T^{(k)}$. □ (Certified exactly — uniform, slice-prepared, and weighted-prior preparations — in `review4_probes.py`, P-F(a)–(c).)

Consequence. The chain (ii) $\implies$ (iii) $\implies$ fixed-$\hat{H}$ form is in-house per horizon *in the uncompressed sense*: the internal representation is exact at integer horizon times on a space of dimension $n \cdot |\mathcal{C}_H|$, which generally exceeds the n^3 bound in (i)'s definition. The $\leq n^3$ compression is the imported correspondence's result for the source's finite-configuration class, under its hypotheses, and bounds one unitary for each time rather than a fixed-$\hat{H}$ form; continuous-t interpolation beyond integer times likewise carries no internal claim. The import therefore retains the compressed representation and the interpretive apparatus — basis independence and the general-observable reading; the Born-rule *form* itself is internal, being the marginal formula of the permutation unitary.

Characterization theorem. *Let $\mathcal{S}$ be a time-homogeneous stochastic process on a finite configuration space $\mathcal{C}_V$ with $|\mathcal{C}_V| \geq 2$ and consider dynamics on accessible timescales $t \ll t_R$ where t_R is the Poincaré recurrence time of any realizing substrate. Then (ii) $\iff$ (iii); and, under (T), (i) holds for $\mathcal{S}$ — as it does for every finite-configuration member of the correspondence's class, P-indivisibility determining not whether the representation exists but whether its dynamics is nontrivially indivisible (Remark below). Condition (C2) is deliberately absent from (iii): the existential realization satisfies it trivially with a static bath ($\tau_B = \infty$), so it is not needed for the equivalence, and its physical content is carried separately by the physical-memory theorem of §3.4 (persistence form without a mixing hypothesis, slow-bath form conditional on it):

(i) $\mathcal{S}$ is mathematically equivalent to unitarily evolving QM (in the sense of the preceding definition).

(ii) $\mathcal{S}$ is non-Markovian on accessible timescales: at some order k with $k\tau_S \ll t_R$,

the conditional mutual information $I(X_{<k}; X_{k+1} \mid X_k)$ is strictly positive — the visible past carries information about the visible future beyond the present.

(iii) For some accessible order k and every finite accessible horizon $K \geq k+1$, $\mathcal{S}$ arises from marginalizing a deterministic bijection on $\mathcal{C}_V \times \mathcal{C}_H$ — the hidden space may grow with the horizon — with (C1) non-trivial coupling, (C3) sufficient capacity, and (C4) history readback at k — with a fixed hidden prior μ_H carried as part of the realization datum (§1.2, Remark on the measure as a realization datum).

The two theorems the equivalence turns on follow, with their own proofs; the characterization's proof comes after them.

Theorem (finite-horizon stochastic–reversible–unitary equivalence). *Fix a finite visible alphabet and a finite accessible horizon K. The following three classes of finite-horizon observable law $P(x_0, \ldots, x_K)$ coincide:*

(S) the finite stochastic laws; (D) the visible marginals of finite reversible deterministic systems under incomplete observation; (Q_{fb}) the laws realizable by finite-dimensional unitary systems with Born-rule readout in a fixed basis, measured projectively in that basis at every step from an initial state diagonal in it, the visible record a fixed function of each outcome — the fixed-basis unitary Born representations. (kernel: `finite_horizon_equivalence`, `S_imp_D` in `OIBridge/Equivalence.lean`; the permutation-unitarity and diagonal-preservation lemmas `permMatrix_mem_unitaryGroup`, `isDiag_Phi` in `OIBridge/EquivalenceChain.lean`)

Proof. (S $\Rightarrow$ D) is the response-table construction of §3.4, which reproduces the whole multi-time law, not merely the one-step matrices (kernel: `S_imp_D`, by a different carrier, a clock with the complete trajectory record, whose initial law may correlate the hidden and visible coordinates). Realizations of this kind without the reversibility requirement are due to Egri et al.: time-indexed deterministic system–ancilla dynamics realize every stochastic process on a finite configuration set over finitely many times with an ancilla encoding the trajectory, under an initial law correlating system and ancilla [64, Prop. 13] or under a fixed ancilla law independent of the system's initial state [64, Prop. 16]. (D $\Rightarrow Q_{\mathrm{fb}}$) is internal and needs no imported correspondence: for a bijection φ on $\mathcal{C}_V \times \mathcal{C}_H$ set $U_\varphi |x, h\rangle = |\varphi(x, h)\rangle$ and take the initial law diagonal. A permutation unitary preserves diagonality, so sequential projective measurements in the configuration basis do not disturb the state, and the measured record reproduces $P(x_0, \ldots, x_K)$ exactly; $U_\varphi = e^{-i\hat{H}}$ for a Hermitian $\hat{H}$, giving a fixed autonomous generator at integer times. ($Q_{\mathrm{fb}} \Rightarrow$ S) is the Born rule: the record of any finite quantum experiment is a finite stochastic law. $\square$ (D $\Rightarrow Q_{\mathrm{fb}}$ certified over 1{,}488 realizations at horizons 3 and 4 in `sdq_probes.py`.)

Remark (what this equivalence does and does not say). The theorem says the three classes **coincide** at the level of finite-horizon fixed-basis observable statistics. A Born-form unitary representation is therefore exact and internal, but

universal: Markov and memory-bearing laws alike lie in Q_{fb}, so representability alone is not the framework's discriminating content. That content is the read-back sector isolated below. C4 forces nonzero predictive memory. The subscript on Q_{fb} marks the remaining boundary precisely. The theorem does not by itself prove that all coherent preparations, noncommuting interventions, and composites are represented simultaneously by one standard local quantum instrument theory. The finite operational gluing theorem crosses much of that boundary for bounded action-labelled experiments, and the uniform continuum theorem gives arbitrary finite experimental accuracy; exact coherent/composite compatibility holds, on a finite carrier with complex matrix kinematics taken as given, exactly under the five completion conditions of the operational-completion characterization (Remark, the operational-extension boundary, below), none supplied by bare finite OI. Which of those conditions hold is an operational question, not the exponent question of §3.1, and the two should not be conflated. Inside Q_{fb} the Born dictionary $p = |U|^2$ is not an open choice at all: it is part of the definition of the class, so the equivalence's quantum side already carries it and nothing here is left unresolved by it. What §3.1's scope clause says is the different and compatible point that the construction does not SELECT the quadratic exponent — the unitaries it supplies are permutations, whose entries are 0 or 1, so the recovered table is the same for $|U|^p$ at every $p > 0$ and no step distinguishes $p = 2$ (check 6, `equivalence_recovery_probes.py`). Why the quadratic functional is the physical one is therefore a frontier item about the dictionary's origin, not a gap in the equivalence. The operational question is instead one of extending the exact description consistently to the full operational algebra.*

Remark (interventions). The same construction indexed by intervention histories realizes any FIXED finite adaptive family (§3.4's comb theorem), which narrows the operational gap for fixed instrument menus. It does not close it: the obstruction of §3.4 — no fixed representation hosts generic combs as the horizon grows, with the dimension forced as $(n_V|\mathcal{A}|)^{K/4}$ — is untouched, and the all-horizons fixed-representation question remains open.

Proposition (finite-substratum operational obstruction — classical-dimension form). Fix one realization: a substratum of $N < \infty$ ontic configurations with its observation map. Its full prepare-and-measure behavior factors as $B = \Xi M$ — preparations as columns of a column-stochastic M (distributions over the N ontic states), effects as rows of $\Xi \in [0,1]^{\cdot \times N}$ — so every behavior matrix the realization generates has nonnegative factorization through N states. For the qubit this is impossible uniformly: qubit prepare-measure behaviors have PSD rank 2 but nonnegative rank growing without bound over finite prepare–measure sets, so for every N there is a finite qubit experiment no N-state model reproduces exactly (Heinosaari–Kerppo–Leppäjärvi–Plávala [55]; an alternative route under procedural-independence assumptions: Garner–Müller [56]). The factorization interface assumed is the ordinary preparation → finite ontic system → measurement one; the same work's coordinated version extends the obstruction against finite shared coordination between the sides [55]. Hence no fixed

finite realization is exactly operationally equivalent to even a single qubit's full theory — the elementary factorization half proved here, the unboundedness imported. The obstruction is independent of C1–C4 and of horizon growth. It does not touch the equivalence theorem — there the realization is chosen per finite-horizon law, and a fixed finite table of statistics always has one — and it is precisely why the equivalence carries its subscript. □

Remark (why the geometry alone cannot do it). Finitely many ontic states do not force a polytope state space, since a convex subset of a simplex need not be a polytope, so state-space geometry alone yields no obstruction. The SIC embedding is a certified instance: tetrahedral a_i, $p_i(r) = (1 + a_i \cdot r)/4$ maps the whole Bloch ball affinely and injectively into Δ_3, so a four-state model with restricted preparations carries the qubit's state geometry intact. What it cannot carry is the sharp effects: the affine extension of $|0\rangle\langle 0|$ takes the value $\frac{1}{2} + \frac{\sqrt{3}}{2} > 1$ at an ontic vertex, no valid response function exists, and the dimension obstruction above is the correct general form (embedding, curvature, effect violation, and the factorization half all certified exactly over $\mathbb{Q}(\sqrt{3})$: `nogo_probes.py`).

Remark (the operational-extension boundary). The exact finite representation theorem and the finite-substratum obstruction answer different questions. At fixed observable horizon, $S \iff D \iff Q_{\mathrm{fb}}$ is exact. For a bounded family of interventions, the gluing theorem below gives one reversible realization for the entire classical action-labelled experiment; the uniform-continuum theorem then covers bounded quantum instrument envelopes to arbitrary requested finite accuracy. What no fixed finite carrier can do is reproduce the entire exact continuum operational theory of even a qubit at all preparations/effects and all precisions, by the classical-dimension obstruction above. The compositional target is a characterization (kernel: `exactAll_iff_physical_general` in `OIBridge/GeneralCarrier.lean`, with necessity `physical_of_exactAll` in `OIBridge/PhysicalCharacterization.lean` and sufficiency `exactAll_of_physical_general` in `OIBridge/GeneralCarrier.lean`; `main_result` in `OIBridge/GeneralCarrier.lean`). An operational theory on a finite carrier, with the complex matrix kinematics the characterization takes as given — operations on the complex matrices indexed by the carrier and, at each ancilla level, by carrier–ancilla pairs, a kinematics whose derivation from operational premises is open — specifies the finite outcome families available on the system and at every finite ancilla level, and the available preparations from the system into each level, subject to six rules (kernel: `FiniteOperationalTheory` in `OIBridge/OperationalAssembly.lean`): (a) the identity is available on the system; (b) coarse-graining the outcome labels of an available family gives an available family, on the system and at every ancilla level; (c) at every ancilla level, an available family followed by available families chosen by its outcome is available, with the pair of outcomes as the joint outcome; (d) attaching a uniformly mixed ancilla of any positive size is an available preparation, as is an available preparation followed by an available single-outcome operation at its level; (e) at every level n some available n-outcome family reads the ancilla, its k-th branch sending every

product $X \otimes Y$ to $X \otimes P_k Y P_k$, with P_k the projector on the k-th ancilla state, a spectator-independence requirement that fixes the branch on every matrix (kernel: `readout_is_localLuders`); (f) an available preparation, followed by an available family at its level and by discarding the ancilla through the partial trace, induces an available family on the system. No rule imposes positivity or trace preservation or extends a system operation to a composite. The five conditions are: (i) valid probabilities (`CompositeOperationalValidity`): at every ancilla level each branch of an available family maps positive semidefinite matrices to positive semidefinite matrices, and the branches' traces sum to the input trace; (ii) trivial-ancilla consistency (`SystemToLevelOne`): every family available on the system is available at ancilla level one, with the one-state ancilla adjoined; (iii) inert spectators (`InertSpectatorCompositionality`): for every family available at an ancilla level, every finite spectator system and every relabelling of the spectator paired with that level's carrier as the carrier of another level, some family available at that other level acts on every product of a spectator matrix and a carrier matrix as the identity on the spectator factor and as the original branch on the other; (iv) full reversible control (`HasCompositeUnitaryControl`): conjugation by every unitary on the carrier of every ancilla level is available as a single-outcome operation; (v) iterated composition (`IteratedAncillaClosure`): attaching a fresh uniformly mixed ancilla of any positive size to the carrier of an ancilla level, applying a family available on the enlarged carrier relabelled as a higher level, and discarding the fresh ancilla gives a family available at the original level. On any nonempty finite carrier, the operational theories whose available families on the system and on every positive ancilla level are exactly the normalized finite Kraus instruments are precisely those satisfying (i)–(v), each independent of the others and of the C1–C4 process, witnessed on the qubit by theories realizing one fixed finite C1–C4 process (kernel: `five_way_minimality` in `OIBridge/RankGapTheory.lean`, `oi_alone_not_qm` in `OIBridge/GeneralCarrier.lean`), so that what the OI architecture must supply is named condition by condition rather than as one lifting hypothesis. Any exact-completion reconstruction beyond a fixed finite carrier must additionally show that its chosen completion retains finite predictive dimension; local tomography/composite completeness and all-level operational purification are sufficient reconstruction routes only after their hypotheses are actually established for that completion. The Bell side is a separate question rather than an analogous one: the graph supplies a causal cone for finite-range substrata, but a measurement-independent deterministic completion with one-step own-wing setting interventions, readouts outside the other wing's cone and a ± 1 outcome in every run is Bell-local. The framework's adopted Bell branch therefore requires ontic parameter dependence, with operational no-signaling required (§3.3).

Theorem (finite operational realization and gluing — ε-form). *Fix a quantum experiment: a system of dimension d with an initial state — alternative preparations entering, when wanted, as outcome-free channels in the family — a finite instrument family $\mathcal{F}$ — each instrument a finite family of CP maps*

$\{\mathcal{I}_o\}$ summing to a channel, each $\mathcal{I}_o$ with finitely many Kraus operators, so conditional states may be mixed — and a finite horizon K; fix a grid size $G \in \mathbb{N}$. There is a finite reversible deterministic embedded realization — visible sector: step counter, intervention record, outcome registers; hidden sector: the unnormalized digitized branch matrix (a density register), a predecessor stack, and K pre-loaded seeds uniform on $\{0, \dots, G-1\}$; one FIXED injective step map φ_a per instrument, extended to a bijection by finite padding — with:

(1) ***Completeness, adaptive.*** *For every adaptive protocol over $\mathcal{F}$ through K, the realized outcome-sequence law is within $K(|\mathcal{O}_{\max}|-1)/G$ of the quantum law in total variation, $|\mathcal{O}_{\max}|$ the largest per-instrument outcome count — K/G for binary-outcome instruments; $G \geq 2K(|\mathcal{O}_{\max}|-1)/\varepsilon$ gives accuracy $\varepsilon/2$. This bound is for exact-arithmetic registers over the instruments' entry ring, where the grid is the only error source. For fixed-point b-bit registers the instrument-level Rounding Lemma below bounds each digitized step on subnormalized cq branches; a bound on the realized outcome law of the clipped and normalized simulator at finite b is open.*

(2) ***Gluing by restriction.*** *For every $\mathcal{F}' \subseteq \mathcal{F}$ and $K' \leq K$, the same realization with the agent confined to $\mathcal{F}'$ and stopped at K' realizes the subexperiment — overlapping families agree on shared subexperiments identically, not approximately, because they run the same maps.*

(3) ***Context-independent implementation.*** *Each labelled instrument is one fixed map, applied whenever chosen, in every context; context enters only through the state. In the constructed realization an outcome's value can depend on the instrument through the threshold order; for $d \geq 3$ and a family containing a Kochen–Specker set, every deterministic account reproducing the exact statistics must assign values contextually, and the realization reproduces the statistics within the stated accuracy. This is implementation-fixedness of labelled instruments, not Spekkens noncontextuality, which concerns operationally equivalent procedures and is not claimed.*

(4) ***Composition.*** *Sequential composition by construction; for spatially separated experiments the product construction implements the joint family by applying the local instruments' CP maps as $\mathcal{I}_a \otimes \mathcal{I}_b$ on the joint branch register, visible sectors in product — necessarily joint for entangled preparations, in the factorizability-failing, locality-retaining pattern of §3.3.*

(5) ***Capacity.*** *With exact-arithmetic registers over the instruments' entry ring, $\log_2 |\mathcal{C}_H| \leq (K+1) \log_2 |V_{\text{reach}}(\mathcal{F}, K)| + K \log_2 G$, where V_{reach} is the finite set of unnormalized branch vectors reachable in $\leq K$ steps; with fixed-point dyadic registers (b fractional bits and $O(\log d)$ integer bits per entry, all counted), $G = \lceil 2K(|\mathcal{O}_{\max}|-1)/\varepsilon \rceil$, and a canonical renormalization — the branch register replaced by $\rho^\sharp = 2^{s(\tilde{\rho})}\tilde{\rho}$ only when its norm falls below the window's lower edge, the rescaling one-sided and upward, hence an exact left shift in fixed point (a downward shift, the only one that could round, never occurs: trace-nonincreasing steps plus the small accumulated error keep the norm under the generous upper*

edge; the exact-ring variant extends the ring by $\frac{1}{2}$); no shift is stored anywhere: the thresholds are scale-invariant, so the realized law is unchanged, and reversibility needs no exponent, the predecessor stack holding the predecessor itself — at a fractional precision of b bits, plus the counted $O(\log d)$ integer bits per entry and no exponent anywhere, the registers give $\log_2 |\mathcal{C}_H| = O(K\, d^2\, b + K \log_2 G)$ for $b \geq \log_2(2K\, C_{d,R})$, the window condition of the Rounding Lemma below, which is $O(K\, d^2 \log(Kd\, |\mathcal{O}|/\varepsilon) + K \log_2 G)$ if $b = O(\log(Kd\, |\mathcal{O}|/\varepsilon))$ bits suffice for accuracy ε, which depends on the open finite-b bound of clause (1) — linear in K up to logarithms (rescaling invariance certified: check 10, `opglue_probes.py`). The lower bound is of a different scope: $\log_2 |\mathcal{C}_H| \geq I^*$ holds universally, for every faithful realization (memory theorem above), and is $\Omega(K)$ on explicit memory-chain families — a constructive upper bound for this realization and a universal floor, deliberately not a matched pair. For the constructed realization on those families, $c_1 K \leq \log_2 |\mathcal{C}_H| \leq c_2 K\, (d^2 b + \log_2 G)$ at fractional precision $b \geq \log_2(2K\, C_{d,R})$.

Rounding Lemma (instrument-level, fixed-point dyadic model). Floating-point entries carry uncounted per-entry exponents, so the model is fixed-point: every register entry is a dyadic rational with b fractional bits and $O(\log d)$ integer bits, all counted; Hermiticity is exact by storing the upper triangle and defining the lower by conjugation; each operation rounds to b fractional bits with absolute error $\leq 2^{-b}$. The lemma is stated for the whole instrument as the classical–quantum map $\mathfrak{I}(\tilde{\rho}) = \bigoplus_o \mathcal{J}_o(\tilde{\rho})$, with $R = \sum_o r_o \leq |\mathcal{O}_{\max}|\, d^2$ the instrument's total Kraus count (for the family, R denotes the family-wide maximum $\max_a \sum_o r_{a,o}$):

$$\sum_o \left\| \hat{\mathcal{J}}_o(\tilde{\rho}) - \mathcal{J}_o(\tilde{\rho}) \right\|_1 \;\leq\; C_{d,R}\, 2^{-b}\, \|\tilde{\rho}\|_1, \qquad C_{d,R} = 48\, R\, d^4.$$

Proof. Kraus quantization, by operator norms: $\|\Delta_\alpha\|_{\mathrm{op}} \leq \|\Delta_\alpha\|_F \leq d\, 2^{-b}$ and $\|M_\alpha\|_{\mathrm{op}} \leq 1$ give, per Kraus term, $\|(M{+}\Delta)\tilde{\rho}(M{+}\Delta)^\dagger - M\tilde{\rho}M^\dagger\|_1 \leq (2d\, 2^{-b} + d^2 4^{-b})\|\tilde{\rho}\|_1 \leq 3d\, 2^{-b}\|\tilde{\rho}\|_1$, summing to $3Rd\, 2^{-b}\|\tilde{\rho}\|_1$ over the instrument. Fixed-point arithmetic with propagation: the first product injects at most $2d\, 2^{-b}$ per entry, the second matmul propagates that with gain $\leq 2d$ and injects again, within-block sums accumulate — componentwise $\leq 7 r_o d^2\, 2^{-b}$ per block — and $\|A\|_1 \leq d^2 \max_{ij} |a_{ij}|$ with the window $\|\tilde{\rho}\|_1 \in [\frac{1}{2}, 4]$ converts and sums the blocks within the stated constant, which carries slack for the clipping below. Window management is one-sided: the register is rescaled upward, by an exact left shift, only when its norm falls below $\frac{1}{2}$; trace-nonincreasing steps grow the norm per step by at most $(1{+}\eta)$, and each upward shift lands it in $[\frac{1}{2}, 1)$, so with $K\eta \leq \frac{1}{2}$ it stays below $e^{1/2} < 2 < 4$ — a downward shift, the only one that could round, never occurs. Telescoping: the exact cq instrument is a trace-norm contraction on Hermitian inputs ($\sum_o \|\mathcal{J}_o(A)\|_1 \leq \|A\|_1$, via $A = A_+ - A_-$ and trace-nonincrease), so along the process σ_t of subnormalized cq branches, $\|\hat{\sigma}_{t+1} - \sigma_{t+1}\|_1 \leq \|\hat{\sigma}_t - \sigma_t\|_1 + C_{d,R} 2^{-b} \|\hat{\sigma}_t\|_1$, whence $\|\hat{\sigma}_K - \sigma_K\|_1 \leq 2K\, C_{d,R}\, 2^{-b}$ — rare outcomes stay weighted by their own small mass, and no per-branch conditioning ever occurs. The simulator does not sample the raw traces: the digitized

Kraus family is not exactly trace-preserving, so it clips and normalizes. Corollary (clipping and normalization).* *For a positive branch with* $s = \sum_o \operatorname{Tr} B_o > 0$ *and raw drift* $E = \sum_o \|\widehat{B}_o - B_o\|_1 \le \eta s$*: clipping (*$C_o = \widehat{B}_o$ *if* $\operatorname{Tr}\widehat{B}_o > 0$*, else* 0*) does not increase the aggregate error, since a clipped outcome's exact block has* $\operatorname{Tr} B_o \le \|\widehat{B}_o - B_o\|_1$*; with* $Q = \sum_o \operatorname{Tr} C_o \ge (1-\eta)s$*, the normalized digitized kernel* $\tilde{\mathfrak{I}}(\rho) = \bigoplus_o (s/Q)\, C_o$ *— exactly the simulator's pre-grid probabilities, the blocks retained up to the irrelevant power-of-two scale — satisfies* $\|\tilde{\mathfrak{I}}(\rho) - \mathfrak{I}(\rho)\|_1 \le \frac{2\eta}{1-\eta}\, s \le 4\eta s$ *for* $\eta \le \frac{1}{2}$*; a parent whose clipped total vanishes takes a fixed deterministic fallback — outcome* o^* *is written and the register set to the reference state* P_0 *at window scale, the predecessor pushed — so the step stays a bijection and no probability mass is lost; the entire mass of such parents is already bounded inside the same error term, so the arbitrary assignment costs nothing further.* □ The Corollary is stated for positive inputs; digitized branches are Hermitian but not positive. Normalization multiplies a parent's blocks by s/Q, which can exceed 1, so clipping and normalization can increase negative mass, and the Corollary's estimate is not established for the simulator's Hermitian branches. A bound on the classical-marginal rounding term at finite b is open.* □ Status: the instrument-level inequality is proved; the per-outcome and cq-aggregate inequalities are checked in exact rational simulation of the b-fractional-bit rounding, Hermiticity enforced by construction, at $b \in \{8, 16, 24\}$, with the telescoping instanced over a K=3 branch ensemble including rare outcomes (checks 14–15, `opglue_probes.py`, positive inputs, check 15 at $b = 16$); check 16 evaluates the clipped-and-normalized simulator kernel against the exact law on one instance ($d = 2$, $K = 3$, $b \in \{20, 24\}$, no window rescaling), with negativity measured on the raw blocks, before normalization, and so does not measure the negative mass of the normalized branches on which the Hermitian case turns.

Proof (construction and bound). The step map φ_a reads the counter t; an outcome-free instrument applies its CP map to the branch matrix and pushes the predecessor onto the stack; an instrument with outcomes computes $q_o = \operatorname{Tr} \mathcal{J}_o(\tilde{\rho})$ exactly in the ring, compares the t-th integer seed by exact threshold ($r \cdot \sum_o q_o < G\, q_{<o}$ cumulatively), writes the outcome to the visible register, sets $\tilde{\rho} \leftarrow \mathcal{J}_o(\tilde{\rho})$ — unnormalized, so the true conditionals are exact ring ratios even when conditional states are mixed — and pushes the predecessor; the stack, not map injectivity, carries invertibility, so multi-Kraus maps cost nothing. Injectivity: the stack makes every step invertible on the reachable set, and a finite injective map extends to a bijection on a finite superset. The bound: each step realizes the grid rounding of the true conditional through cumulative integer thresholds, off by at most $(|\mathcal{O}|-1)/G$ in total variation — $1/G$ for two outcomes, and — because the branch register is carried unnormalized — the true conditionals are computed exactly in the ring, so in the exact-arithmetic regime the grid is the only error source; the chain rule over K steps gives (1). In the finite-precision regime the comparison at the level of subnormalized cq branches telescopes: one digitized instrument step differs from the exact step

by at most $C_{d,R}\,2^{-b}$ in trace norm on the full outcome direct sum, and the exact instrument is trace-norm contractive on Hermitian inputs, so the branch errors add over K steps (the instrument-level Rounding Lemma). The step from that branch-level bound to the outcome law of the clipped and normalized simulator is open (clause (1)). Adaptivity is free: an adaptive protocol's trajectory probability is the product of the same per-step conditionals along its realized history, and (2) is the observation that the construction never asks which protocol is running. *Certified* at $(d, K, |\mathcal{F}|) = (2, \{2, 3\}, 4)$ — projective Z and X, the Hadamard intervention, and the 3-4-5 rotation, whose non-dyadic branch probabilities make the grid error strictly positive — with the K/G bound and $\sim 1/G$ scaling observed at $G \in \{8, 128, 1024\}$; operational noncommutativity reproduced at total variation $\frac{1}{2}$ between the XZ and ZX outcome-string laws, with Z-repeatability exact; restriction identity; per-instrument injectivity over all 672 reachable states; capacity floor — exact arithmetic over $\mathbb{Q}(\sqrt{2})$; the general multi-Kraus case certified via the density register against a two-Kraus-per-outcome instrument with exactly mixed conditionals (checks 11–12) — `opglue_probes.py`. □

Corollary (single controlled dynamics). The per-instrument family assembles into one map: on the enlarged hidden sector carrying a read-only program register — the protocol table — alongside the visible record, the seeds, and the branch register, the policy-controlled step $\Phi(x) = \varphi_{\pi(x_{\mathrm{vis}})}(x)$, the instrument selected by the protocol from the retained record, is a single reversible deterministic map realizing the entire adaptive experiment; and one $\Phi_{\mathcal{F},K}$ serves every protocol over $\mathcal{F}$ at once, the protocol entering as the preparation of the program register rather than as a change of dynamics. Invertibility: strip the appended outcome from the record, recompute the control from the restored record, undo the controlled step. One law, initial conditions vary: a single reversible dynamics realizes the experiment at finite resolution (check 13, `opglue_probes.py`: one map, two protocols including an adaptive one, all steps injective, laws exact). Quantum mechanics enters as the instrument data and as the state held in the hidden register, so this shows that finite-accuracy quantum statistics are compatible with one fixed reversible law; the construction uses no property of the substratum beyond finiteness and reversibility. Resource accounting: the capacity clause counts branch register, stack, and seeds — the controlled model, intervention labels arriving as visible inputs. Here the read-only program register is additional state: an arbitrary adaptive policy over $\mathcal{F}$ through K costs up to $\Theta(|\mathcal{O}_{\max}|^K \log_2 |\mathcal{F}|)$ bits of table — exponential in K, charged separately and deliberately outside the linear-in-K estimate. One dynamics; the program is paid for as an initial condition, not obtained free. □

Remark (what the gluing theorem does and does not deliver). At finite resolution it crosses most of the gap between Q_{fb} and operational quantum experiments: incompatible measurements, adaptive interventions, preparation and instrument statistics, fixed per-instrument mechanisms, and gluing by construction. It is an **operational realization theorem**, not a uniqueness theorem: the same reversible machinery can realize non-quantum finite instrument fam-

ilies, so the theorem by itself does not identify the standard local quantum instrument category as the only coherent completion. The exact-continuum boundary is separately sharp: no fixed finite realization can reproduce the full prepare-and-measure theory of even one qubit exactly, while the constructed quantum-shadow family is dense on every finite test. Thus the finite realization direction is closed at the stated accuracy. The uniqueness side of the coherent/composite lift is settled by the coherent-completion classification (kernel: `sameData_unitary_or_transpose` in `OIBridge/JordanClassification.lean`; for Hermitian accessible menus spanning the full matrix algebra and containing the unit at a common index, and separating positive-semidefinite state families generating the positive cone): completions carrying the same complete operational data are equivalent up to unitary gauge and one global antiunitary orientation reversal. Its existence side is settled by the operational-completion characterization (Remark, the operational-extension boundary, above): within the complex matrix kinematics it takes as given, exact finite operational quantum mechanics on a carrier and its composites is equivalent to five named conditions, none supplied by bare finite OI, so what remains is orientation, the physical questions of whether an extended OI mechanism supplies full reversible control, which the substratum's stated interventions do not supply (kernel: `substratumGen_not_control` in `OIBridge/StructuralClosure.lean`, on the two-state carrier), and of which of the other conditions the substratum supplies, and a derivation of that kinematics, not the size of the family and not a lifting theorem. Bell is not an extra rung of that lift: §3.3 separately fixes the deterministic Bell branch as measurement-independent but ontically parameter-dependent, with operational no-signaling. The Kochen–Specker inheritance (§3.2) remains conditional exactly at that operational-algebra layer.*

The operational-completion theorem also admits a physically layered formulation. Keeping the original OI core unchanged, define quantum-complete OI (OI^{+}) as well-formedness together with observational independence, reversible richness, and observer recursion. Within the same complex matrix kinematics, observational independence is equivalent to inert spectators on every finite carrier; reversible richness implies sufficient reversible control, and conversely for well-formed theories on a nonempty carrier; observer recursion implies iterated composition, and conversely when the identity and the relative ancilla readout are available at every level, which sufficient reversible control and inert spectators supply. Hence, on every nonempty finite carrier,

$$\mathrm{OI}^{+} \iff \text{exact finite endomorphic operational QM.}$$

Each added principle is independently necessary, on the qubit, relative to the OI core, well-formedness, and the other two (kernel: `oiPlus_independence` in `OIBridge/CompletedOI.lean`). This is a characterization of a quantum-complete extension of OI, not a claim that bare OI entails the added principles; the OI-core clause becomes logically redundant once the reversible-control condition is present. On a general carrier OI^{+} consists of well-formedness and the

three added principles, with no OI-core clause; the OI-core clause is defined on the qubit, where it does not change the class (kernel: `carrier_general_oiPlus`, with directions `qm_of_oiPlus` and `oiPlus_of_qm`, and `oiPlus_qubit_iff` in `OIBridge/CarrierGeneralOIPlus.lean`). A primitive-source refinement restates the three added principles at the level of physical implementations and embedded observers — implementation locality, phase-free richness — one continuously driven transition and the exchanges of distinguishable states, with no phase operation — and embedded observation, with no closure of the implementations under the adjoint and no inverse accessibility assumed — equivalent to the same operational quantum mechanics on every nonempty finite carrier, inverse accessibility following on any well-formed theory (kernel: `carrier_general_oiPlusMin`, `oiPlusMin_iff_qm` in `OIBridge/MinimalRepertoire.lean`; `carrier_general_oiPlusPos` in `OIBridge/PositivePackage.lean`, `inverseAccessibility_of_lieRank` in `OIBridge/MicroscopicReversibility.lean`). A substratum-source form completes that refinement: the concrete OI substratum supplies the structural architecture — closure under composition and coarse-graining, stability under an uncoupled spectator, relabelling invariance, reversibility, the exchanges of distinguishable states, and, as a stated intervention principle rather than a consequence of its finite states and bijective dynamics, the phases — but, under finite bijective read-write dynamics, not the continuously tunable state-mixing operation; with that one operation added as an empirical controllability resource it is equivalent to the same operational quantum mechanics on every nonempty finite carrier, while finite reversible read-write dynamics, even with a nontrivial reversible read-write coupling, does not itself generate quantum state mixing (kernel: `substratum_plus_control_qm`, with the converse `qm_generated_by_substratum_extension`, and `readWriteOperator_supplied` and, on the two-state carrier, `substratumGen_not_control`, in `OIBridge/StructuralClosure.lean`, `readWriteControl_independent` in `OIBridge/ReadWriteControl.lean`). Its typed form removes the endomorphic qualifier: under the carrier-general typed operational interface — identity, composition, coarse-graining, relabelling, uniform finite attachment, discard, and factor readout, an interface that itself carries no quantum content — the same equivalence holds with exact finite-dimensional typed operational quantum mechanics, operations between any two nonempty finite carriers included, so the remaining qualifier is finite-dimensional rather than endomorphic (kernel: `typed_determined_iff` in `OIBridge/TypedCompletion.lean`, `typed_determined_of_oiPlusMin` in `OIBridge/TypedPositive.lean`). The finite theory admits a canonical infinite-region completion without a continuum limit: the finite regions of the fixed-spacing lattice carry full matrix algebras joined by compatible injective unital embeddings with observables of disjoint regions commuting, the norm completion of their union is a quasilocal C^*-algebra on which every consistent family of finite-region density matrices is a state and the reversible finite-range substratum update is an isometric $*$-automorphism, and any system carrying the same local stages is canonically and uniquely $*$-isomorphic to it, intertwining the dynamics; this removes the finite-region

cutoff for the algebra, the state space, and the OI-induced discrete dynamics rather than characterizing all infinite-dimensional quantum instruments or all locality-preserving quantum dynamics; the lattice spacing stays fundamental, no Hilbert-space representation and no superselection sector is selected at the level of the laws, the restriction to the substratum-induced dynamics is a theorem rather than a convention — a phase automorphism at one site preserves locality without arising from any reversible finite-range bijection — and, while the update is joined to the identity by an exact strongly continuous path of $*$-automorphisms of that algebra, with no one-parameter-group law established and no generator exhibited, a continuous-time Hamiltonian law remains additional structure (kernel: `closure_iUnion_stage`, `quasiState_unique` in `OIBridge/QuasilocalAlgebra.lean`, `canonEquiv`, `phaseQ_ne_heisQ` in `OIBridge/QuasilocalCharacterization.lean`; `driveQ_isContinuousPath`, `driveQ_smul`, `driveQ_one_eq_heisQ` in `OIBridge/SecondOrderDrive.lean`). Within the structural architecture the substratum supplies, with the phase intervention stated, exact finite endomorphic operational quantum mechanics holds exactly when one continuous layer flow of an involution of the configuration space moving some configuration is an available operation at every level and every intermediate time, the executability of that flow being an assumption not derived by configuration-level generation, as the phase intervention is an assumption on the substratum class and not a consequence of the finite states with the bijective dynamics (kernel: `derivedOI_qm_iff_layerFlowExecutable'` in `OIBridge/DerivedQ3.lean`, `substratumTheory_not_layerFlowExecutable` in `OIBridge/LiftAudit.lean`). The stated substratum dynamics with the observer's read and write access do not source that executability, the layers being available at time one and not at intermediate times, and the observer-level lift, in every formulation stated that lands in the operational interface, derives neither a non-monomial admissible operator nor the layer flow, the site-space wave-operator formulation being unresolved rather than excluded (kernel: `obs_not_layerFlowExecutable` in `OIBridge/ExecSource.lean`, `coherentLiftClass_eq_substratumClass` in `OIBridge/LiftSource.lean`).

Passive observation, kept apart from the characterization. Four statements about passive observation stand beside the equivalence above and enter none of its hypotheses or conclusions. Complete passive observation of a finite-dimensional observable algebra in block-diagonal form — observation by a passive instrument, one whose branches leave the block projectors fixed — is possible exactly when the algebra is commutative: some passive instrument separates its states exactly when the algebra is commutative, and every passive instrument induces a classical stochastic observation of the algebra's center (kernel: `complete_passive_iff_commutative`, `central_classification` in `OIBridge/CentralObservation.lean`; `no_complete_passive_observation` in `OIBridge/PassiveObservation.lean`). Quantum noncommutativity cannot coexist with complete passive observation. This is theory-insensitive: passive incompleteness holds in every finite operational theory on a carrier with two or more states, so it carries no information about whether the OI core is realized;

the implication from the core to passive incompleteness holds vacuously, and its converse fails, witnessed by a theory that is passively incomplete and lacks the core (kernel: `passivelyIncomplete_of_card`, `oiCore_to_passive_vacuous`, `passivelyIncomplete_without_oiCore`, `passive_nondiscriminating` in `OIBridge/PassiveIndependence.lean`). Passive incompleteness is therefore not evidence for a hidden ontology, and nothing here says that quantum mechanics rests on observation incompleteness: the containment in the other direction, that exact finite quantum mechanics realizes the sealed core, is a consequence of full control and nothing more (kernel: `qm_implies_oiCore` in `OIBridge/CompletedOI.lean`, through `realizesSealedOICore_of_control` in `OIBridge/OIRealization.lean`). A passive internal observer, whose outcome is a function of a register read on block-diagonal states, can only read an existing record: each of its branches is the record projection $\rho \mapsto P_o \rho P_o$, with outcome law $\mathrm{tr}(P_o \rho P_o)$ (kernel: `internal_branch_eq_blockPart`, `internal_outcome_law` in `OIBridge/InternalObserver.lean`), and creating a genuinely new record changes the joint system, the recorder that writes the register being not passive on the record-block algebra (kernel: `recordInstr_writes`, `recordInstr_not_passive`, `no_full_passive_self_record`). All four statements are finite-dimensional, and none bears on the equivalence.

The uncertainty relations. The uncertainty principle [58] has two standard quantitative forms, both inequalities in the commutator: the preparation relation $\sigma_A \sigma_B \geq \frac{1}{2}|\langle [A, B] \rangle|$ for a state and two observables A, B [59], and the error–disturbance relations of measurement [60, 61]. The first statement of the preceding paragraph is a qualitative finite-dimensional consequence of the same non-commutativity for observation without disturbance [62]: an instrument that leaves every state of the algebra unchanged on average obtains information only about the algebra's center. The preparation relation does not discriminate the OI core either: it holds for every density matrix and every pair of Hermitian matrices, hence in every finite operational theory in the present sense, including the theory that is passively incomplete and lacks the core. Observational incompleteness in the sense of the OI core concerns the hidden sector of an embedded observer, not the joint sharpness of non-commuting observables. Under (C4) it is present already for a single fixed readout, where no choice between measurements arises: every faithful deterministic completion of the visible process carries readout-relevant hidden predictive memory. The visible laws of fixed-readout processes are, by $S \iff D \iff Q_{\mathrm{fb}}$, exactly the finite stochastic laws, and no uncertainty relation constrains that class (kernel: `S_imp_D` for the realization of every finite stochastic law, `D_imp_Qfb` and `Qfb_imp_S` for the converse, assembled in `finite_horizon_equivalence` in `OIBridge/Equivalence.lean`). The relations bind only where non-commuting observables are operationally available, and that availability belongs to the coherent-instrument completion rather than to the finite-law equivalence: in the OI^+ characterization above, the full non-commutative operational structure follows from well-formedness and the three added principles, the OI-core clause being redundant once reversible control is present, and the operational realization theorem realizes non-

quantum instrument families as well. Whether embedded observation selects a non-commutative operational structure is the selection question.

Lemma (adaptive diamond telescoping). Let π be an adaptive protocol of horizon K and π' the protocol with each instrument I_t replaced by I'_t. Then the outcome-sequence laws obey $\mathrm{TV}(P_\pi, P_{\pi'}) \le \sum_{t\le K} \max_a \|I_t^a - I_t'^a\|_\diamond$. *Proof.* Hybrid: swap one step at a time. An instrument is a channel into a classical–quantum register, the adaptive continuation is itself a channel, and the diamond norm is monotone under pre- and post-composition and under tensoring — so each single swap moves the final classical law by at most that step's diamond distance; summing the hybrids gives the bound. □

Theorem (uniform continuum realization — mutual ε-density). *Fix d, an outcome bound m, a horizon K, and $\varepsilon > 0$. Let N be a δ-net, $\delta = \varepsilon/(2K)$, in diamond norm on the compact set of $\le m$-outcome instruments on $\mathbb{C}^d$ (preparations included as channels; a standard volumetric bound gives $|N| \le \exp(C\, m\, d^4 \log(1/\delta))$, the set having real dimension $\le 2md^4$), and apply the gluing theorem to $\mathcal{F} = N$ with $G = \lceil 2K(m{-}1)/\varepsilon \rceil$. The single resulting finite reversible realization R satisfies: (i) for every adaptive quantum protocol π of horizon $\le K$ with arbitrary continuum instruments there is an R-protocol $\hat\pi$ — the net rounding — with $\mathrm{TV}(P_\pi^{\mathrm{QM}}, P_{\hat\pi}^R) \le K\delta + K(m{-}1)/G \le \varepsilon$, by the telescoping Lemma plus clause (1); (ii) conversely every R-protocol runs genuine quantum instruments (net members), so its law is within $K(m{-}1)/G \le \varepsilon/2$ of a quantum law. The two operational theories are therefore mutually ε-dense at horizon K, in the finite-test metric the density Corollary below makes explicit. In the controlled model the mechanisms are per-instrument, so the branch register, stack and seeds hold no per-instrument state (the autonomous program register is charged separately: single-dynamics Corollary above). In exact arithmetic the reachable branch vectors, and with them clause (5)'s exact-ring capacity bound, grow with the net; a hidden-state capacity independent of $|N|$ is obtained with fixed-point registers, whose finite-b accuracy bound is open (clause (1)). The description complexity of the controlled transition law scales with the instrument net and is outside this capacity measure; the net enlarges the visible intervention record by $\lceil \log_2 |N| \rceil = O(m\, d^4 \log(2K/\varepsilon))$ bits per step.* □ *(Telescoping and off-net density certified on the qubit instance — the 3-4-5 against the 5-12-13 rotation, and a mean-angle off-net target realized through its nearer net member within the predicted bound — checks 8–9,* `opglue_probes.py`*.)*

Corollary (finite-test density of deterministic realizations in operational quantum theory). Along any $K_n \to \infty$, $\varepsilon_n \to 0$, the quantum-shadow realizations R_n^Q of the theorem satisfy: every finite-horizon adaptive quantum protocol's law is the limit of its net-shadows' realized laws, and every R_n^Q-law is within $\varepsilon_n/2$ of a quantum law — so, in the finite-test (Hausdorff-on-behavior-sets) topology, $Q_{\mathrm{operational}}^{(d,m)} \subseteq \overline{D_{\mathrm{finite}}^{(d,m)}}$ at each fixed system dimension d and outcome bound m — the scope the theorem fixes; the union over (d, m) is a further limit, not taken here — with two-sided ε-density holding for the constructed subclass $\{R_n^Q\}$. As classes this is density, not identity: D_{finite} is strictly larger — the response-

table construction realizes arbitrary finite stochastic behavior, quantum or not — so its closure contains non-quantum operational theories too; the additional structure that makes one controlled family of preparations, coherent instruments, and composites a single quantum operational theory on a fixed finite carrier is characterized, within complex matrix kinematics taken as given, by the five completion conditions of §3.4. The exact continuum theory stays unreachable at any fixed stage (the classical-dimension Proposition). □

Summary (the equivalence at maximum proved strength). Four claims, each with its exact status. (1) $S \iff D \iff Q_{\mathrm{fb}}$ — proved, all finite laws. (2) $M_t > 0 \iff$ readout-relevant hidden predictive memory in every faithful completion — proved: clauses (a)–(c) of the theorem below proved in the kernel for a joint law of history, present and hidden state, with the reading of a realization's time-t law as that joint law not formalized; the biconditional restates the pushforward identity (a), its converse applying (a) in one faithful completion, which exists ($S \Rightarrow D$), and both directions using the vanishing criterion for conditional mutual information, which is not formalized. (3) No fixed finite classical carrier reproduces full qubit operational behavior — proved under the stated operational interface, the unbounded-dimension half imported [55]. (4) Every d-dimensional quantum experiment of horizon K has a finite reversible deterministic ε-realization — proved in exact arithmetic, covering general instruments, mixed conditionals, incompatible measurements, and adaptive protocols, with explicit controlled-model resources and one controlled dynamics serving all protocols; at finite b the instrument-level Rounding Lemma bounds each step, and the bound on the realized law is open. Per (3), the equivalence cannot be strengthened to exact class identity at any fixed carrier; a fixed finite instrument family is realized exactly by the intervention dilation above, with a product prior carrying the quantum conditionals (Extension, response-table form), and the accuracy ε enters with a rational uniform seed and with uniformity over the instrument continuum. What remains is selection.

Theorem (unavoidable hidden predictive memory). *Let $\mathcal{S}$ be a visible process and let $\mathcal{R}$ be any faithful deterministic realization of it on a horizon K — a bijection φ on $\mathcal{C}_V \times \mathcal{C}_H$ with an initial law on $\mathcal{C}_V \times \mathcal{C}_H$, which may correlate the coordinates, under which the visible trajectory has the law of $\mathcal{S}$. Fix $1 \le t \le K-1$. Write $f_x(h) = \pi_V(\varphi(x,h))$ for the next visible state, and for a positive-probability visible history p ending in x write μ_p for the induced hidden posterior $P(H_t = \cdot \mid X_{<t} = p, X_t = x)$. Then:*

(a) ***Pushforward identity.*** *$P(X_{t+1} \mid p, x) = (f_x)_{\#}\mu_p$ exactly.*

(b) ***Distinguishability floor.*** *For positive-probability histories p, p' sharing the endpoint x, the observed gap obeys $\delta = \|P(X_{t+1} \mid p, x) - P(X_{t+1} \mid p', x)\|_{\mathrm{TV}} \le \|\mu_p - \mu_{p'}\|_{\mathrm{TV}}$.*

(c) ***Capacity floor.*** *$M_t := I(X_{<t}; X_{t+1} \mid X_t) \le I(X_{<t}; H_t \mid X_t) \le \log_2 |\mathcal{C}_H|$.*
(kernel: `unavoidable_hidden_predictive_memory`, `distinguishability_floor`, `capacity_floor` in `OIBridge/HiddenMemory.lean`, for a joint law of history,

present and hidden state; reading a realization's time-t law as that joint law is a step not formalized)

Proof. (a) Determinism makes X_{t+1} a function of (X_t, H_t), so conditioning on p and pushing μ_p through f_x is the conditional law. (b) A deterministic map is a channel and total variation is monotone under channels. (c) $X_{<t} \to H_t \to X_{t+1}$ is a Markov chain given X_t, by the same functional dependence; data processing gives the first inequality and $I \le H(H_t) \le \log_2 |\mathcal{C}_H|$ the second. □ (All three checked over every bijection at $(n_V, |\mathcal{C}_H|) = (2,2), (2,3), (3,2)$ under the uniform prior — $1\{,\}464$ cases — in `memory_probes.py`, (a) and (b) in exact rational arithmetic and (c) in floating point.)

Remark (what this strengthens). The equivalence of §3.4 is existential: some finite reversible realization exists. That statement is satisfied by Markov processes too, so it cannot by itself distinguish the framework's regime. The theorem above is universal — it constrains every deterministic completion — and it is the sense in which observation incompleteness forces hidden structure rather than merely permitting it: **if the visible process remembers its past, then every deterministic completion reproducing it must carry that memory in the hidden sector, with distinguishability surviving to readback and capacity at least M_t.** And the statement is a biconditional: if in some faithful realization two positive-probability histories sharing an endpoint induce differing pushforwards $(f_x)_{\#}\mu_p \neq (f_x)_{\#}\mu_{p'}$, then by (a) those pushforwards *are* differing next-step conditional laws, so $M_t > 0$ by the vanishing criterion for conditional mutual information — hence, since a faithful realization exists (`S_imp_D`), $M_t > 0$ holds exactly when every faithful deterministic realization — of any horizon $K \ge t+1$, the least on which M_t is defined — contains readout-relevant hidden predictive memory at t. The conditions follow rather than being assumed: (C1) because if f_x were constant on the union of the supports of μ_p and $\mu_{p'}$ the two pushforwards would coincide, so readout-relevant memory makes f_x non-constant on positive-probability hidden states; (C2-structural) as the retention required by (b) — the slow-bath form (C2-slow) one mechanism among several (§1.3); (C3) by the data-processing argument of (c) applied to the whole visible future (kernel: `c3_necessity` in `OIBridge/C3Necessity.lean`), a bound that holds in every faithful realization, Markov ones included. In the bijective class reversibility fixes two of the theorem's terms: the posteriors of distinct positive-probability histories ending at x have disjoint supports, so the right side of (b) is 1, and $I(X_{<t}; H_t \mid X_t) = H(X_{<t} \mid X_t)$ for every law. The retention in (b) is therefore supplied by reversibility for Markov and non-Markovian laws alike; what is specific to non-Markovian laws is (a) with the visible witness.

Remark (a variational question). Over reversible deterministic completions $M_{\mathrm{det}} := \inf_{\mathcal{R}} I(X_{<t}; H_t \mid X_t)$ does not equal M_t in general at any hidden dimension: every bijective completion has $I(X_{<t}; H_t \mid X_t) = H(X_{<t} \mid X_t)$ (Remark above), which exceeds M_t by $H(X_{<t} \mid X_t, X_{t+1})$ — for an i.i.d. fair coin at $t = 1$, $M_1 = 0$ while every reversible completion carries 1 bit. At

$(n_V, |\mathcal{C}_H|) = (2, 3)$ and $t = 2$, the uniform prior on $\mathcal{C}_V \times \mathcal{C}_H$ gives sixteen distinct laws of $(X_0, \ldots, X_3)$ with $M_t > 0$, and $M_{\text{det}} = M_t$ for two of them, with residuals up to 2/3 bit (`memory_probes.py`). The agreeing laws are those with $H(X_{<t} \mid X_t, X_{t+1}) = 0$, and each residual equals that entropy by the identity above, so it is the same under every hidden prior that realizes the law. Without the bijectivity requirement the infimum is attained: in every deterministic completion the whole visible future is a function of (X_t, H_t), so $I(X_{<t}; H_t \mid X_t) \geq I(X_{<t}; X_{>t} \mid X_t)$ (kernel: `c3_necessity_via_hidden` in `OIBridge/C3Necessity.lean`, stated there for bijective realizations; the same data-processing step for non-injective and staged completions is proved on paper), with equality for the canonical predictive quotient (the predictive-completion theorem below), whose hidden state at t is the visible future tail. The infimum over deterministic completions is therefore $I(X_{<t}; X_{>t} \mid X_t)$, equal to M_t exactly when $I(X_{<t}; X_{>t+1} \mid X_t, X_{t+1}) = 0$; for X_0, X_1, X_2 i.i.d. fair bits and $X_3 = X_0$, $M_1 = 0$ while every deterministic completion carries 1 bit at $t = 1$.

Remark (the conditions are diagnostics, not hypotheses). The reduction extends one step further. (C4) is not an independent condition on the realization either: it asserts a readback gap *mediated through the hidden state*, and in a faithful realization every correlation is so mediated, so the mediation clause is automatic and (C4) coincides with condition (ii) of the characterization theorem. Exhaustive enumeration confirms it — the (C4) gap and $I(X_{<k}; X_{k+1} \mid X_k) > 0$ agree on all 24, 720 and 720 bijections under the uniform prior at $(n_V, |\mathcal{C}_H|) = (2, 2), (2, 3), (3, 2)$, with no disagreement (`primitive_probes.py`). Taken together with the dependencies above, this does not license dropping readback from the right-hand side. The bare form

accessible non-Markovianity $\iff$ a per-horizon finite reversible deterministic realization exists

is **false**: the process-dilation construction of §3.4 realizes Markov laws as well, so the right-hand side is satisfied by processes with no memory at all. What the reduction does license is the *universal* statement — the theorem above — in which readback appears not as a hypothesis but as an unavoidable feature of every faithful realization. What C1–C4 supply is not hypotheses but a *diagnostic checklist* describing what such a realization looks like — which is precisely what the physical applications need, since one cannot inspect a cosmological horizon or a Boltzmann brain for "a realization exists" but can check coupling, capacity, readback and persistence directly. The substance of the theorem lives in the construction (§3.4) — the response-table realization, the hidden space growing with the horizon, the prior carried as realization datum — and is untouched by this reading.

Theorem (canonical predictive quotient — a purification ingredient, substratum form). *Let P be a finite-horizon law on visible trajectories. Define the tail object $T(P)$: hidden states the stage-indexed pairs (t, τ) of a time and a positive-probability length-$(K-t)$ visible future-tail, prior P itself at stage* 0,

dynamics the shift $(t, \tau) \mapsto (t{+}1, \tau_{2:})$*. Then: (existence)* $T(P)$ *completes* P*, and adjoining a history ledger — the emitted prefix, with each tail's chain closed into a cycle — makes it a reversible (bijective) realization; (universality) every faithful deterministic realization* R *of* P *carries, on its prior-supported reachable part, the canonical morphism* $q_R : (x, h) \mapsto (x,$ *its visible future-tail*$)$ *— a morphism of realizations here being stage-preserving, visible-preserving (*$\pi_V \circ q = \pi_V$*), and dynamics-intertwining (*$q \circ \varphi_R = \sigma \circ q$*,* σ *the shift) — which pushes the prior forward to* P *and is the unique morphism* $R \to T(P)$*; (canonical quotient)* $T(P)$ *is the canonical terminal predictive quotient of the category of faithful realizations — itself a staged predictor, bijective only once the ledger is adjoined — and any two faithful realizations of the same* P *induce the identical canonical quotient, so the predictive content of every completion is unique — while the hidden fibers above the quotient may differ: full completions are NOT claimed unique up to reversible relabeling, and same-law realizations with non-relabel-equivalent fibers exist (*`fiber_freedom`*, certified).* □ *Proof.* Determinism makes the visible future from (x, h) a single string, so q_R is well-defined; one step of φ shifts that string, giving the intertwining; faithfulness makes the pushforward P. Uniqueness of the morphism: visible-preservation propagated through the intertwining forces the image's visible trajectory to equal the source's, and a $T(P)$-state IS its remaining visible trajectory, so any morphism sends each state to its tail — without visible-preservation the claim would fail, since law symmetries can permute equal-probability tails; the definition excludes exactly that. Terminality: the same argument against any tail-injective target. Reversibility of the ledger extension: the prefix-and-cycle construction is a permutation — injective along each chain, chains disjoint, terminals rotated to roots. □ (All clauses certified exhaustively — intertwining, pushforward, canonicity across same-law realizations, quotient idempotence, ledger reversibility, fiber freedom, and the predictive floor $\log_2 |\text{tails}_t| \geq I(X_{<t}; X_{>t} \mid X_t)$ — over all $1\{,\}464$ φ-realizations at $(n_V, |\mathcal{C}_H|) = (2,2), (2,3), (3,2)$ under two priors, $2\{,\}928$ instances, in exact rational arithmetic except the predictive floor, which is checked in floating point: `purification_probes.py`.)

Remark (relation to the purification axiom). In the Chiribella–D'Ariano–Perinotti reconstruction [53], purification demands existence and essential uniqueness — up to reversible transformations of the purifying system — within the operational theory. The theorem above supplies the substratum-level existence half and, toward uniqueness, the canonical predictive quotient every completion shares — an ingredient, not the axiom: CDP's uniqueness (any two purifications related by a reversible transformation on the purifier alone) is strictly stronger, and the certified fiber freedom shows completions here are not unique as wholes. For the operational-reconstruction route, two questions remain: translate this substratum construction into the operational composite language, and determine whether quotient-level uniqueness is strong enough to play purification's reconstruction role. The result is an in-house ingredient toward that lift, not yet the all-level operational purification principle itself.

Remark (which condition is primitive). Of the three structural conditions, only (C4) is logically independent. (C1) follows from it — a readback gap requires hidden mediation, and zero-coupling realizations exhibit no gap at any order — and (C3) holds in every faithful realization by data processing applied to the whole visible future, with the bound $\log_2 |\mathcal{C}_H| \geq I^*$ attained by some realizations. Exhaustive enumeration in `primitive_probes.py` checks the first at small sizes in exact rational arithmetic, and checks the second in its per-order form $\max_k M_k \leq \log_2 |\mathcal{C}_H|$ over realizations with a readback gap, in floating point (§1.3). The equivalence could therefore be stated with (C4) alone. It is not, deliberately: (C1) and (C3) are independently *checkable* — on the adopted ontic-parameter-dependent branch of §3.3 Bell violation requires cross-wing coupling and so checks (C1) directly, and capacity arguments use (C3) directly — and a characterization that lists checkable necessary conditions alongside its primitive is more useful than a minimal one, in the way that axiom lists for groups are conventionally redundant. The dependency is recorded at each condition's definition (§1.3).*

Remark (two notions of memory). Non-Markovianity of the process — condition (ii) — and P-indivisibility of the transition-matrix family are distinct, and the distinction mirrors the quantum-side gap between process-tensor non-Markovianity and CP-divisibility (§3.2). Markov implies P-divisible (chain factorization), so P-indivisibility is the strictly stronger property; the strictness is witnessed by an explicit family — X_1 uniform independent of X_0, $X_2 = X_0 \oplus X_1$ — with maximal conditional memory ($I(X_0; X_2 \mid X_1) = 1$ bit) whose matrix family factors through $\Lambda = I$ (`review4_probes.py`; up to flipping the final bit, the process of [5], Eq. (15)). The framework's accessible-timescale claim is (ii). P-indivisibility enters as the stronger property in two ways: unconditionally at the recurrence scale from a non-permutation one-step witness (§2.3), and empirically — P-indivisible dynamics are documented in open-system experiments, the imported correspondence reads Bell-class dynamics as indivisible, and (ii) itself is certified by directly observed memory effects.

Proof. (iii) $\implies$ (ii): the readback lemma of §2.3 — (iii) includes (C4), whose gap forces $I(X_{<k}; X_{k+1} \mid X_k) \geq p_0 \delta^2 / \ln 2 > 0$ at an accessible order. (ii) $\implies$ (iii): existence, per accessible horizon containing the witness order k, by the finite-horizon process-dilation theorem above, which reproduces the full multi-time law; the realization inherits (C4) at k directly — the ledger-conditioned kernels reproduce $\mathcal{S}$'s differing past-conditionals exactly (`c4_backflow_probes.py`) — and satisfies the remaining conditions: (C1) because a Markov-breaking kernel dependence requires hidden influence at that step (with none anywhere, every visible transition is a fixed function of the visible present and the process is Markov, contradicting (ii)); (C2) trivially in the construction — the bath is static, $\tau_B = \infty$, satisfying $\tau_S \ll \tau_B$; for *physical* hidden sectors the physical-memory theorem's contrapositive (§3.4, the C2 process form) supplies the necessity direction — observed conditional memory bounds the mixing residual below with no mixing hypothesis (persistence form), and, conditional on the conditional-mixing-curve hypothesis with finite $\tau_B(\varepsilon_0)$, bounds the bath's

mixing time (timescale form) (Remark, §3.4; ETH [17, 18] the motivation; no smallness of ε at the coupling time is assumed); (C3) from the data-processing bound in per-process form ($m \geq 2^{I^*}$). Both directions are proved here in prose and neither is formalized in the kernel; since (C4) coincides with (ii) in every faithful realization (Remark, *the conditions are diagnostics, not hypotheses*), what (ii) $\implies$ (iii) adds to (ii) is the existence of a realization, and (iii) $\implies$ (ii) is the readback lemma. Finally, a unitarily evolving representation is available for $\mathcal{S}$ regardless — internally by the permutation unitary of any realization (the equivalence above), and under (T) by the imported correspondence [11, 12], which represents it by a unitarily evolving quantum system, one unitary for each time on dimension $\leq n^3$ as in (i) — the one external import of the quantum-representation layer, sitting outside the equivalence; the fixed-$\hat{H}$ form is the in-house one, for realized processes at integer horizon times (Lemma above). □

Remark (the horizon restriction is forced). The witness-anchored quantifier in (iii) is not a courtesy. Any order k satisfying (iii) is a memory witness: a (C4) gap at k in a faithful realization is a differing next-step conditional of $\mathcal{S}$ itself (universal theorem, (a)). The order is bound within (iii) and not taken from (ii): quantified over every order at which (ii)'s conditional mutual information is positive, (iii) would hold vacuously for a Markov law, which has none. A horizon ending before the first memory witness restricts $\mathcal{S}$ to a Markovian law, and by the universal theorem's biconditional no faithful realization of a memory-free law carries a readback gap — so the unrestricted every-horizon form of (iii) is false, while every horizon containing a witness supports the construction. The strong per-witness form is exact: $M_t > 0 \iff$ for every $K \geq t+1$ there is a faithful realization of the law through K with readback biting at t — the forward direction by dilation of the K-law, whose restriction contains the t-witness, the converse because a readback gap in any faithful realization *is* a differing next-step conditional (universal theorem, (a)).

Remark (retired direction). (i) alone does not imply (ii): a diagonal Hamiltonian satisfies the definition with $T(t) = I$ for all t — perfectly P-divisible — and a nearly degenerate $\hat{H}$ is permutation-close throughout an accessible window (`review3_probes.py`). The equivalence therefore runs (ii) $\iff$ (iii) with representability following for the source's finite-configuration class; (ii) is certified empirically by directly observed memory effects in open-system dynamics, and the further reading of Bell-class dynamics as temporally indivisible is the imported framework's — the necessity argument uses neither that reading nor any converse.

Remark (existential and universal readings). The equivalence is existential: (ii) holds iff *some* per-horizon (C1)–(C4) realization exists. The necessity analysis is universal and per-condition: any deterministic realization of a non-Markovian process has hidden influence at some step (C1), memory capacity $m \geq 2^{I^*}$ (C3), and read-back differing past-conditionals (C4); C2's necessity holds in its persistence form for every such realization and in its timescale (slow-bath) form within the conditional-mixing class. The two readings are logically independent.

Remark (representation, not explanation). The equivalence is a representation theorem: every accessible non-Markovian law *admits* a per-horizon C1–C4 realization. It does not, by itself, explain why the physical universe exhibits such a law; that explanatory burden — the specific substratum satisfying C4 quantitatively, with the horizon's capacity and timescales — is carried by the cosmological construction and tested there.

Remark (consolidation). The equivalence is per-horizon, and the universal all-time form is settled *negative*: a bijection on a finite set has finite order L, so any fixed realization with a fixed prior has an exactly L-periodic multi-time law ($\varphi^{k+L} = \varphi^k$) — a nonperiodic target admits no exact all-time realization by one fixed finite substratum. For the rooted visible maps the all-time question has an exact answer: on a finite $\mathcal{C}_V$, a family $(\Gamma_t)_{t\geq 0}$ of real square matrices equals, at every $t \geq 0$, the rooted maps $\Gamma_t(a,j) = \mu_H\{h : \pi_V\varphi^t(a,h) = j\}$ of some finite $\mathcal{C}_H$, one bijection φ of $\mathcal{C}_V \times \mathcal{C}_H$ and one probability distribution μ_H on $\mathcal{C}_H$ shared by every root, not required to have full support, if and only if $\Gamma_0 = I$, every Γ_t is row-stochastic, and $\Gamma_{t+M} = \Gamma_t$ for some $M > 0$ and all t (kernel: `pper_of_finiteRootedRealizable` in `OIBridge/RootedClassification.lean` for the forward direction, from finiteness and bijectivity; `pper_has_responseRealization` in `OIBridge/RootedClassificationAllTime.lean` for the converse, by a response-table construction; the two directions assembled in `finiteRootedRealizable_iff_pper`). The open questions are (a) exact all-time gluing for recurrent families of multi-time laws — which compatibility conditions on the per-horizon realizations suffice — and (b) the physically operative form: quantitative approximation of the target on windows $t \ll t_R$ by one fixed substratum before its recurrence. Form (b) coincides with the accessible-timescale scoping of §4.5.

Remark (scope: time-homogeneity). The restriction to time-homogeneous processes is load-bearing for the fixed-$\hat{H}$ form. A non-autonomous quantum evolution such as $U(t) = e^{-i\sigma_x t^2}$ (generated by $H(t) = 2t\,\sigma_x$) has P-indivisible computational-basis populations, but $\cos^2(t^2)$ is not almost-periodic and therefore not any fixed-$\hat{H}$ ancilla-marginal — (ii) without homogeneity does not deliver a fixed-$\hat{H}$ form. Autonomy is not homogeneity: a single fixed bijection applied every step can still produce time-dependent visible kernels through a hidden clock — the §3.4 construction is the explicit example. Homogeneity is an additional property of the visible process; the imported correspondence assumes no homogeneity and supplies one unitary for each time, with a time-dependent $\hat{H}(t)$ where $U(t)$ is differentiable [11, 12], not a fixed generator. The fixed-$\hat{H}$ form itself is constructive for realized processes — the realization's permutation unitary with $\hat{H} = i \log U$ (Lemma above) — at the horizon's integer times; the almost-periodicity obstruction of this Remark concerns the all-real-time reading.

At the level of finite-horizon fixed-basis observable laws, embedded observation and unitary Born-rule representation are not merely *compatible* — they are mathematically *equivalent* ($S \iff D \iff Q_{\mathrm{fb}}$, the theorem above). The scope restriction is the content: the equivalence identifies the representation

class exactly and stops short of operational quantum mechanics: the instrument algebra is reached, on a finite carrier within complex matrix kinematics taken as given, exactly when the five operational completion conditions hold (Remark, the operational-extension boundary; Remark below), none of them supplied by the fixed-basis layer, and cross-horizon representation consistency is bounded by the classical-dimension obstruction.

Remark (scope: representability, not selection of the relating evolution). The equivalence $S \iff D \iff Q_{\mathrm{fb}}$ is a representability theorem at the finite observable-law layer: it fixes which laws admit a fixed-basis unitary Born representation, not which unitary evolution relating two times is the physical one. For dilated processes that distinction has content. One visible family can admit distinct coherent unitary lifts — each reproducing the visible law at every time, each composing exactly across times — whose relating evolutions differ, because a lift may be deformed by a time-dependent right action that preserves everything the visible law sees. Ordinary coherence therefore does not by itself make the relating evolution unique. A deformation constant in time cancels from every relating evolution, so the nonuniqueness is carried by time dependence specifically. Uniqueness needs a selection condition beyond visible-law representability and ordinary coherence; this paper neither supplies one nor says what form one must take. The uniformly invisible deformations of a lift are two-sided: the time-dependent right action just described, by the anchored class, and a left action by unitaries acting within each visible output's hidden fibre; each is maximal among uniform actions of its kind, and neither is seen by the visible law. Invisibility does not exhaust the freedom: two lifts of one visible family can fail to be related by any uniformly invisible right deformation and still agree on every relating evolution, so lift freedom and freedom in the relating evolution are not the same freedom; and a pair outside that right class can differ in its relating evolution while being related by a left in-fibre move, so quotienting by the right class alone does not determine the relating evolution. At a single time the residue after the two-sided action is classified exactly: two lifts are related by it exactly when, for each visible output, the Gram matrix of the hidden states into which they send the visible inputs — the fibre-Gram data — agrees up to the anchored phases; the visible law is exactly the diagonal of that data, and every positive semidefinite, rank-bounded, identity-summing family with the prescribed diagonal is realized. So what the visible law leaves free at one time is exactly the off-diagonal fibre-Gram data modulo phases, and that freedom is neither empty nor inert: on a single visible law two lifts can lie in different two-sided classes and differ in their relating evolution. At one time that residue is not seen by the fixed-basis layer. Take a lift with trivial ancilla at one time and form the fixed-basis datum whose evolution is the lift's unitary, with uniform initial law and identity readout: its finite-horizon trajectory laws are determined by the visible law alone, and remain so after the unitary is composed on either side with contractively scaled partial permutations; two such data have equal trajectory laws exactly when their visible laws are equal; and each such law is fixed-basis realizable and stochastic, hence in-

side $S \iff D \iff Q_{\mathrm{fb}}$ (kernel: `bridge_traj`, `rooted_eq_iff_slice_eq`, `realData_traj_stochastic`). The fixed-basis correspondence therefore acts on the quotient by visible equality, and two lifts of one visible law in different two-sided classes are not separated by it. The trivial ancilla is part of the statement: an ancilla attached as a tensor factor with its own unitary leaves the rooted family unchanged (kernel: `padData_rooted`), whereas an ancilla carried between steps as hidden basis states, in a lift that is not a tensor product, can separate two lifts of one visible law that share every column entering the visible law, already in the two-step law (exact computation). Across time the question is two-part, and both parts are open: what selects or constrains the trajectory of fibre-Gram data through time, and what fixes the relating evolution within a given trajectory — the second not fixed even by the complete trajectory, since a time-dependent invisible right deformation preserves every fibre-Gram matrix at every time and changes the relating evolution. The fibre-Gram data is a coordinate on the space of lifts, not a physical quantity, and nothing here says what selects it. None of this alters $S \iff D \iff Q_{\mathrm{fb}}$ as the exact finite observable-law correspondence, and none of it makes embedded observation and quantum mechanics inequivalent: the statement is that the visible law does not by itself fix the relating evolution. **This is an open frontier of the correspondence, not a local technical caveat.** Three statuses should be kept apart: finite observable-law representability is *established*; exact finite operational quantum mechanics is *characterized conditionally*, within complex matrix kinematics taken as given and under the explicit completion principles of the operational-completion characterization of §3.4, none of which bare embedded observation supplies; and selection of a unique relating evolution from the coherent lifts of one visible family is *open*. This paper neither closes the third nor claims it cannot be closed.

Remark (scope of the equivalence). The characterization theorem, with the finite-horizon equivalence, supplies Hilbert-space representations with unitary dynamics and Born-form transition probabilities: internally the fixed-basis permutation-unitary representation of the finite-horizon visible law, and under (T) the compressed form of (i) for its transition data. Items (i)–(iii) comment on further structures of operational quantum mechanics, item (iv) on classical P-indivisibility, and item (v) on the external dependencies.

(i) Tensor product structure (visible–hidden). The Stinespring route (§3.2) constructs $\mathcal{H} = \mathcal{H}_V \otimes \mathcal{H}_H$ and derives the CPTP channel Φ by partial trace. The visible–hidden tensor product is therefore given by the construction, not postulated.

(ii) Tensor product structure (visible-sector subsystems). Spatially separated visible regions $V_A, V_B \subset V$ inherit an exact **dynamical causal separation** from the dependency graph: after k updates, a local state can depend only on the graph ball of radius k (§3.3). This establishes the causal-cone part of locality without the stronger simple-cubic nearest-neighbor stipulation. It does not alone establish statistical product structure after conditioning, local

tomography, or one common tensor-product instrument category. Bell adds a sharper restriction: if the setting interventions, applied at one step, each read and write only their own wing, each wing's readout lies outside the other wing's graph cone, a ± 1 outcome is recorded in every run, and the pre-setting ensemble is measurement-independent, determinism gives Bell factorization and $|S_{\mathrm{CHSH}}| \leq 2$. The framework therefore does not use graph locality as the hidden-variable explanation of quantum Bell violation; it adopts ontic parameter dependence while requiring operational no-signaling.

(iii) State update, measurement, and multi-time predictions. Projective measurement in the emergent description corresponds to conditioning on a visible-sector outcome and re-marginalizing over the hidden sector. The Lüders rule $\rho \to P_k \rho P_k / \mathrm{Tr}(P_k \rho)$ transcribes Bayesian updating on the classical substratum (Lemma 3) for the represented diagonal sector — which is the transition-statistics layer, not the instrument algebra: the general operational update for noncommuting interventions belongs to that algebra, reached, on a finite carrier within complex matrix kinematics taken as given, exactly under the operational completion conditions (§3.4); its extension to arbitrary coherent density matrices belongs to the operational bridge (§3.2). No independent measurement postulate is required for the represented statistics. Multi-time correlation functions of the passive law are standard calculations within the delivered formalism ($\mathcal{H}_V$, $U(t)$, Born rule); weak values and Leggett-Garg protocols are intervention-rooted and belong to the intervention dilation (§3.4 Remark; §3.2 target) — the delivered formalism underdetermines them. The multi-time statement carries one qualification. The correspondence is engineered to reproduce rooted two-time transition data, which under-determines multi-time joint statistics [22]; the substratum, by contrast, fixes a unique multi-time law. Agreement of the substratum's conditioned multi-time statistics with the quantum formalism's intervention-rooted predictions depends on the open dynamical question of hidden-sector mixing recorded as ledger item (v) (§4.5); hidden-sector mixing is not the Bell-violation mechanism: the adopted Bell branch relinquishes ontic parameter independence (§3.3). Residual pre-division memory, if present, would appear as deviations from quantum multi-time predictions — a falsifiable texture of the same standing as the residual dissipation $\mathcal{D}$.

(iv) Classical P-indivisibility and quantum mechanics. The representation clause is not a claim that all P-indivisible processes are "secretly quantum" in some metaphysical sense. Classical systems exhibiting P-indivisibility — renewal processes, semi-Markov processes on finite state spaces — admit a unitary description, constrained, under the phase-locking hypotheses (G1)–(G4), up to shift, rephasing and the antiunitary conjugate, whose consequences separate into two layers: continuous transition data constrain *generator structure* beyond the raw data up to that gauge — energy-gap structure, under the phase-locking hypotheses — while *operational* interference and entanglement predictions require in addition the instrument and observable structure of the operational bridge, characterized conditionally and with its physical sourcing open (§3.2, §3.4). The processes can be realized as marginalization of deterministic dynam-

ics under C1–C4 (necessity theorems). The framework locates the observer's process inside the correspondence's domain, where the imported theorem supplies a unitarily evolving representation for every finite-configuration member — including divisible and Markovian ones. P-indivisibility does not gate that supply; it distinguishes representations whose dynamics is genuinely indivisible from those that are trivially unistochastic (§3.4). The physical content of the unitary is not that of a mere mathematical embedding: under the phase-locking hypotheses (G1)–(G4), its Hamiltonian is determined by the transition data up to the stated gauge freedom — energy shift, rephasing, and the antiunitary conjugate $H \to -H^*$ (phase-locking lemma) — arising from the *actual* dilation structure of the system (§3.2).

(v) The external dependencies, scoped. The quantum-representation layer imports one external result: the Barandes stochastic-quantum correspondence [11, 12], which establishes that every finite-configuration process in its class — a broad class in which Markov chains are special cases — can be regarded as a subsystem of a composite unistochastic process on dimension $\leq n^3$, so that the stochastic and unitary descriptions are inter-translatable in compressed form. The import is load-bearing only where the internal results stop: existence of a fixed-basis unitary Born representation is internal (the equivalence theorem — the permutation unitary of any realization), as are the dilation construction, the per-time ancilla-marginal form on dimension n^2 (§3.1), the C1/C3/C4 necessity theorems, and the C2 physical-memory bound (persistence form without a mixing hypothesis; slow-bath form conditional on the mixing hypothesis, which ETH motivates); the import retains the interpretive apparatus that travels with it, under (T), the $\leq n^3$ bound standing as the source's own result. This gives the dependency a clean falsification profile: if the correspondence were shown to be incomplete or to require additional hypotheses, the framework would lose the interpretive apparatus — and nothing else, since existence, the per-time form on dimension n^2, equivalence, and necessity are internal. Skepticism about the import therefore bears on the interpretation claims only; the framework's own results do not route through it. Beyond the textbook results [15, 16], two further external results are imported where they are used: the result of [55] that for every N some finite qubit prepare-and-measure experiment has no N-state model reproducing it exactly, which supplies the unbounded-dimension half of the finite-substratum operational obstruction (§3.4), and canonical typicality [31], which the structural observer-selection theorem of §4.6 uses. The premises collected in the posit ledger (§4.5) are declared inputs, not imports.

3.5 Generative and limitative faces of the observation map

The characterization theorem establishes what embedded observation *produces*. The same trace-out map has a complementary structural property — what it *fails to fix* — and stating it makes precise a limitation that recurs in the gravitational and Standard-Model sectors (the residual-input status of spatial isotropy

in the framework's gravitational and Standard-Model treatments).

Lemma (rest-frame selection). *Let the embedded-observation partition $S = V \sqcup B$ be realized across a causal horizon $\mathcal{H}$, with induced coarse-graining map T. The horizon fixes a simultaneity foliation of the substratum, hence a distinguished timelike direction n up to sign (the frame in which the update steps are simultaneous); and let T determine this foliation, so that any Λ leaving T invariant maps the foliation to itself. Then the invariance group G_T of T within the local frame transformations $O(3,1)$ is contained in the stabilizer of the timelike line $\mathbb{R}n$, which is isomorphic to $O(3) \times O(1)$, and its orthochronous part $G_T \cap O^+(3,1)$ is contained in the stabilizer of n, the little group $O(3)$ of a timelike vector. Consequently, under this hypothesis, boosts, which move n, are not invariances of T. Which transformations fixing n are invariances of T is not fixed by n; it depends on the substratum dynamics and on the lattice geometry of the partition.*

Proof. T is defined by the horizon partition, which fixes the simultaneity foliation and hence n up to sign. By hypothesis a frame transformation $\Lambda \in O(3,1)$ that leaves T invariant maps the foliation to itself, i.e. $\Lambda n = \pm n$. These transformations preserve the spacelike complement $n^\perp$ and act on it orthogonally, so they form a group isomorphic to $O(3) \times O(1)$. An orthochronous Λ maps n, taken future-directed, to a future-directed vector, so $\Lambda n = n$; the stabilizer of a timelike vector in $O(3,1)$ is isomorphic to $O(3)$ (the standard little-group classification). A nontrivial boost relative to n maps n to a timelike vector other than $\pm n$ and so is not an invariance. Fixing n is necessary for an orthochronous invariance, not sufficient: the hypotheses constrain T only through n, and the inference from fixing n to invariance fails already for lattice maps. A nearest-neighbour step kernel on the cubic lattice, with spacetime increments $n \pm e_i$ each of weight $1/6$, selects n: its invariance group within $O(3,1)$ is exactly the octahedral group O_h of order 48, which fixes n and no nonzero spatial vector. □

Corollary (boost/rotation inequivalence). Under the lemma's hypothesis, because the horizon partition defining T selects the timelike direction n but no spatial direction, the emergent theory inherits from the partition a physical rest frame and no physical spatial axis; whatever spatial axes it carries come from the substratum dynamics and the lattice geometry of the partition. Lorentz-violating operators therefore have structurally inequivalent status by sector: boost-sector violation is tied to the map-selected object n (and is correspondingly exposed to map-level structure — the local-boost-invariance prerequisite of the framework's horizon-thermodynamic construction, the custodial mechanism of its lattice Standard-Model construction), whereas rotational-sector violation is referred to no map-selected object and is inherited from the substratum dynamics and the lattice geometry of the partition. This is the structural origin of the boost/rotation asymmetry recorded in the framework's Lorentz-violation accounting; it explains the asymmetry without resolving either sub-sector.

Converse. *If the orthochronous invariance group $G_T \cap O^+(3,1)$ is a maximal*

compact subgroup of $O^+(3,1)$, then it is the stabilizer of a unique future-directed unit timelike vector n, so T selects a rest frame; combined with the lemma, such a group equals the stabilizer of the horizon's n. This is the standard maximal-compact-subgroup structure of the Lorentz group: the maximal compact subgroups of $O^+(3,1)$ are the conjugates of the rotation group $O(3)$, and they stand in bijection with the points of the rest-frame space $O^+(3,1)/O(3) = H^3$, each fixing exactly one future-directed unit timelike vector. In the full group $O(3,1)$ the maximal compact subgroups are the conjugates of $O(3) \times O(1)$, each preserving exactly one timelike line. The statement is one-way, and its hypothesis is not established for the coarse-graining map of the horizon partition, since the lemma bounds G_T from above only. Rest-frame selection does not force the full little group: the cubic step kernel selects n with invariance group O_h, a proper subgroup of $O(3)$. Which rotations fixing n are invariances of T, and whether the emergent description is spatially isotropic, are open; spatial isotropy enters as empirical input E4 (posit-ledger item (vi) of §4.5).

The two faces. The characterization theorem and this lemma are two properties of the *same* map T: what it produces (its image: the quantum kinematics) and what it leaves invariant (its symmetry, under the lemma's hypothesis: contained in the transformations preserving the line $\mathbb{R}n$, with no boosts). They are not opposite operations (generation versus limitation) but the *same* operation, a projection of the observed statistics under information missing to the embedded observer; what differs is the *type* of missing information. In the quantum case the missing information is the hidden-sector *state*, which is dynamically resampled (the bath thermalizes) and therefore fluctuates; averaging over a fluctuating quantity is productive and yields the memory-bearing non-Markovian statistics that quantum mechanics represents. In the isotropy case the "missing" information is the *orientation* of the lattice constitution, a fixed global parameter that is not resampled; there is no ensemble to average over, and the observer's ignorance of the lattice axes is not a statistical mixture. The discriminant is whether the missing information fluctuates: fluctuating missingness is averaged into emergent structure, fixed missingness is inherited. The quantum/isotropy contrast is attributed to fluctuating-versus-fixed missingness, not to a dichotomy of operations.

Two qualifications bound this subsection. First, the lemma is one-directional and near-elementary (it follows from the little-group classification once the horizon is granted to select n and T to determine the horizon foliation); it is far less deep than the characterization theorem — proved in-house, the imported correspondence retaining only the compressed form under (T) (§3.4) — and it is labelled a lemma for that reason. Second, it is a structural statement about the map's reach, not a derivation of a physical law: it explains *why* the framework can constrain the boost sector and must take spatial isotropy as input, but it neither supplies that isotropy nor forbids its violation. (Whether the fixed orientation could, in some régime, be effectively resampled — restoring isotropy generatively with a calculable residual anisotropy of the $\ell = 4$ cubic-harmonic form — is a separate open question pursued in the book's open-problems treat-

ment; the present lemma is agnostic on it.)

4.1 Interpretive Consequences

The degrees of freedom involved in quantum experiments — photons, electrons, slits, detectors — are all visible-sector objects. Their quantum behavior is a downstream consequence of the trace-out: the quantum representation, once established by the theorem, is available for all visible-sector dynamics — the operational theory's unique selection remaining open (§3.4).

In the double-slit experiment, a finite set of apparatus settings — both slits open, one slit closed, a which-path record coupled — with the detection statistics over finitely many outcomes under each setting defines a finite-horizon action-labelled family, the settings being its action alphabet. By the intervention dilation (§3.4), every such family has a finite reversible realization with one fixed prior, in which the substratum occupies a single configuration at every step and the visible transition law is indexed by the setting; coupling a which-path record is an intervention that changes it. The detection statistics are an input of that construction, and a derivation of the two-slit interference pattern from the substratum dynamics is open. Within the framework's ontology the Everettian measure problem does not arise: the system evolves as a single reality; "branches" are features of the compressed description. The Born rule, often treated as an independent postulate, is here part of the exact fixed-basis representation dictionary ($S \iff D \iff Q_{\mathrm{fb}}$, §3.4), rather than an additional stochastic axiom. What remains open is the common coherent operational extension of that dictionary to all interventions and composites.

Observer covariance and nested observers. Different observer partitions $V \subsetneq S$ yield their respective visible stochastic laws and fixed-basis quantum representations. The construction below relates two such marginals to one joint law of the same substratum. This supplies an observer-relative account of the recorded information; a full nested-agent protocol retains its separate operational and consistency premises.

Consider the Wigner's-friend setup: two disjoint partitions $V_W, V_F \subset S$ (Wigner and Friend) with $V_W \cup V_F \subsetneq S$, and shared external hidden sector $H = S \setminus (V_W \cup V_F)$. The framework assigns three descriptions, all derived from the same φ by the same trace-out procedure:

- **Friend's description.** Trace over $S \setminus V_F = V_W \cup H$. The framework's proposal is that, under C1–C4 applied to $(V_F, S \setminus V_F)$, Friend's reduced description is quantum-mechanical; what is established is that it is a finite-horizon law admitting the fixed-basis quantum representation of §3.4 whether or not the conditions hold, with memory where (C4) holds. Friend observes definite outcomes in her visible sector, including any system she measures.

- **Wigner's description.** Trace over $S \setminus V_W = V_F \cup H$. Friend is part of Wigner's hidden sector. The proposal and the established statement of Friend's case apply equally to Wigner's reduced description, for $(V_W, S \setminus V_W)$. Wigner does not see a definite Friend-outcome within his own description — Friend's state is traced out.
- **Joint description.** The marginal transition matrix $T_{WF}(\alpha, \beta \to \alpha', \beta') = |\{h \in H : \pi_{V_W \cup V_F}(\varphi(\alpha, \beta, h)) = (\alpha', \beta')\}|/|H|$ exists and is unique given φ. Its marginals reproduce Wigner's T_W and Friend's T_F respectively.

The three descriptions are simultaneous rather than competing. Friend's claim that she has a definite outcome, and Wigner's description of Friend as a traced-out subsystem, are statements in different reduced descriptions, not conflicting claims about the substratum state. At the substratum level, Friend, Wigner, and the system all have definite configurations; neither Friend nor Wigner has epistemic access to the complete state, and this epistemic limitation produces their respective reduced descriptions through the same mechanism in §2–§3. This gives an observer-relative interpretation of the common substratum and its marginal laws. It does not by itself resolve the full Frauchiger–Renner protocol: the proposed completion's operational probabilities, probability-one certainty inferences, consistency and single-outcome premises must be assessed separately. Multiple partitions alone neither eliminate the joint law above nor establish failure of a Frauchiger–Renner premise.

4.2 Scope

The characterization theorem's equivalence is (ii) $\iff$ (iii), with (iii) carrying (C1), (C3) and (C4) and (C2) outside it, both directions in-house; the quantum representation follows for the source's finite-configuration class, and directly observed memory effects in open-system dynamics certify (ii), Bell data bearing instead on the spatial-composite branch (§3.3); representability alone is strictly weaker (diagonal-$\hat{H}$, §3.4 Remark). The theorem does not identify which physical systems satisfy the conditions; this is an empirical question.

4.3 The Status of the Discreteness Scale

ϵ does not smuggle in a quantum assumption, but the reason is foundational rather than field-theoretic. Finiteness of the *visible* configuration space is a lemma of the observation axiom itself: a single tokened differentiation registers determinate — and therefore finitely structured — content, so what is distinguishable at the locus per moment is finite; invoking holographic bounds at the foundational layer would be circular, those being downstream physical results. Finiteness of the hidden sector is posited in the forward construction, as item (ii) of the posit ledger (§4.5), with the deep-sector scope stated there; it is not derived from Lemma 1, whose argument concerns what is distinguishable at the locus per moment, and a boundary-entropy bound on the boundary layer $\mathcal{C}_B$

would rest on the holographic input that this layer excludes. In the reconstruction direction it is *constructed* rather than assumed (§3.2's dilation supplies a finite hidden space). Holographic entropy bounds [2] are compatible with the finiteness assumed here and are not premises of it.

4.4 Relation to Prior Work

't Hooft [1]: Differs in mechanism (epistemic trace-out vs. information loss), generality (any embedded observer vs. particular rules), and Bell placement (ontic parameter dependence with measurement independence retained vs. superdeterminism). **QBism/relational QM:** Share observer-relative epistemic states; this framework provides a structural *why* and quantitative predictions.

Relation to prior stochastic-quantum programs. Three earlier programs sought to derive quantum mechanics from classical-looking substrates. **Nelson's stochastic mechanics [32]** postulates a continuous Brownian-motion-like process on configuration space with drift determined by the imaginary part of the log wave function; the Schrödinger equation follows from the Fokker-Planck structure. Wallstrom [33] showed that Nelson's program cannot uniquely select the quantization condition $\oint p \cdot dq = nh$ — the stochastic process admits a broader class of solutions than standard QM, and the quantization must be reimposed by hand. The present framework is not subject to this critique because it does not derive QM from a continuous stochastic process on ψ or on configuration space; its exact identification is of the quantum REPRESENTATION — universal at the fixed-basis layer, with the common coherent operational lift stated separately — and proceeds through the characterization theorem for embedded observers on a deterministic bijection, with stochasticity emerging from the trace-out over the hidden sector rather than being fundamental. **Adler's trace dynamics [34]** derives QM from statistical mechanics of matrix-valued dynamical variables with a trace-class Hamiltonian, with the canonical commutation relations emerging as expectation values over a specific equilibrium ensemble. The framework here requires neither matrix-valued variables nor an equilibrium-ensemble derivation; the substratum is a finite configuration space with a deterministic bijection, and the emergence mechanism is the Stinespring dilation applied to observer-marginalized transition probabilities. **Barandes's stochastic-quantum correspondence [11, 12]**, used as a technical tool in §3.1, is not a competitor but a building block: the correspondence maps its finite-configuration tuple class — a broad class in which Markov chains are members and P-indivisibility marks the nontrivially indivisible dynamical sector, not a membership criterion — to unitary quantum evolutions in compressed ($\leq n^3$) form under (T); the present framework supplies the physical mechanism (embedded observation under C1–C4) that produces the memory-bearing processes occupying that nontrivial sector — P-indivisibility in the stronger regimes established separately — and supplies the fixed-basis representation itself internally ($S \iff D \iff Q_{\mathrm{fb}}$, §3.4), retaining the import for compression and interpretation. The three earlier programs identified emergent QM as a legitimate

research target; what the present framework adds is an exact finite-horizon representation equivalence, $S \iff D \iff Q_{\mathrm{fb}}$, holding for every finite law, a characterization (ii) $\iff$ (iii) whose conditions are diagnostics of a realization, (C4) coinciding with (ii) in every faithful realization (§3.4), and a constructive derivation chain that does not require independent quantization conditions or equilibrium ensembles.

Relation to the post-2022 observer-essentiality wave. A substantial body of work since 2022 has independently established that incorporating observers as physical degrees of freedom produces structural changes in the formalism of quantum field theory in gravitational settings, with results that converge on the foundational claim of the present framework — that observation cannot be treated as an external operation on physics — while differing in mechanism. **Chandrasekaran, Longo, Penington, and Witten [35]** showed that adding an observer to QFT in a gravitational subregion promotes the von Neumann algebra of observables from type III to type II_∞, with a well-defined trace and entropy; Witten [36] extended the construction to de Sitter space. **Maldacena's recent dS observer-cancellation [37]** demonstrates that the unphysical phase of the de Sitter sphere partition function cancels exactly when the observer-with-clock degrees of freedom are properly incorporated, removing a long-standing obstruction to a clean dS thermodynamic interpretation. **Harlow, Usatyuk, and Zhao [38]** argue that the closed-universe Hilbert-space dimension is determined by the observer's degrees of freedom rather than being a free parameter; the **AAIL construction [39]** provides a complementary calculation in which the relevant dimension scales as $d_{Ob} \cdot e^{2S_0}$, refining HUZ's heuristic bound. **DEHK partition-relativity [40]** establishes that the algebra of observables is partition-dependent and that physical predictions are encoded in equivalence classes under partition refinement, paralleling the present framework's commitment to partition-defined observation. **Slagle and Preskill [41]** constructed an explicit emergent-QM model in which boundary quantum dynamics arise from a classical bulk lattice with stochastic dynamics, demonstrating the logical possibility of QM-from-classical at the cost of an exponentially long extra spatial dimension and explicit stochasticity.

The present framework converges with these programs on the foundational substance — observation is not external to physics; the algebra of observables depends on the observer; the partition structure carries physical content — while differing in mechanism and scope. CLPW obtain a type II algebra by adding observer degrees of freedom to QFT; the present framework derives the characterization of embedded observation directly from the deterministic substratum. Its observable algebra on each finite region is the full matrix algebra on the region's configuration space, and its infinite-region completion is a quasilocal C^*-algebra with no state or representation selected (§3.4). No von Neumann closure of that algebra, no relation to CLPW's observer-dressed algebra and no identification of CLPW's entropy with a substratum quantity is derived; whether the construction reproduces CLPW's algebra and its entropy is open. Maldacena requires an explicit observer-with-clock; here the partition trace-

out is the *defining* operation, so the observer is present from the outset, but the de Sitter sphere partition function and its phase are not computed, and whether the trace-out reproduces the cancellation is open. HUZ argues that dimension is observer-determined; in the present framework the representation dimensions are finite counts fixed by the partition — the permutation unitary acts on $\mathcal{H}_V \otimes \mathcal{H}_H$, spanned by $\mathcal{C}_V \times \mathcal{C}_H$ (§3.2) — and AAIL's $d_{Ob} \cdot e^{2S_0}$ scaling is not derived. Slagle-Preskill demonstrates the logical possibility of QM-from-classical with an extra spatial dimension and stochastic dynamics; the present framework uses neither — deterministic dynamics on a finite substratum, no extra dimension — and obtains an exact fixed-basis representation equivalence on every finite horizon ($S \iff D \iff Q_{\mathrm{fb}}$, §3.4). A continuous-time Hamiltonian law is additional structure (§3.2), and exact finite operational quantum mechanics follows only with the added principles of the quantum-complete extension (§3.4). The present framework's distinguishing feature is that it derives this content from a finite deterministic substratum.

Witten's Type II algebra program achieves finite entropy without finite-dimensional Hilbert spaces; this is compatible, since the framework's predictions depend on finite entropy (the physical content of Lemma 1), not on dimensional finiteness per se.

4.5 Assumptions, Limitations, and Falsifiability

The theorem requires Lemma 1, the stochastic-quantum bridge (§3.1 and §3.2; two routes reach its transition-statistics layer, neither reaching the operational instrument algebra), and genericity conditions (the phase-locking hypotheses (G1)–(G4): non-degenerate spectrum and gaps, non-vanishing overlaps, triangle genericity), which hold for all but a measure-zero set. The P-indivisibility proof uses total finiteness (the total-finiteness posit, §1.2 Remark, §4.5 ledger) via the recurrence argument ($\varphi^N = \mathrm{id}$). The readback lemma (§2.3) provides the quantitative route to accessible non-Markovianity, with P-indivisibility retained at the recurrence scale, establishing observable conditional memory on the timescales accessible to laboratory experiments — physically indistinguishable from a timeless identity for all experiments well below the recurrence time.

Falsifiability. The framework's structural claim is that *the hidden sector is a classical substratum* — a finite set with a deterministic bijection, finiteness being the total-finiteness posit (§1.2 Remark, §4.5 ledger item (ii)). Bäcker, Beyer, and Strunz [28] classify the memory of a two-step open-system process as classical when the second step can be written as a quantum instrument realizing the first step, followed by channels conditioned on the instrument's classical outcome. The classification is a property of the process maps on all density matrices; applying it to the substratum therefore requires the coherent-preparation and instrument bridge of §3.4, which is open. The substratum alone does not fix the classification. Restricted to inputs diagonal in a fixed basis, every two-step process admits a classical-memory decomposition. On coherent inputs the permutation unitary of §3.4 carries visible coherence into the hidden register, and

the classification depends on the realization rather than on the visible law: the law $X_1 = 0$, $X_2 = X_0$ has a ledger realization (§3.4) with classical memory and a swap realization, $\varphi(x, h) = (h, x)$ with an initialized hidden slot (§1.2), whose two-step maps are $\rho \mapsto |0\rangle\langle 0|$ and $\rho \mapsto \rho$. Every Kraus operator of the first map has one-dimensional range, so every processing conditioned on its outcome is entanglement-breaking, and the identity map is not; this swap realization admits no classical-memory decomposition. By the gluing and uniform-continuum theorems of §3.4, a finite reversible realization reproduces the statistics of any finite quantum-memory experiment to any stated accuracy. A memory-witness experiment [28, 29, 30] is interpretable as a test of the ontology only given two inputs not supplied here: the coherent-preparation and instrument bridge, and an outcome-independent rule identifying the fundamental open-system processes — those arising from embedded observation — as distinct from engineered quantum environments. It is a falsifier only given a third: a constraint, derived from the completed model, on what eligible processes must show — every eligible realization classical-memory in the sense of [28], or another explicit witness bound. The swap realization shows that the bridge alone does not supply such a constraint, and none is established here; no memory-witness outcome therefore falsifies the ontology at present. Were such a constraint established, the test would be sharpest in the *C2-marginal regime* ($\tau_S \sim \tau_B$ rather than $\tau_S \ll \tau_B$), where non-Markovianity is $\mathcal{O}(1)$ rather than suppressed; candidate systems are strongly-driven superconducting qubits with two-level-system baths, biological electron transport with vibrational coupling, and spin-boson dynamics at strong coupling, with the classical-vs-quantum memory distinction surveyed in [30].

Theorem-level falsifiability. The characterization theorem is mathematical — the fixed-basis representation and the hidden-memory theorem are internal (§3.4), with Barandes retained for compression and Stinespring as supporting structure — so no experiment falsifies the theorem itself; what is empirically falsifiable is the framework's physical identification of nature with its deterministic finite ontology. No ontology-level memory falsifier is established until the completed model supplies the constraint on eligible processes stated above.

Laboratory tests of the characterization theorem may be possible in tabletop systems where the hidden sector capacity is tunable.

The posit ledger. The two-axiom presentation of §1.2 names the foundation's primitives; the full argument consumes more inputs than two, and they are collected here so they are countable in one place. The list is complete in one specified sense: it is the audited inventory of the explicit premises that the derivations of this paper consume, including those it takes from the companion substratum construction, as these texts stand in the same revision of the repository. It does not enumerate every physical assumption one might make; the named hypotheses of the matter, gauge and gravity sectors are carried where they enter; and the empirical motivation recorded for an item does not make that item logically necessary. The framework's substantive posits, beyond Axiom 1 (tokened differentiation, indubitable), are: (i) Axiom 2 — recurrence — owned as a posit with

a non-derivability countermodel; (ii) total finiteness of the bijective representative of the status Remark of §1.2 — Lemma 1 delivers finiteness of the observer's distinguishable states, and the extension to the whole representative is A1 of the companion substratum construction — there scoped as the gauge choice of the minimal finite representative, the cardinality of a deep sector being unobservable at the fixed visible/hidden interface when that sector is a finite factor prepared independently in a normalized state and evolved by its own permutation, with product dynamics, which leaves every reduced visible channel, and hence every transition probability, unchanged (deep-sector enlargement, developed there) — consumed here by the recurrence step of §2.3 and by §4.6, whose conclusions hold on the finite representative; their transfer, where expressible through the transition probabilities, to every representative reached from it by relabelling the hidden states, by such a finite decoupled enlargement or, under the alphabet-size hypothesis that systems with the same coupling structure and dynamics class differing only in alphabet size have the same emergent transition probabilities, by a change of alphabet that keeps the coupling structure and dynamics class is stated and proved in the companion substratum construction, a deep sector that the dynamics couples to the remaining degrees of freedom being outside that transfer (status Remark of §1.2), while the recurrence-scale conclusion of §2.3 is a statement about finite representatives, the remaining uses of the posit being listed in the status Remark of §1.2; (iii) the record-preservation and eventual-image arguments that ground determinism (Lemma 3) close the hidden-sector margin using one *empirical* input — the observed statistics, in the uniform-prior doubly stochastic scope — and one structural principle, the restriction of a finite map to its eventual image, which covers registered histories whose states lie in that image, so bijectivity is part-empirical rather than a pure lemma (a reader who declines the recovery may instead adopt reversibility as the standard information-conservation postulate of unitary quantum mechanics and classical Hamiltonian flow [42, 43], at no cost to the downstream results — see the remark following Lemma 3); (iv) the measure selection is a maximal-entropy principle ("the maximal-entropy invariant measure compatible with the observer's records") — a standard but substantive epistemic postulate, not derived; (v) the hidden-sector mixing hypothesis (ETH-motivated) conditioning the slow-bath form of the C2-necessity direction, for a system class (emergent descriptions of linear-mod-q automata) on which the standard ETH evidence base is silent, and whose exact linear rule conserves a quadratic invariant for every symmetric translation-invariant coupling and, at prime q not dividing the lattice size, for $d \geq 2$ splits its largest energy shell into exponentially many cycles [SM §4.2], so that numerical motivation comes from the rounded candidate of [SM §4.2], which behaves chaotically at the sizes measured, while whether the realized hidden sector satisfies the mixing hypothesis remains open — this is the same open dynamical question as the typicality selection in (iv), one conditional wearing two hats; (vi) spatial isotropy, entering as empirical input E4 of the reconstruction; (vii) linearity of the substratum update (A5), stronger than the affine restriction that amplitude-scale gauge invariance yields in the form that every first difference of the update is independent of its base point,

since such differences telescope to a constant plus an additive map; and, for the spatial results — the causal cone and Bell ceiling of §3.3, the observer selection of §4.6 and the companion reconstruction — the spatial constitution, which the characterization theorem of §3.4 does not use: (viii) site factorization — the configuration space is identified with a product of local configuration spaces over a set of sites; given that identification the dynamics determines its coupling graph, the dynamics alone has not been shown to select the identification, and the coupling graph, its causal cone and the regions of §3.3 and §4.6 are read relative to it; (ix) bounded coupling degree (A3), as one degree bound across the family of finite representatives — at a single finite lattice the bound holds automatically, so the item has content only in family form, where it gives the cone of §3.3 bounded growth and is used by §4.6; (x) the nearest-neighbour range — on the reference vacuum/uniform state class the coupling graph G_φ^{ref} of §3.3 is the simple-cubic graph — on which the form of the wave rule, its dispersion and the six-link carrier of the companion construction rest, and which the cone of §3.3 does not require; (xi) center independence (A4), in two clauses: the update commutes with lattice translations up to gauge, and the update at a site contains no explicit copy of that site's present value; (xii) background independence (A6), covariance of the dynamics under site-dependent internal-index transformations with the link coupling transformed alongside, which in the form formalized imposes no restriction on the rule and fixes the interface through which the gauge symmetry is read locally; and, for the continuous path of $*$-automorphisms from the identity to the update in §3.2 and §3.4, (xiii) the second-order form of the update, a posit of the selected lattice model, not derived and not an axiom of OI: each site of (viii) carries a previous and a current value in a finite abelian group, and one step is $x(t+1) = F(x(t)) - x(t-1)$, in phase-space form $(x(t-1), x(t)) \mapsto (x(t), F(x(t)) - x(t-1))$, with each F_i reading the current values at finitely many sites and each site read by finitely many F_i (kernel: `ruleDynamics` in `OIBridge/SecondOrderDrive.lean`), the update displayed in §3.2 being its instance with $q = 2$ and F the nearest-neighbour sum; and, for the Bell analysis of §3.3, (xiv) measurement independence, $p(\Omega \mid a, b) = p(\Omega)$ — the response-relevant pre-setting variable Ω is uncorrelated with the setting generators' outputs — a posit about the initial condition, not derived. Fourteen items, of unequal weight, and not fourteen independent physical assumptions: (viii) is foundational and (ix)–(xi) and (xiii) are stated relative to it; (x) implies the bound of (ix) for G_φ^{ref}, and with (vii) for G_φ; the second clause of (xi) depends on the identification of (viii) as well as on the rule; (xii) restricts no rule; and (xiv) enters only the Bell analysis of §3.3. Part I of the companion methodology paper gives the complementary view: an eight-rung analysis of the primitive that recovers the proper part, the existence of a dynamics (conditionally), bijectivity and per-moment finiteness as lemmas of the two-axiom base and isolates recurrence (Axiom 2) as the one irreducible non-lemma; the total finiteness of item (ii) and items (iv)–(xiv) lie outside the layer it treats. The two accounts describe one structure — the rung analysis counts what is *recovered*, this ledger counts what is *assumed* — and agree on the substance, including that the determinism recovery closes its hidden-sector margin with an empirical input and

that finiteness is delivered only per-moment. The two-axiom statement remains correct about what is *primitive*; this ledger states what is *assumed*, and the framework's claims are conditional on the full list.

Status ledger (theorems and conditions). The table transcribes the standing status of each load-bearing claim of this paper's foundational argument, with its canonical site; the matter-sector flags (the chirality hypothesis H-χ' and the Casimir-chain carrier input) are carried at their sources in the companion matter paper and in the book's open-problems ledger. It introduces no new claims; a row may not state a status stronger than its cited site, and any apparent strengthening is an error in the table, not a result.

Claim	Status	Site
(i) under (T): quantum representation for every process of the class, (ii) or not; P-indivisibility bears only on whether the represented dynamics is nontrivially indivisible (the representation itself universal, §3.4)	Imported ([11, 12]); accessible timescales; one unitary per time, no fixed generator; (i) does not imply (ii) (diagonal-$\hat{H}$; slow clock; `review3_probes.py`)	§3.4

Claim	Status	Site
(ii) $\Longrightarrow$ (iii): process $\Longrightarrow$ deterministic realization	Proved constructively, per accessible horizon — finite-horizon process dilation reproducing the full multi-time law; the realization inherits (C4); C1/C3/C4 legs unconditional; (C2) is not part of (iii), its static-bath form holding in the construction, and its necessity for physical hidden sectors is the C2-necessity row (persistence form without a mixing hypothesis, timescale form conditional on it); arbitrary-kernel form via the response-table extension; Consolidation: universal exact all-time form impossible (finite-order periodicity); all-time realizability of rooted visible maps classified (identity root, row-stochastic, periodic; kernel); recurrent multi-time-law gluing and pre-recurrence window approximation open	§3.4

Claim	Status	Site
Fixed-$\hat{H}$ form for realized processes	Proved, uncompressed — the realization's permutation unitary, $\hat{H} = i \log U$, exact at integer horizon times for uniform, slice, and weighted priors (`review4_probes.py`, P-F(a)–(c)); the external correspondence's $\leq n^3$ bound is for one unitary per time, with no fixed generator, for the source's finite-configuration class	§3.4
(iii) $\Longrightarrow$ (ii): realization $\Longrightarrow$ accessible non-Markovianity	Proved — the readback lemma: a (C4) gap forces $I(X_{<k}; X_{k+1} \mid X_k) \geq p_0 \delta^2 / \ln 2$ on the window; bare-recurrence form from a non-permutation rooted map at t_R	§2.3
CP-indivisibility of $\{\Phi_t\}$	Proved for the framework's diagonal-preserving (permutation-dilation) channels; fails for general CPTP families (Hadamard boundary)	§3.2
C1 necessity	Proved — any realization of a non-Markovian process has hidden influence at some accessible step (none anywhere $\Longrightarrow$ Markov); a non-permutation one-step T is the sufficient first-step witness, not an equivalence	§3.4, §1.3

Claim	Status	Site
C2 necessity	Non-mixing (persistence) form proved: in the coupling-step model of the physical-memory theorem (§3.4) — next visible state determined by the current visible and hidden states, any hidden map between couplings — observed conditional memory $M_k > 0$ at an accessible order k places some realized hidden conditional at that coupling time at least $\varepsilon_{\min}(M_k, n_V)$ from its reference prior π_H (physical-memory theorem, realized-ε contrapositive); in a bijective realization the retention of (C2-structural) holds for every law. Slow-bath form conditional on the hidden-sector mixing hypothesis; the quantitative bound $\varepsilon \geq \delta/(k-1)$ is proved, its constant saturated in the certification suite (`c2_mixing_probes.py`); ETH is physical motivation, not a premise	§3.4, §1.3
C3 necessity	Proved — data-processing bound in per-process form, $m \geq 2^{I^*}$	§3.4, §1.3

Claim	Status	Site
C4 necessity	Immediate given observed non-Markovianity (any realization of a non-Markov process reads differing past-conditionals); separation certified — (C1)–(C3) alone admit zero-memory realizations (the coin)	§1.3, §2.3

4.6 Structural Exclusion of Equilibrium-Phase Observers

An immediate corollary of (C1)–(C4) is a structural selection rule on where in a finite bijective (S, φ) observers can exist. The rule replaces a postulate often imported from elsewhere — the "past hypothesis" of low initial entropy, in its role of selecting observer-compatible configurations — with a theorem conditional on the effective-mixing hypothesis (EM) below and on the rate condition $\tau_{\mathrm{mix}} \ll \tau_S$ stated in the observer-selection corollary; under (EM) it is derived from the framework's own definitions, with (C2) read in its timescale form (C2-slow) — record persistence and readback being suppressed there, not excluded — for the coarse-grained description that the effective channel generates (Scope (ii)), and without (EM) it is a conjectural application.

Setup. Let V be a partition with hidden sector $B = S \setminus V$. Define the coupling-graph neighborhood $\Sigma_{\mathrm{loc}}(V)$ as the subset of S reached from V by at most $\tau_S/\Delta t$ steps of φ, where Δt is the update step. The exact Koopman operator of φ — and of its restriction to any invariant subsystem — is a permutation operator: unitary, with modulus-one spectrum and exact recurrence. It has no dissipative gap; exponential mixing cannot hold for it. Mixing, where it holds, is a property of a *compression*: let $\mathcal{K}_\Sigma := P_\Sigma \mathcal{U}_\varphi P_\Sigma$ denote the coarse-grained effective channel on the observable algebra of a local region Σ. **Effective-mixing hypothesis (EM).** *The local effective channel $\mathcal{K}_{\Sigma_{loc}(V)}$ mixes, with mixing time $\tau_{mix}(\Sigma_{loc}(V))$ set by its (dissipative) spectral gap.* (EM) is an explicit physical hypothesis — motivated by local thermalization phenomenology, not derived from φ — with the same status as the conditional-mixing curve of (C2).

Theorem (structural observer selection — conditional on (EM)). Under the effective-mixing hypothesis, for any partition V of bounded coupling-graph diameter, the hidden-sector memory time of the coarse-grained description generated by iterating $\mathcal{K}_{\Sigma_{\mathrm{loc}}(V)}$ obeys

$$\tau_B(V) \leq \tau_{\mathrm{mix}}(\Sigma_{\mathrm{loc}}(V)) + O(\varepsilon),$$

where ε absorbs typical-set and finite-size corrections. In particular, if $\Sigma_{\mathrm{loc}}(V)$ is in its local equilibrium macrostate in the sense of canonical typicality [31], $\tau_B(V)$ is bounded by the local thermal decorrelation time.

Proof. Correlations between a V-localized observable f and a B-localized observable g at time t require information to propagate from V to B at time 0 and back to influence the V-correlated component of g at time t. By the bounded per-step range of φ (its causal cone), this round trip stays within $\Sigma_{\mathrm{loc}}(V)$ for $t \le \tau_S$. In the description generated by iterating $\mathcal{K}_{\Sigma_{\mathrm{loc}}}$ the n-step channel is $\mathcal{K}^n_{\Sigma_{\mathrm{loc}}}$, and under (EM) its dissipative gap — which on an equilibrium macrostate sets the local thermal decorrelation rate — bounds the decay of such coarse-grained correlations exponentially at rate $1/\tau_{\mathrm{mix}}$; canonical typicality [31] supplies the local-equilibrium *state*, not the temporal mixing, which is exactly what (EM) posits. Hence $\tau_B(V) \le \tau_{\mathrm{mix}}(\Sigma_{\mathrm{loc}}) + O(\varepsilon)$. □

Corollary (anti-Boltzmann-brain). A Boltzmann brain is by construction a local fluctuation within a neighborhood Σ_{loc} that is itself in local thermal equilibrium — that is what "fluctuation from equilibrium" means. The exclusion is the conditional chain, in the coarse-grained description generated by iterating $\mathcal{K}_{\Sigma_{\mathrm{BB}}}$ (Scope (ii)): (EM) together with $\tau_{\mathrm{mix}}(\Sigma_{\mathrm{BB}}) \ll \tau_S$ implies the candidate partition fails (C2-slow), and hence — by the fast-bath erasure lemma of §2.3, under its uniform-erosion hypothesis — has its order-k readback gap suppressed as $e^{-(k-1)\tau_S/\tau_B}$ for every $k \ge 2$; that gap is what (C4), the condition the characterization actually requires, asserts. For a numeric illustration under a further generic-thermal estimate — (GT): the region's effective-channel correlation time is set by the thermal scale, $\sim \hbar/(k_B T_{\mathrm{local}}) \cdot O(1)$, an estimate that fails for protected slow modes (conserved-quantity bottlenecks, activation barriers, glassy or critical slowing) — the bound is at most $\sim 10^{-12}$ s for temperatures warm enough to support structured matter. The illustration rides (GT); the exclusion itself rides only the chain. Any finite-information observer has τ_S bounded below by its own internal processing time — for biological or synthetic BBs of any currently-envisioned type, $\tau_S \gtrsim 10^{-6}$ s. The slow form (C2-slow), $\tau_S \ll \tau_B$, therefore fails by at least six orders of magnitude for any BB whose local environment satisfies (GT) — and because (EM) posits genuine mixing rather than mere speed, the record the persistence form (C2-structural) requires is degraded with it: records are scrambled before readback up to a residue suppressed exponentially in $\tau_S/\tau_{\mathrm{mix}}$. The chain bounds the readback gap: by the fast-bath erasure lemma, the failure of (C2-slow) with uniform erosion bounds the order-k gap by $\delta_0\, e^{-(k-1)\tau_S/\tau_B}$ for every $k \ge 2$ — Boltzmann-brain observers so described carry at most exponentially suppressed readback at those orders, and since they fail (C2-slow), within that description they are not observers in the sense of (C1)–(C4) with (C2) read in its timescale form; the fixed-basis quantum representation they trivially possess ($S \iff D \iff Q_{\mathrm{fb}}$) carries no discriminating content. The exclusion is structural, not measure-theoretic — it follows from the incompatibility of (C2-slow) with fast-mixing local equilibrium, not from arguments about the ratio of BBs to "ordinary" observers in an

anthropic sample.

Corollary (observer-selection rule, with its rate condition). The theorem above bounds $\tau_B(V) \leq \tau_{\text{mix}}(\Sigma_{\text{loc}}(V)) + O(\varepsilon)$, while (C2)-slow requires $\tau_S \ll \tau_B$; those do not compose. Contradicting (C2-slow) additionally requires $\tau_{\text{mix}} \ll \tau_S$ — the condition the Boltzmann-brain corollary states explicitly. A slowly mixing equilibrium with $\tau_S \ll \tau_{\text{mix}}$ is entirely compatible with $\tau_S \ll \tau_B \leq \tau_{\text{mix}}$, so the valid form is that a sufficiently fast-mixing equilibrium supports no observer satisfying (C2) in its timescale form (C2-slow); for (C2-structural) and (C4) it gives suppression, not exclusion (§2.3). This matters for the de Sitter route, where $\tau_{\text{thermal}} \sim \hbar/(k_B T_{\text{GH}}) = 2\pi/H$ is Hubble-order and hence enormously longer than laboratory τ_S: a de Sitter thermal state does not erase observer memory on accessible timescales. Under $\tau_{\text{mix}} \ll \tau_S$: For (S, φ) in its equilibrium macrostate, canonical typicality [31] implies every local subsystem is also in its local equilibrium. By the theorem — under (EM) for each region's effective channel, in the coarse-grained description it generates — $\tau_B(V)$ is bounded by the local τ_{mix} for every bounded-diameter partition V. No bounded-diameter partition supports (C2-slow) in that description. That every relevant equilibrium region satisfies (EM) at the required rate is itself a physical hypothesis, to be established model by model; the theorem converts it into the selection rule and does not supply it. In that description, with (C2) read in its timescale form, the framework's observers therefore exist only in the non-equilibrium phase of (S, φ). The low-entropy initial condition commonly imported as a "past hypothesis" to select observer-compatible configurations is replaced here, within that description, by a structural selection rule: observers exist where (C1)–(C4), with (C2) read as (C2-slow), are satisfiable, and (C1)–(C4), so read, are not satisfiable in the coarse-grained description of the equilibrium phase.

Scope. The theorem uses bounded coupling, finiteness of S, the canonical-typicality result [31], and the effective-mixing hypothesis (EM) — the last an additional conditional physical premise, established model by model; it does not follow from isotropy or bounded-coupling regularity. Three scope notes. (i) *Locality of V.* The theorem assumes V has bounded coupling-graph diameter. Adversarially non-local partitions — cherry-picking distant subsystems that happen to be correlated — are excluded by that assumption; bounded coupling degree does not by itself bound the diameter of V, and the finite response time of a physical observer motivates the assumption. (ii) *Ergodicity.* The gap the argument needs belongs to the coarse-grained effective channel $\mathcal{K}_\Sigma$, not to φ itself: the exact Koopman operator of a finite permutation has modulus-one spectrum and cannot mix, so no spectral gap on typical orbits of φ is available or assumed. A gap for $\mathcal{K}_\Sigma$ is exactly the content of (EM) and does not follow from statistical isotropy or bounded-coupling regularity. The cycle structure of φ does not settle (EM), which concerns the one-step compression: a single cycle does not make the exact compressions relax (below), and orbits of length at most 2 do not prevent a gap (on four states with cells $\{0,1\}$, $\{2,3\}$, the transposition $(0\,2)$ gives $\mathcal{K}_\Sigma$ with eigenvalues 1 and 0). The exact linear rule of

[SM §4.2], whose largest energy shell for $d \geq 2$ splits into exponentially many cycles, is therefore not excluded by its cycle structure; whether it or the rounded bijection satisfies (EM) is open. The exact n-step compressions $P_\Sigma \mathcal{U}_\varphi^n P_\Sigma$ are compressions of a permutation, recur, and need not equal $\mathcal{K}_\Sigma^n$: for the cyclic shift on five states with a coarse variable marking one of them, they equal $\mathcal{K}_\Sigma$ for $1 \leq n \leq 4$ and the identity at $n = 5$, while $\mathcal{K}_\Sigma^n$ relaxes. The theorem and its two corollaries are therefore statements about the coarse-grained description generated by iterating $\mathcal{K}_\Sigma$; that these iterates govern the exact compressions over the accessible window is not established, and the corollaries do not assert that the exact visible law carries no accessible memory. (iii) *The cosmological realization.* The de Sitter horizon is thermal in the static-patch sense at Hawking temperature, but the globally expanding system is not in equilibrium with respect to any static-observer partition. The horizon's expansion sets $\tau_B \sim H^{-1}$ structurally, not thermally, which is the OI-native mechanism making human-timescale laboratory observation compatible with (C2).

---

5. CONCLUSION

The paper's central result is stronger than a finite-accuracy analogy and narrower than an unqualified claim that C1–C4 uniquely generate operational quantum mechanics. At finite observable horizon the stochastic laws, finite reversible deterministic realizations, and fixed-basis unitary/Born representations are exactly the same class:

$$S \iff D \iff Q_{\mathrm{fb}}.$$

Thus the reduced law of a finite embedded observer has an exact quantum representation, and every finite quantum record has a finite reversible realization. The representation statement is universal; it is not the selector. The content of observation incompleteness lies in history readback. C4 forces conditional memory and readout-relevant hidden predictive structure in every faithful completion. C1 and C3 are necessary diagnostics of that mediation and its capacity, while C2-structural, the persistence of the record to readback, holds by reversibility in every bijective realization and constrains coarse-grained descriptions of the hidden sector, with C2-slow one sufficient physical mechanism.

The spatial theory adds an exact causal-cone result. Because the coupling graph is defined by one-step dynamical dependence, k updates can carry influence no farther than graph distance k. Bell locality then follows when settings are interventions, applied at one step, that each read and write only their own wing on a common setting-independent pre-setting ensemble, **and** each wing's ± 1 outcome, recorded in every run, is read outside the other wing's graph cone; by §3.3 the resulting deterministic completion has $|S_{\mathrm{CHSH}}| \leq 2$. Reproducing quantum correlations therefore requires giving up either ontic parameter independence, which the intervention and readout conditions secure through the cone lemma, or measurement independence. §3.3 gives up ontic parameter independence; for

the specific SM/GR target the proposed implementation is preparation-indexed state-dependent adjacency, whose compatibility with operational no-signaling is part of the open hypothesis H-Bell. The superdeterministic alternative is not adopted: the manuscript proves only the model-independent necessary lower bound $I_{\mathrm{ont}} \geq 0.0309$ bits, while Hall–Branciard's optimized causal benchmark is about 0.080 bits. The PR-box construction remains the correct negative control for dropping the locality/intervention structure, not a counterexample to unitary representability.

The reverse direction is equally concrete. Every finite stochastic law has a finite reversible process dilation, and the operational gluing theorem shows that for any fixed finite dimension, finite instrument menu, finite horizon, and $\varepsilon > 0$, one finite reversible OI realization covers the whole corresponding quantum experimental envelope to accuracy ε. A single fixed finite carrier cannot equal the full continuum operational theory at all precisions, but the framework does not require an ontically infinite carrier; the continuum is the mathematical idealization of its finite-resolution physical description.

Accordingly, the phrase *equivalence of quantum mechanics and embedded observation* has a precise theorem-level meaning: **exact bidirectional equivalence of finite observable stochastic laws and their quantum representations, with history readback selecting the memory-bearing sector.** One boundary remains explicit rather than hidden in the slogan. Full equality with the standard local operational quantum theory requires a common compatible representation of the complete coherent instrument/composite hierarchy. Bell compatibility is a separate spatial-composite condition: the adopted deterministic branch keeps measurement independence but requires ontic parameter dependence with operational no-signaling. The existing classical action-labelled comb construction already covers finite menus of classical controls, so this residual is specifically the coherent/noncommuting and compositional layer. On a finite carrier, within complex matrix kinematics taken as given, that layer is characterized exactly by five completion conditions, none supplied by bare finite OI (§3.4); under the substratum's stated interventions, (iv), full reversible control, is not supplied (§3.4); whether an extended OI mechanism supplies it, which of the other conditions the substratum supplies, and a derivation of the kinematics are open, and none of these is a defect in the exact stochastic–quantum equivalence proved here.

Code and Data Availability

The verification code for §3.2 is under `papers/oi_lattice_code/coherence/`: `coherence_enumeration.py` (the (C1) counterexample, the exhaustive small-system enumeration behind the counts quoted in the Remark on the failure of (C1), and the Lemma 1 boundary bounds) `coherence_theorem_tests.py` (the Weyl transfer rule and normal form of Lemma 2, the separability threshold exhaustively at $s = 2, 3$, and both directions of Corollary 1 — a negative partial transpose wherever $w < \kappa$, an explicit measure-and-prepare form wher-

ever $w = \kappa$ — for $q = 2$ and $q = 3$), and `eb_pindivisibility_tests.py` (the strict one-way containment of complete dephasing in entanglement breaking, and an explicit reversible dilation whose one-step channel is entanglement-breaking while its visible process is P-indivisible). All are deterministic, depend only on NumPy and SciPy, and exit non-zero on failure; no result in this paper rests on sampling or on a random seed.

The repository is archived on Zenodo with concept DOI 10.5281/zenodo.19060318, which resolves to the latest version; per-release version DOIs are minted at release time, and a reader reproducing a specific claim should cite the version DOI of the release carrying it rather than the concept DOI. Paths above are relative to the repository root, which is also the root of the archived snapshot. The code is also available on GitHub at https://github.com/amaybaum/incompleteness and is released under the MIT License.

ACKNOWLEDGEMENTS

AI assistance in preparing this work is disclosed at
https://github.com/amaybaum/incompleteness#ai-assistance.

REFERENCES

[1] G. 't Hooft, *The Cellular Automaton Interpretation of Quantum Mechanics* (Springer, 2016).

[2] N. Bao, S. M. Carroll, and A. Singh, "The Hilbert space of quantum gravity is locally finite-dimensional," *Int. J. Mod. Phys. D* **26**, 1743013 (2017).

[3] H.-P. Breuer and F. Petruccione, *The Theory of Open Quantum Systems* (Oxford University Press, 2002).

[4] Á. Rivas, S. F. Huelga, and M. B. Plenio, "Entanglement and non-Markovianity of quantum evolutions," *Phys. Rev. Lett.* **105**, 050403 (2010).

[5] Á. Rivas, S. F. Huelga, and M. B. Plenio, "Quantum non-Markovianity: characterization, quantification and detection," *Rep. Prog. Phys.* **77**, 094001 (2014).

[6] P. Pechukas, "Reduced dynamics need not be completely positive," *Phys. Rev. Lett.* **73**, 1060 (1994).

[7] F. A. Pollock, C. Rodríguez-Rosario, T. Frauenheim, M. Paternostro, and K. Modi, "Operational Markov condition for quantum processes," *Phys. Rev. Lett.* **120**, 040405 (2018).

[8] P. Strasberg and M. Esposito, "Stochastic thermodynamics in the strong coupling regime: An unambiguous approach based on coarse graining," *Phys. Rev. E* **95**, 062101 (2017).

[9] B. Bylicka, M. Johansson, and A. Acín, "Constructive method for detecting the information backflow of non-Markovian dynamics," *Phys. Rev. Lett.* **118**, 120501 (2017).

[10] S. Milz, M. S. Kim, F. A. Pollock, and K. Modi, "Completely positive divisibility does not mean Markovianity," *Phys. Rev. Lett.* **123**, 040401 (2019).

[11] J. A. Barandes, "The Stochastic-Quantum Theorem," arXiv:2309.03085 (2023; v2, 2026).

[12] J. A. Barandes, "The Stochastic-Quantum Correspondence," *Philosophy of Physics* **3**(1):8 (2025).

[13] S. Calvo, "On the Stochastic-Quantum Correspondence," arXiv:2601.18720 (2026).

[14] L. S. Pimenta, "Divisible and indivisible Stochastic-Quantum dynamics," arXiv:2505.08785 (2025).

[15] W. F. Stinespring, "Positive functions on C-*algebras,"* Proc. Amer. Math. Soc.* **6**, 211–216 (1955).

[16] M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information* (Cambridge University Press, 2000).

[17] K. Brandner, "Dynamics of microscale and nanoscale systems in the weak-memory regime," *Phys. Rev. Lett.* **134**, 037101 (2025).

[18] K. Brandner, "Dynamics of microscale and nanoscale systems in the weak-memory regime: A mathematical framework beyond the Markov approximation," *Phys. Rev. E* **111**, 014137 (2025).

[19] J. Mehl, B. Lander, C. Bechinger, V. Blickle, and U. Seifert, "Role of hidden slow degrees of freedom in the fluctuation theorem," *Phys. Rev. Lett.* **108**, 220601 (2012).

[20] S. Gröblacher, A. Trubarov, N. Prigge, G. D. Cole, M. Aspelmeyer, and J. Eisert, "Observation of non-Markovian micromechanical Brownian motion," *Nature Communications* **6**, 7606 (2015).

[21] S. W. P. Kim, C. von Keyserlingk, and A. Lamacraft, "Measurement-induced phase transitions in quantum inference problems and quantum hidden Markov models," arXiv:2504.08888 (2025).

[22] J. Doukas, "On the emergence of quantum mechanics from stochastic processes," arXiv:2602.22095 (2026).

[23] J. S. Bell, "On the Einstein Podolsky Rosen paradox," *Physics Physique Fizika* **1**, 195–200 (1964).

[24] J. A. Barandes, M. Hasan, and D. Kagan, "The CHSH Game, Tsirelson's Bound, and Causal Locality," arXiv:2512.18105 (2025).

[25] A. Fine, "Hidden Variables, Joint Probability, and the Bell Inequalities," *Phys. Rev. Lett.* **48**, 291 (1982).

[26] T. Le, F. A. Pollock, T. Paterek, M. Paternostro, and K. Modi, "Divisible quantum dynamics satisfies temporal Tsirelson's bound," *J. Phys. A* **50**, 055302 (2017).

[27] B. S. Cirel'son, "Quantum generalizations of Bell's inequality," *Lett. Math. Phys.* **4**, 93–100 (1980).

[28] C. Bäcker, K. Beyer, and W. T. Strunz, "Local disclosure of quantum memory in non-Markovian dynamics," *Phys. Rev. Lett.* **132**, 060402 (2024); arXiv:2310.01205.

[29] A. Yosifov, A. Iyer, V. Vedral, and J. Sun, "On the emergence of quantum memory in non-Markovian dynamics," arXiv:2507.21907 (2025).

[30] R. Gangwar and U. Sen, "Genuine and Non-Genuine Quantum Non-Markovianity: A Unified Information-Theoretic Review," arXiv:2603.28277 (2026).

[31] S. Goldstein, J. L. Lebowitz, R. Tumulka, and N. Zanghì, "Canonical Typicality," *Phys. Rev. Lett.* **96**, 050403 (2006); arXiv:cond-mat/0511091.

[32] E. Nelson, "Derivation of the Schrödinger equation from Newtonian mechanics," *Phys. Rev.* **150**, 1079 (1966); *Quantum Fluctuations* (Princeton University Press, 1985).

[33] T. C. Wallstrom, "Inequivalence between the Schrödinger equation and the Madelung hydrodynamic equations," *Phys. Rev. A* **49**, 1613 (1994).

[34] S. L. Adler, *Quantum Theory as an Emergent Phenomenon* (Cambridge University Press, 2004).

[35] V. Chandrasekaran, R. Longo, G. Penington, E. Witten, "An algebra of observables for de Sitter space," *JHEP* **02** (2023) 082; arXiv:2206.10780.

[36] E. Witten, "Algebras, Regions, and Observers," *Proc. Symp. Pure Math.* **107**, 247 (2024); arXiv:2303.02837.

[37] J. Maldacena, "Real observers solving imaginary problems," arXiv:2412.14014 (2024).

[38] D. Harlow, M. Usatyuk, Y. Zhao, "Quantum mechanics and observers for gravity in a closed universe," *JHEP* **02** (2026) 108; arXiv:2501.02359.

[39] T. Akers, R. Akhmedov, R. M. T. Iyer, M. Loyola, "Closed universes and the entropy bound," arXiv:2501.04305 (2025).

[40] L. Donnelly, L. Eccles, S. Hollands, T. Kibe, "Partition relativity in algebraic quantum field theory," arXiv:2405.21010 (2024).

[41] K. Slagle and J. Preskill, "Emergent Quantum Mechanics at the Boundary of a Local Classical Lattice Model," *Phys. Rev. A* **108**, 012217 (2023); arXiv:2207.09465.

[42] S. L. Braunstein and A. K. Pati, "Quantum Information Cannot Be Completely Hidden in Correlations: Implications for the Black-Hole Information Paradox," *Phys. Rev. Lett.* **98**, 080502 (2007).

[43] S. W. Hawking, "Breakdown of Predictability in Gravitational Collapse," *Phys. Rev. D* **14**, 2460–2473 (1976).

[44] V. I. Arnold, *Mathematical Methods of Classical Mechanics*, 2nd ed., Graduate Texts in Mathematics **60** (Springer-Verlag, New York, 1989).

[45] K.-J. Engel and R. Nagel, *One-Parameter Semigroups for Linear Evolution Equations*, Graduate Texts in Mathematics **194** (Springer-Verlag, New York, 2000).

[46] A. Einstein, B. Podolsky, and N. Rosen, "Can quantum-mechanical description of physical reality be considered complete?" *Phys. Rev.* **47**, 777–780 (1935).

[47] A. Valdés-Hernández, L. de la Peña, and A. M. Cetto, "Bipartite entanglement induced by a common background (zero-point) radiation field," *Found. Phys.* **41**, 843–862 (2011).

[48] L. de la Peña, A. M. Cetto, and A. Valdés-Hernández, *The Emerging Quantum: The Physics Behind Quantum Mechanics* (Springer, 2014).

[49] K. Gödel, "Über formal unentscheidbare Sätze der Principia Mathematica und verwandter Systeme I," *Monatshefte für Mathematik und Physik* **38**, 173–198 (1931).

[50] D. H. Wolpert, "Physical limits of inference," *Physica D* **237**, 1257–1281 (2008).

[51] L. Hardy, "Quantum theory from five reasonable axioms," arXiv:quant-ph/0101012 (2001).

[52] L. Masanes, M. P. Müller, "A derivation of quantum theory from physical requirements," *New J. Phys.* **13**, 063001 (2011).

[53] G. Chiribella, G. M. D'Ariano, P. Perinotti, "Informational derivation of quantum theory," *Phys. Rev. A* **84**, 012311 (2011).

[54] A. M. Gleason, "Measures on the closed subspaces of a Hilbert space," *J. Math. Mech.* **6**, 885 (1957); P. Busch, "Quantum states and generalized observables," *Phys. Rev. Lett.* **91**, 120403 (2003).

[55] T. Heinosaari, O. Kerppo, L. Leppäjärvi, M. Plávala, "Simple Information Processing Tasks with Unbounded Quantum Advantage," arXiv:2308.07727 (2023; v4 April 2024), *Phys. Rev. A* **109**, 032627 (2024). Classical-dimension unboundedness for qubit prepare–measure behaviors via the PSD-rank /

nonnegative-rank separation; shared randomness simulates individual tasks, while the coordinated classical resource across the family is itself unbounded.

[56] A. J. P. Garner, M. P. Müller, arXiv:2004.06136; superseded by M. P. Müller, A. J. P. Garner, "Testing Quantum Theory by Generalizing Noncontextuality," arXiv:2112.09719, *Phys. Rev. X* **13**, 041001 (2023). Embedding of probabilistic/operational models under procedural-independence / no-restriction assumptions — the alternative route.

[57] M. J. W. Hall and C. Branciard, "Measurement-dependence cost for Bell nonlocality: Causal versus retrocausal models," *Phys. Rev. A* **102**, 052228 (2020).

[58] W. Heisenberg, "Über den anschaulichen Inhalt der quantentheoretischen Kinematik und Mechanik," *Z. Phys.* **43**, 172–198 (1927).

[59] H. P. Robertson, "The uncertainty principle," *Phys. Rev.* **34**, 163–164 (1929).

[60] M. Ozawa, "Universally valid reformulation of the Heisenberg uncertainty principle on noise and disturbance in measurement," *Phys. Rev. A* **67**, 042105 (2003).

[61] P. Busch, P. Lahti, and R. F. Werner, "Proof of Heisenberg's error-disturbance relation," *Phys. Rev. Lett.* **111**, 160405 (2013).

[62] P. Busch, " 'No information without disturbance': Quantum limitations of measurement," in W. C. Myrvold and J. Christian (eds.), *Quantum Reality, Relativistic Causality, and Closing the Epistemic Circle* (Springer, 2009).

[63] H.-P. Breuer, E.-M. Laine, and J. Piilo, "Measure for the degree of non-Markovian behavior of quantum processes in open systems," *Phys. Rev. Lett.* **103**, 210401 (2009); arXiv:0908.0238.

[64] G. Egri, M. Gömöri, B. Gyenis, and G. Hofer-Szabó, "Trajectory of probabilities, probability on trajectories, and the stochastic-quantum correspondence," arXiv:2602.23491v1 (2026).